SECUREKLOUD

Delivering Cloud Excellence With Trust

Key Concepts of Cloud Computing

An essential guide on Cloud Transformation: Purpose, Benefits, and Best Practices



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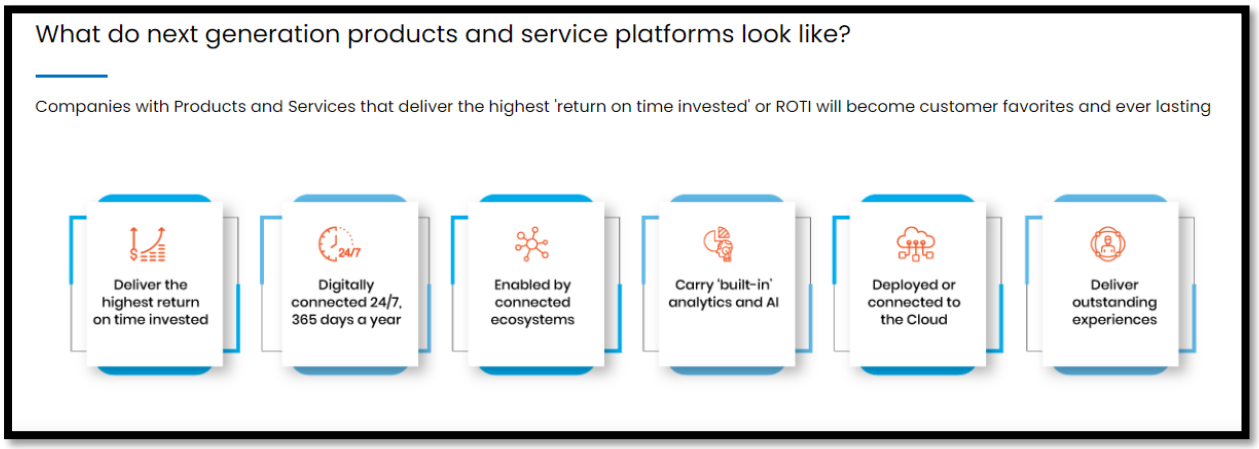
1. Introduction

Many people have often heard about cloud migration and cloud transformation. Well, these two terms define two different meanings. Cloud Migration means transforming an organization's digital assets, services, database, software, applications, and all IT resources that are moved into the cloud either partially or completely. Cloud migration is a significant step in Cloud transformation as it is a broader concept.

Whereas Cloud Transformation means an entire cycle of digital innovation from end-to-end improvements in an organization's past, present and future applications & its infrastructure, data and analytics, cloud security, cost optimization, software programs, and internal/external stakeholders data, the whole is transformed digitally on a cloud.

6 in 10 CxOs say cloud computing has improved security, as 60% of C-Suite executives have found cloud transformations have helped them in cost savings, scalability, ease of maintenance, and speed, according to Oracle.

Let's See for What Purpose Cloud Transformation Is Required and Its Benefits and Best Practices to make It a Success:



What is Cloud Transformation?

Cloud transformation is the process of transforming an organization's data, applications, software, or its entire IT infrastructure to the cloud. While similar to cloud migration, cloud transformation goes more in-depth with the modernization and optimization of an organization's data management, analytics, and cybersecurity to align with the company's overall digital transformation objectives.

Why Cloud Transformation?

Cloud transformation is a powerful and effective way to complete a metamorphosis of your entire organization across all areas – from technology and processes to people. It enables your business to

streamline operations and drive growth and innovation, such as the development of better customer experiences as well as new business models and products.

This, in turn, helps your business become more competitive and resilient, increasing revenue.

What's not to like? Done right, cloud transformation enables and accelerates digital transformation because it enhances and improves organizations' operations, processes, and services – making it easier and faster to digitalize. The flexibility and scalability of cloud easily supports changing business priorities while reducing the risk of wasted IT resources, which previously put a damper on investments in new digital services.

Understanding the anatomy of digital transformation:

While the cloud is a catalyst for digital transformation, migrating tools and applications to the cloud is more than just a lift-and-shift. Learn more about our digital transformation services and how organizations worldwide use the power of the cloud to redefine their business functions.

Purpose Of Cloud Transformation:

When an organization starts over to think about its cloud transformation journey, it must be made clear like scalability they are at present and its future business goals. Some of the purposes could be:

1. For small-scale businesses: the focus could be on developing business domains over more geographical boundaries and building strong foundation capabilities for effortless scalability.

2. For large institutions: the scenario might be different. It could be creating a basis for building workloads, modernization of data analytics, and a strong mechanism to support huge IT infrastructure and services in the cloud environment.

3. For mid-scale organizations: the purpose of cloud transformation could be quick scalability, higher performance, greater agility, adoption of approaches for richer external capabilities, better strategy formulation, and more.

So the purpose could vary from organization to organization but the main focus of every organization is fulfilled by cloud transformation such as business goals alignments, metrics establishment for monitoring and optimization, and tapping on data governance solutions that can drive good decision-making strategies.

Benefits Of Cloud Transformation:

There are no limits to leveraging so many benefits of cloud transformation for organizations. Let's have a look at those:

1. Excellent Operational Efficiency: Once you begin your cloud transformation journey, an authorized person will be able to access data in just one click, boost the flow of data processes, easily facilitate real-time communication, eliminate data silos, and enhance data

analytical abilities across the whole organization. Furthermore, the authorities could only focus on bringing innovative tech solutions rather than worrying about data infrastructure and its maintenance.

2. Hassle-Free Data Security: Cyber-attacks are not a new term for many organizations these days. The chances of data theft, phishing, and malware are turned into the biggest concern for all organizations at every scale. Furthermore, when the data is on-premises, the chances of cyber-attacks are more than that stored on the cloud. With cloud transformation, the data is secured via encryption, and layers of security are added to make it difficult for attackers to gain access. Other than this, the cloud hosting features offer multiple data backup options to decrease downtime significantly in case of system failure.

3. Cost Management at Its Best: An organization always bears high prices for getting the right resources at the right time at the right place even if it fails to use those resources at their highest potential. By having cloud transformation services, a business can not only terminate capital expenses but also remove the cost of having physical infrastructure at every scale. The expenses for all its maintenance can also be reduced and the border aspects could be discovered more quickly.

4. Easy Scalability Opportunities: The rise of competition is always a topic to consider for every organization. And to keep up that pace of scaling opportunities for staying in the marketplace is always necessary. Cloud transformation services help to establish a key differentiating factor for easy scaling and managing all the resources for months and years.

5. Seamless Collaboration: For multiple time zones, multiple teams and multiple work projects could be run under Cloud Transformation simultaneously. For any organization's success, the most important part is having seamless collaboration that boosts productivity across many cross-functional teams and they can get access to any information anytime.

Without investing any amount of time or money on additional complex infrastructure, organizations could easily develop, test and deploy their applications on multiple platforms by enabling cloud transformation. Advanced prototype functions will help to set up the right cloud computing resources that fit right into the cloud environments.

With The Help of Proper Strategic Planning and Its Execution, Here Are Some Best Practices To Make Sure That The Cloud Transformation Process Is Successfully Implemented:

1. An in-depth understanding of cutting-edge competition helps to grow different perspectives in strengthening an organization and eradicating the weak points. It encourages the future allocation of resources for optimum performance.

2. Entire acknowledgment of goals and strategies for bringing any kind of transformation to a business model. Organizations must determine their present and future perspective goals and roadmap before beginning any new journey. It helps to smoothen the workflow strategically with documentation of every system and process.

3. Know high-quality cloud hosting Platforms for a business cloud transformation roadmap. There are plenty of options available for cloud hosting but choosing the best one that fits business goals is less. So, for long-term valuable outputs, administrative support, technology compatibility, newest updates, easy deployments, and stability, analyze the best hosting platforms and ask your cloud service provider to execute the same.

4. Qualitative metrics for monitoring and optimization are essential to consider for a successful cloud transformation after its deployment. Real-time metrics measurement helps to keep track of all records and enhances the user experience at a better level. Organizations can focus on consistently value-adding with proper metrics to evaluate system performance and eradicate issues at prior.

5. Leveraging Data Governance Solutions is important before introducing cloud transformation in businesses. The current data governance policies and processes should be made clear and robust data governance policies/processes should be configured for their methods, responsibilities, and ways to incorporate new cloud-based data transformation.

6. Unleashing security gaps and compliances should be the beginning move to minimize the common risk associated with cloud transformation. A user must be prepared in advance for resolving any issues coming into the cloud environment and respond more quickly to such issues.

Get Into The Future Of Cloud Transformation With Aarav Solutions.

Gartner estimates that in 2023, the global end-user spending on public clouds will reach over $599 billion, up from $421 billion in 2021 and nearly $500 billion in 2022. Unlock the new growth opportunities with cloud transformation and scale more effectively to meet up the customer demand with Aarav Solutions expertise solutions as a cloud consulting services provider. Explore the unlimited potential in the most cost-efficient way and get top-notch digital transformation for seamless capabilities, robust services, automated tools, exclusive insights, security analytics, error scanning, data compliances, and more.

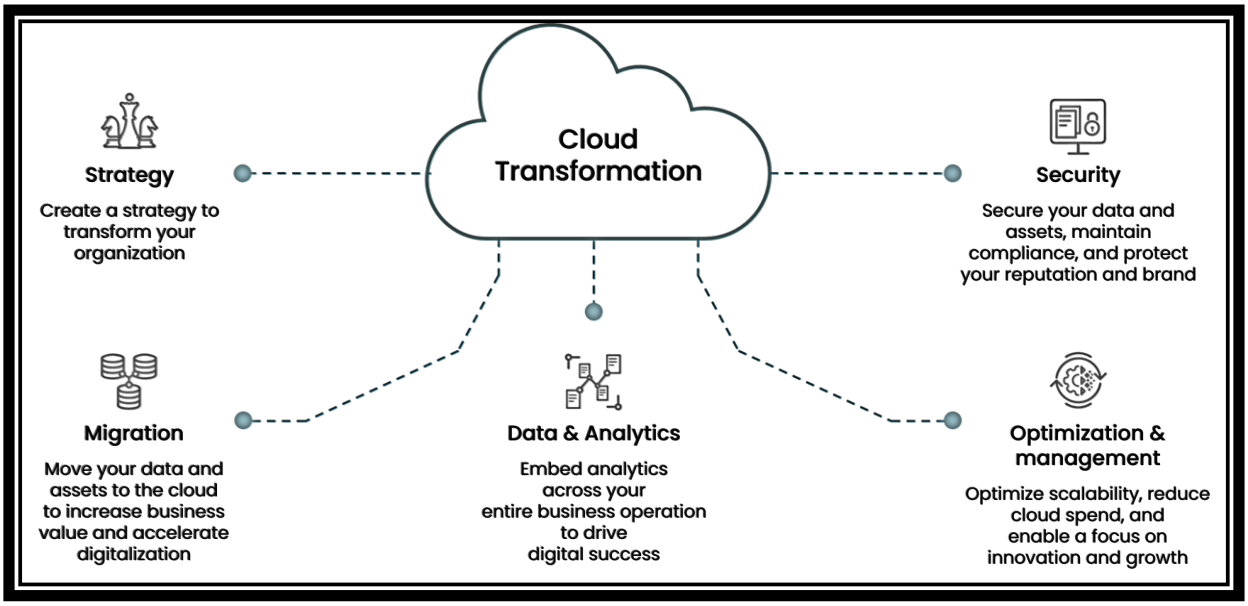
1.1.1 What's the Difference Between Cloud Transformation and Cloud Migration?

Cloud migration is one piece of cloud transformation. Cloud migration is focused solely on the act of moving applications, data, and other components from on-premises to a cloud-based infrastructure.

Cloud transformation, however, is the enablement of a complete evolution of a business. To achieve this, organizations focus on cloud strategy, migration, management and optimization, data and analytics, and cloud security to become more competitive and resilient.

This is why, as our research shows, more than 50% of "digital champions" – what we call the 7% of organizations that are successful with digital transformation – believe the cloud is a game changer that helps disrupt their business model for the better.

Components of Cloud Transformation



CLOUD STRATEGY: As a precursor to strategy, it's important to first assess your status and undergo a benchmarking exercise to see how your organization stacks up to your competitors, and understand where your organization's capabilities and gaps are relative to a maturity curve. Benchmarking is an essential but often overlooked key to knowing where to invest your time, effort, and resources to maximize your returns.

But it's not all about the technology. To leverage the results of the above analysis, your strategy will require a deep integration of both IT and business strategy. This means either aligning your enterprise cloud strategy to existing business plans or devising an overarching strategy to leverage the power of the cloud for your business goals.

Once you've developed your cloud strategy, it's crucial to create a strategic roadmap to guide the process. Just as no two companies are completely alike, no two roadmaps are completely alike, and great care must be taken during this stage to ensure your investment yields commensurate returns.

SELECTING A CLOUD PLATFORM PROVIDER: If your organization is a large business, you'd be wise to choose one of the top-tier cloud platform providers, such as Amazon Web Services (AWS), Microsoft Azure, or Google Cloud Platform (GCP).

All of them provide dedicated servers, serverless compute, and storage nodes, as well as numerous platform-monitoring tools to simplify cloud adoption.

They also offer a broad range of integrations with third-party providers, further expanding the functionality and potential of your cloud infrastructure to drive business growth. Many companies retain expert help in the cloud assessment and platform selection stage.

ONE CLOUD OR MULTIPLE CLOUDS?: By using a multi-cloud strategy, you can leverage best-of-breed hosting infrastructure for specific business workloads, improving performance and service delivery quality while driving innovation across the business. In addition, you can reduce operating costs by choosing the platform that offers the best value for the money with your target workload. Moreover, you can improve operational resilience against IT outages by diversifying your IT operations across multiple cloud hosting platforms.



However, you might want to stay with one cloud or collapse a multi-cloud set-up so you don't overtax your operations, development, and security teams who would have to manage multi-cloud applications and systems. Also, leveraging just one cloud for your data makes tracking costs and managing vendors a lot simpler.

CLOUD MIGRATION: Before you begin, you'll need to decide which systems to move first, since enterprises usually transition one portion of their data center systems to the cloud at a time. Some companies choose to first migrate their email and messaging, which already have built-in cloud support, while others migrate systems whose hardware is reaching its end-of-life.

Regardless, an effective cloud consulting services provider can help your organization successfully execute your cloud migration and make it move faster by using an accelerator tool. For instance, Trianz' EVOVE (powered by CompilerWorks) utilizes high levels of automation and reusable components to drive accelerated and high-accuracy migrations of legacy data platforms to a modern architecture.

CLOUD DATA AND ANALYTICS: Cloud analytics leverages advanced cloud-based analytics processing, streamlining business intelligence processes, data integration and analysis, and the visualization of data insights for decision-making purposes. Digital champions weave advanced analytics into their operational processes and work culture – shaping their strategy, business requirements, business model design, data structure, measurement, and reporting to digitally transform their operations.

Because of this, digital champions tend to lead their competitive peer group in key operating metrics across marketing, service, sales, and internal operating processes. They plan and execute their new data-driven business models to promote loyalty in their customer base and drive growth in profitability at a rate faster than their competitors'. Embedding analytics ubiquitously across the entire business operation drives this digital success.

CLOUD OPTIMIZATION AND MANAGEMENT: The cloud unlocks a whole new world of possibilities through scalability and simplified operations management, where server

infrastructure can adapt alongside your business. Compared to traditional on-prem infrastructure, the cloud delivers substantial cost benefits at scale at the enterprise level. Its real-time analytics, virtual R&D, and design capabilities help accelerate new product/service and internal IT application launches.

The right consulting services provider can work with you to manage your cloud infrastructure more efficiently and effectively to increase ROI and improve the scalability of your cloud infrastructure. They can assist you with managing cloud consumption, capacity, performance, and cost through continuous monitoring; dynamic, scalable operations; control and visibility of OpEx spend, compliance, outcomes, and SLA-driven governance; and built-in security compliance.

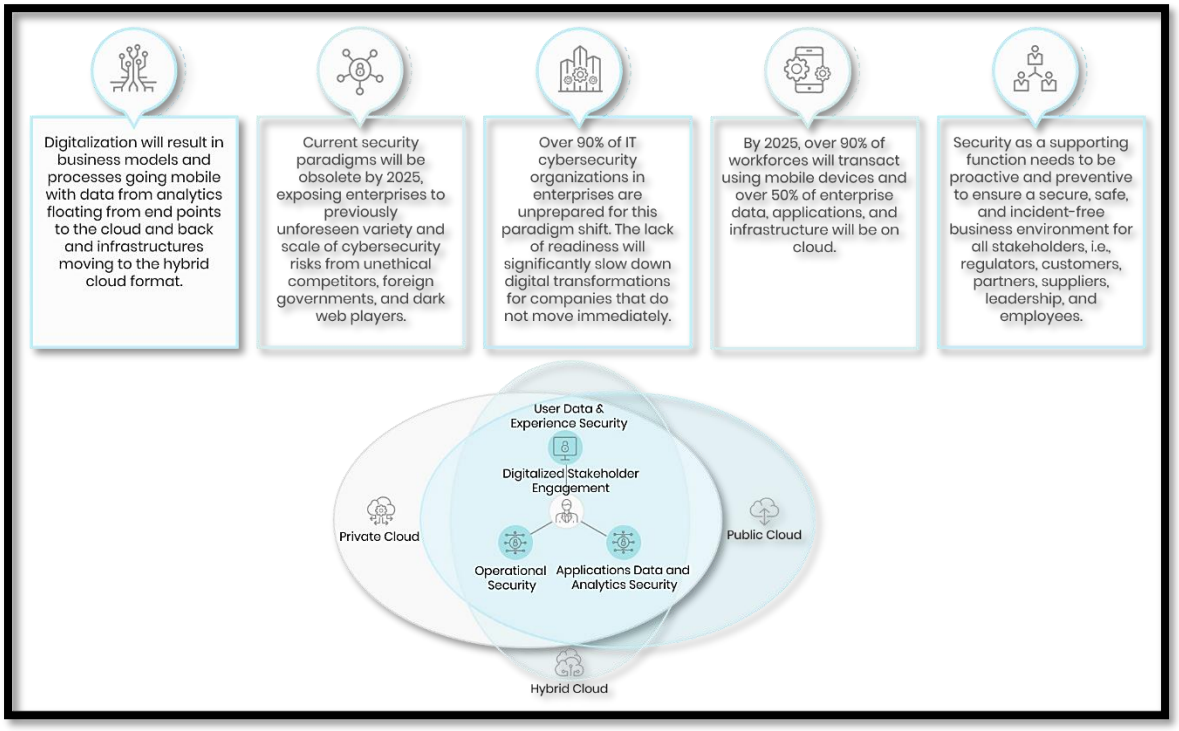
CLOUD SECURITY: Cloud security is an essential piece of a successful cloud transformation. Therefore, it needs to be a priority and preplanned component of your overall strategy and roadmap. The first step for many is a cloud security assessment to determine your enterprise's cloud security posture. The goal is for the organization to understand its current state, gaps, and intended future state.

Key facets of a security program that should then be recommended include the following:

- Governance, risk management and compliance to manage your enterprise GRC effectively while staying on top of innovation.
- Secure SDLC consulting to reduce vulnerabilities and detect flaws early in every aspect of your software development lifecycle.
- Cloud security strategy integrating cloud strategy within your cybersecurity posture to guide your journey — from security to relationship management.
- Identity and Access Management (IAM) strategy to fit your business, from assessing the past and understanding the present, to designing the future.
- Continuous audit and assessment to increase enterprise value and reduce audit fatigue.

A robust cloud security program enables the organization to develop a security compliance culture and protects you against threats to IP and customer data.

THE ROLE OF CYBERSECURITY



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1.1.2 Cloud Transformation: A Game Changer

With the right approach and preparation, cloud transformation can take your organization to an entirely different level. By leveraging the cloud to disrupt your business, you're not only modernizing your technology, but also reinventing your whole organization – driving innovation, customer growth, and improvement in your bottom line, as well as enabling your business to become more resilient, agile, and disaster-proof.

2. Cloud Based Security Solutions for Businesses

(SMB or Large Enterprise)

With more and more businesses adopting cloud infrastructure to safeguard their critical data, there is also a continued increase in concerns over data privacy and security.

Despite trends such as cloud computing, mobility and virtualization which are enabling various ventures to enhance their agility and flexibility at a fast pace, APTs (Advanced Persistent Threats) like Hydraq and Stuxnet are also bringing forth widespread challenges in the protection of information.

With the numerous security threats looming at such a significant rate, it is now essential for businesses to predetermine data breaches and prevent the attacks frm penetrating their systems.

Currently, businesses should implement elaborate cloud security measures to establish a strong data encryption and protection posture.

They need to be versatile and use technologies in their setting like file integrity monitoring, intrusion detection, two-factor authentication and much more.

Also, cloud security is an evolving challenge which can be addressed only if the technologies and tools are structured to grow along with it.

In such scenarios, numerous organizations are looking for a solution to safeguard their cloud architecture and subsequently relying on various solutions to help them curb breaches.

To help you navigate and identify the right cloud security solution, herein is an overview of what these systems do along with a list of options for you to choose from.

What Are Cloud Security Solutions?

Cloud security solutions safeguard data, networks and access by securing data, workloads and apps that are hosted on Cloud infrastructure.

The same servers, software and apps that used to be located on-premises in the previous decades, are now migrated to cloud services, but they still need the same security protection level as before.

Firms use such tools in enforcing policies, protecting information transmitted through or stored in cloud applications. These provisions offer secure access points, like gateways and sandboxes, for only approved users to only launch and subsequently run cloud applications in a safe environment.

Administrators can set permissions, manage governance and monitor anyone accessing and using the applications. Numerous cloud security solutions offer data loss prevention controls and encryption to safeguard further data and documents stored within cloud applications.

Numerous cloud security solutions operate like secure email gateways and secure web gateways. While they may offer similar access points, they also serve to safeguard entire application networks as opposed to just email documents or server data.

Why Choose Cloud-Based Security Solutions?

Notwithstanding the size of your business, whether a small enterprise or a large corporation, provided you have access to the internet, you remain susceptible to cyberattacks.

As such, there is a need to stay vigilant and safeguard your network from malicious attackers, and among the best ways to guarantee this is to adopt cloud security solutions since they are essential for protecting cloud-hosted data.

These solutions help you secure your data as follows:

- Cloud security features improve the monitoring and tracking of attacks in a global scale compared to non-cloud security solutions. They offer signature and real-time firewall updates, blocking unsafe traffic protection etc.
- Offers 24/7 live tracking and security of websites, APIs, serverless apps, containers etc through continuous monitoring and web application attack protection.
- Aside from network and application scanning, cloud security solutions also enhance application speed performance through resources that can scale-up on demand.

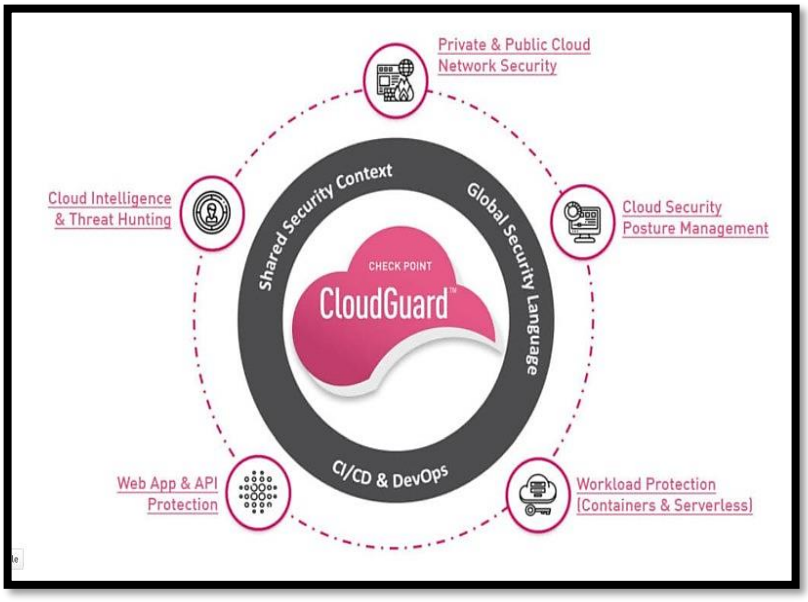
Having identified the benefits of cloud security solutions, here are some of the best cloud security solutions in the market today.

A. Check Point CloudGuard

Checkpoint is one of the most trusted vendors and manufacturers of network firewalls and other security products. This Israeli company focuses on creating security solutions for any business market.

CloudGuard, Checkpoint's product for securing multi-cloud environments, is a context-based cloud security engine supporting many different cloud environments such as AWS, Azure, Vmware, Oracle etc.

Rather than having multiple components of security control products, CloudGuard takes a unified approach to threat prevention and posture management from a unified platform.



The product includes also API protection so malicious actors can't abuse your services. It also comes with both private and public cloud security and monitoring, and it actively hunts for new threats on the horizon. Notably, CloudGuard supports also security for containerized and serverless apps.

Teams can take advantage of the DevSecOps that's offered. This allows for seamless and continuous integration of good security practices with software in development, lessening attack surface.

B. CloudPassage Halo

CloudPassage aims to be a unified security platform for all of your cloud assets such as Public IaaS (Infrastructure as a Service), Cloud Servers and Containers.



It works on the principle that proactive security is a better approach than vendor-level endpoint security.

This safety-first philosophy helps prevent zero-day attacks even if the vendor can't issue a hotfix quickly enough.

The Halo product comes in three different services as below:

- Cloud Secure: Cloud security posture management (AWS, Azure, Google cloud).
- Server Secure: Cloud workload protection platform (Linux and Windows)
- Container Secure: Security for Docker, Kubernetes and CD

It works on the Infrastructure-as-a-Service (IaaS) level (for both private and public cloud infrastructures) and also covers hybrid environments.

Likewise, CloudPassage works at the bare metal level to ensure server-level compliance with industry standards. Beyond that, it supports most containerization, giving users three layers of cloud security.

CloudPassage can be used by employees all the way from junior-level engineers through the C-suite.

For example, Cloud Security Engineers and DevOps Engineers can easily review reports to help them determine faster, more secure workflows and procedures.

Cloud Architects can use the all-in-one solution to easily manage multiple layers of security. Even executives can benefit from data-intensive reports on audit compliance and form a better line of communication with their DevOps teams.

C. Threat Stack Cloud Security Platform

Threat Stack's platform is designed to both secure your cloud and give your team the most information both observable and concise.

This solution is mostly focused on monitoring your security, what they call as "observability" in DevOps terms.



With a special focus on containerized cloud applications, their security platform operates on five layers of monitoring and observability: the application itself, container orchestration (how your container behaves as it launches), the container itself, the hosting server, and a customized cloud security management console.

By offering observable security suggestions and shortcomings, Threat Stack helps teams identify vulnerabilities before hackers do.

Even if a security incident does arise, the platform can help Cloud Security Engineers figure out what's going on faster and respond more quickly and accurately.

International regulatory and legal compliance is also tracked, helping ensure that the customer is fully covered.

Finally, Threat Stack prides its platform on the fact that regardless of what the "flavor of the month" is with containerization, it only has one focus: cloud security.

D. Symantec Cloud Workload Protection

Security giant Symantec's cloud product goes beyond security. Supporting all major cloud computing platforms with "adapters" for their product (AWS, Azure, Google), Cloud Workload Protection (CWP) can automatically scale up and down as workload resources scale up and down as well.

CWP in short is for protecting your workload instances that run on the popular Cloud Providers. To do this, an agent must be installed on each workload instance in order to monitor and enforce the security policies to the instance.

Security-wise, the software has a three-pronged approach. First, it simply "listens" to incoming and outgoing traffic and records anomalies. Second, it analyzes situations using a recommendation engine, providing real-time feedback and even automatic reconfiguration to prevent attacks. Third, it "acts" by automatically deploying whatever is necessary, provided it has the permissions to do so.

A key CWP feature is that everything on the workload is controlled and secured by the agent, such as zero-day threats, file integrity monitoring, host-based IPS, vulnerability protection etc.

With everything "baked in" through native cloud adapters, developers can "set it and forget it" until there's an incident requiring manual attention.

E. Datadog

Datadog sells and maintains only its flagship product, unlike some of its competitors, which is centered around monitoring (for performance, security etc.)

Marketed as "Cloud Monitoring as a Service" (CMaaS), Datadog's goal is to combine proven security practices with modernized reporting infrastructure.

Regardless of how servers hosting a cloud-based application are configured, Datadog creates a "deep-dive" view of what's really happening behind the scenes.

Anyone authorized to can use the platform to see real-time statistics on database queries, server cluster uptime, resource usage, and even suggestions for improvements as development continues.

Individual container instances can be debugged if an application is experiencing problems. All of these operations are performed at the GUI level.



Regarding security monitoring, Datadog helps you identify potential threats to systems in real-time. Your applications and network can send logs to Datadog which parses and enriches these logs in order to extract security information and metrics.

F. Hytrust

HyTrust offers three different product categories for security:

- CloudControl: For workload protection (advanced access control, policy enforcement, forensics etc).
- DataControl: Workload encryption at-rest.
- Key Management: For managing the encryption keys of workload encryption service.

HyTrust workload encryption is designed for applications handling sensitive data in the cloud. Its workload encryption services work with both in-house, physical servers and all major cloud platforms.

Rather than simply encrypt databases and salt them, HyTrust's platform automatically granularly encrypts each container and virtual machine and contents within them.

With so many layers of data control, it's almost impossible for hackers to steal this information even if there is a security incident.

Regardless of where an application is hosted, HyTrust provides a full suite of keying tools, including complete key management and even cloud-based rekeying on the fly. It can also be easily ported from one cloud provider to another.



G. PaloAlto Prisma

Given that seven out of the top ten Fortune companies utilizing PaloAlto's Prisma security platform, there's a reason the product is so well-known. This is that Prisma truly does it all, from analytics reporting to threat prevention.



Prisma's Posture Management services allow for both regulatory compliance measures and threat detection.

With a combination of a user and entity behavior analysis engine and/or signature-based antimalware crawler, cloud apps remain secured. Working from a server level, virtual machine level, and container and headless level, no layer is forgotten.

Cloud Network Security that supports full and partial network segregation allows for network engineers to truly define secure data transmission.

Other features, like identity management, layer seven (web application network layer) level security, and API protection as well as a sleek reporting infrastructure make this truly an all-in-one cloud security solution.

H. Fortinet

Fortinet's cloud security products place an emphasis on application portability on any cloud platform. This gives development teams the most breathing room regarding changing hosting providers, should the need arise.

With Fortinet, Cloud Engineers can have full visibility of the current state of security of their applications. With their Security-as-a-Service API, developers can easily hook existing products into Fortinet's native cloud adapters. With its unique application layer tools, even Office365 data is protected.

Fortinet offers a unique "Cloud Security Hub" with all the reports and analytics Cloud Architects need.

Using FortiCASB you can have SaaS visibility and control and using FortiCWP you can deploy IaaS visibility and control as well.

I. Cisco Cloud Security

While most of its competitors sell all-in-one solutions, Cisco allows teams to pick and choose individual cloud security components and products if they simply need to complete an already-existing partial security solution.



For example, some cloud security products offered by Cisco include:

- Cisco Umbrella: for secure cloud access
- Cisco Cloudlock: protection of SaaS applications
- Cloud Email Security: blocks and remediates email threats
- Stealthwatch Cloud: monitors IaaS instances
- AppDynamics: application performance monitoring

Teams who develop SaaS apps can utilize Cisco's SaaS application security. Threats can be easily detected and thwarted, as Cisco provides real-time security alerts. Setting security aside, teams can also take advantage of AppDynamics, which allows developers to easily view performance and resource metrics.

With multi-cloud network protection and workload protection, teams developing apps on public, private, and hybrid clouds can all benefit.

In addition, Cisco provides email security, assistance with securing remote workers' machines, and many other tertiary security solutions.

J. Lacework



Lacework's cloud security platform takes a top-down approach to securing all types of cloud-based apps, containerized or not, and on any major platform.

The "Lacework Polygraph" establishes "normal" baseline behavior, which is used to later determine potential behavior anomalies. This allows for potential attacks to be detected even if there's no signature match.

Lacework's overall infrastructure monitoring makes it easy for Security Engineers to identify shortcomings and fix them. Artificial intelligence makes alerts more informative than average and doesn't clog developers' inboxes.

Finally, Lacework has a suite of applications to assist with security. For example, their "Configuration and Audit Control" application allows developers to set compliance and legal requirements for which the tool will regularly scan.

Workload and container security, advanced threat protection, and vulnerability monitoring all can be viewed granularly, as well. Lacework's unique "DevSecOps" tool makes adding cybersecurity to the workflow easy.

K. CipherCloud

CipherCloud is a cloud security solution company established in 2010. It operates across three distinct models—PaaS, SaaS and IaaS.

CipherCloud offers a single platform that secures sensitive customer information across multiple private and public cloud applications without compromising on performance or functionality.



Features:

- It provides services across multiple sectors, including telecommunications and government. It safeguards essential cloud applications such as OneDrive, Google Drive, among others.
- It also facilitates data loss prevention, cloud computing as well as related security, cloud encryption gateway, among others uses.

L. Qualys

Qualys is a secure cloud security solution provider which was started in 1999. It provides security to your devices and compliance assets as well as related services. It facilitates data protection through identifying compromised assets and handling them accordingly.



This solution secures your web apps and devices while assisting you to stay compliant through its exclusive cloud-only solutions (needs no software or hardware). The firm assesses threat data to ensure nothing penetrates your system.

Features:

- Qualys provides end-to-end solutions such as cloud infrastructure security, endpoint security, web app security, DevSecOps, compliance, among others—ensuring your teams stay in sync with one another.

- Offers reliability and security across both private and public clouds, threat protection, vulnerability management, monitoring, file integrity, among other utilities.

M.Sophos

A renowned security firm, Sophos was started in 1985 and offers cloud solutions such as firewall, encryption, web and mobile security, antimalware, among others. It offers a cloud-based console known as Sophos Central.

Features:

- Sophos Central offers runtime protection against attacks such as ransomware, mitigating exploits within the web, preventing external DLLs to load, plugins, java applications, among others.

- Provides security solutions designed to suit email, web, mobile, wireless, web servers and other platforms.

N. Proofpoint

Proofpoint. although best known for its email security solution, is another cloud-based security provider which offers protection against numerous cybersecurity threats. Proofpoint's services are customized for all sizes of businesses.

When considering security vectors (holes within a network that provide a leeway for bad actors to access the system), one of the weakest points is email. Proofpoint precisely works on email security management so customers can ensure their systems are free of threats.

Aside from ensuring nothing risky gets in, Proofpoint also safeguards even outgoing data. Furthermore, it also guarantees that while it preserves to avoid loss of data, it comprises no keys for decrypting any information.

Features:

- Offers email, SaaS, protection against attacks from typical email attachments as well as a mobile solution from targeted cyber vulnerabilities.
- It safeguards sensitive and essential business data by providing cloud email security solutions to all sizes of businesses and brands.

O. SiteLock

SiteLock was started in 2008 and secures more than 21 million websites across the world. SiteLock is a cloud-based security firm which safeguards sites from attacks, malware and other cyber threats.

Features:

- SiteLock provides website protection by scanning vulnerabilities, identifying and eradicating malware.
- It safeguards against attacks such as SQL, DDoS and XSS.

SiteLock offers dynamic and static caching, load balancing and a global Content Delivery Network (CDN), thereby boosting and enhancing website performance.

2.3 Final Word

Different cloud security products boast different pricing and features. When choosing an exclusive security solution, it is essential you identify your particular needs and then select a solution that best covers these needs.

However, and most importantly, ensure that the solution you ultimately settle for offers you adequate monitoring and support.

What's more, if you are a new startup and are yet to identify the precise cloud security solution you can invest in, you can fundamentally start with selecting an effective cloud hosting program. Adequate cloud hosting can also work well in offering you added security from attackers.

3. Data Centre Hyperscalers

As the demand for data booms, hyperscaling explodes in conjunction with that demand. Here

are the Top 10 data centre hyperscalers across the planet:

If you're reading this, you have data centres to thank for it...

Data centres are the backbone of the digital economy, providing the infrastructure necessary to store, process and transmit vast amounts of data.

Hyperscale data centres are massive facilities built by companies with vast data processing and storage needs.

These firms may derive their income directly from the applications or websites the equipment supports, or sell technology management services to third parties.

Hyperscale data centres are significantly larger than enterprise data centres and significantly outperform them, too, making them an attractive choice for data centre operators.

In the modern era, many companies are building and maintaining their own.

Hyper scales are designed to be highly efficient and increasingly - sustainable; many using renewable energy sources and advanced cooling technologies.

As the demand for data booms, data centre hyperscaling explodes in conjunction with that demand.

And it's all thanks to you, reader…

Here's the list of the Top 10 data centre hyperscalers based on their 2023 operational data centre capacity:

10. Oracle

Oracle is a major player in the enterprise software and cloud computing markets, offering a range of services to businesses and organisations around the world. The company has built a large network of data centres to support its various services, with a focus on energy efficiency and sustainability. Oracle is also exploring new technologies such as AI and machine learning to optimise its data centre operations. The company is planning to build new data centres in the United States, which will be powered by renewable energy sources.

Oracle's 2023 operational data centre capacity: 400 MW

9. OVHcloud

OVHcloud is a major player in the cloud computing market in Europe, offering a range of services to businesses and organisations of all sizes. The company has built a large network of data centres to

support its various services, with a focus on energy efficiency and sustainability. OVHcloud is also exploring new technologies such as AI and machine learning to optimise its data centre operations. The company is planning to build new data centres in Europe, which will be powered by renewable energy sources.

OVHcloud's 2023 operational data centre capacity: 600 MW

8. Tencent

Tencent is a major player in the online gaming and social media markets in China. The company has built a large network of data centres to support its various services, with a focus on scalability, reliability, and security. Tencent is also committed to using renewable energy sources to power its data centres. The company is planning to build new data centres in China, which will be powered by renewable energy sources.

Tencent's 2023 operational data centre capacity: 700 MW

7. Huawei

Huawei is a major player in the telecommunications and cloud computing markets, offering a range of services to businesses and organisations around the world. The company has built a large network of data centres to support its various services, with a focus on energy efficiency and sustainability. Huawei is also exploring new technologies such as AI and machine learning to optimise its data centre operations. The company is planning to build new data centres in China, which will be powered by renewable energy sources.

Huawei's 2023 operational data centre capacity: 1,000 MW

6. Alibaba

Alibaba is a major player in the e-commerce and cloud computing markets in China. The company has built a massive network of data centres to support its various services, with a focus on scalability, reliability, and security. Alibaba is also exploring new technologies such as AI and machine learning to optimise its data centre operations. The company is planning to build new data centres in China, which will be powered by renewable energy sources.

Alibaba's 2023 operational data centre capacity: 1,200 MW

Here are the Top 5, of the Top 10 data centre hyperscalers in the world...

5. Apple

Apple is a major player in the consumer electronics market, but the company also operates a large network of data centres to support its various services such as iCloud and Apple Music. Apple is committed to using renewable energy sources to power its data centres, and the company has also

developed innovative cooling technologies to improve energy efficiency. Apple is planning to build a new data centre in Denmark, which will be powered entirely by renewable energy.

Apple's 2023 operational data centre capacity: 1,500 MW

4. Microsoft

Microsoft is a major player in the cloud computing market, offering a range of services through its Azure platform. The company has invested heavily in building out its data centre infrastructure, with a focus on energy efficiency and sustainability. Microsoft is also exploring new technologies such as liquid cooling and AI to further improve the efficiency of its data centres. The company is planning to build new data centres in China, which will be powered by renewable energy sources.

Microsoft's 2023 operational data centre capacity: 2,200 MW

3. Meta

Meta, formerly known as Facebook, is a major player in the social media and online advertising markets. The company has built a massive global infrastructure of data centres to support its various services, with a focus on energy efficiency and sustainability. Meta is also exploring new technologies such as AI and machine learning to optimise its data centre operations. The company is planning to build a new data centre campus in Singapore, which will be one of the largest in the region.

Meta's 2023 operational data centre capacity: 2,500 MW

2. Amazon Web Services (AWS)

Amazon Web Services (AWS) is the world's largest cloud computing provider, offering a wide range of services to businesses and organisations of all sizes. AWS has built a massive global infrastructure of data centres, with a focus on scalability, reliability, and security. The company is also committed to sustainability, with a goal of powering its data centres with 100% renewable energy. AWS is planning to deploy an additional 500 MW by 2031 in key markets, accounting for close to 70% of the newly deployed colocation.

AWS' 2023 operational data centre capacity: 3,000 MW

1. Google

Google is a leader in the development of artificial intelligence and machine learning technologies, which are increasingly being used to optimise data centre operations and improve energy efficiency. The company owns and operates data centres all over the world, designed to be highly efficient and sustainable, using renewable energy sources and advanced cooling technologies. In total, Google has nearly 30 owned data centres either in operation or under development across 10 countries throughout the world. Google is also investing in new technologies such as liquid cooling and AI to further improve the efficiency of its data centres.

Google’s 2023 operational data centre capacity: 3,500 MW

4. Shared Responsibility Model

The **Shared Responsibility Model** is a security and compliance framework that outlines the responsibilities of **cloud service providers (CSPs)** and **customers** for securing every aspect of the cloud environment, including hardware, infrastructure, endpoints, data, configurations, settings, operating system (OS), network controls and access rights.

In its simplest terms, the Shared Responsibility Model dictates that the cloud provider—such as Amazon Web Service (AWS), Microsoft Azure, or Google Cloud Platform (GCP)—must monitor and respond to security threats related to the cloud itself and its underlying infrastructure. Meanwhile, end users, including individuals and companies, are responsible for protecting data and other assets they store in any cloud environment.

Unfortunately, this notion of shared responsibility can be misunderstood, leading to the assumption that cloud workloads – as well as any applications, data or activity associated with them – are fully protected by the cloud provider. This can result in users unknowingly running workloads in a public cloud that are not fully protected, making them vulnerable to attacks that target the operating system, data or applications. Even securely configured workloads can become a target at runtime, as they are vulnerable to zero-day exploits

Shared Responsibility across the three cloud service delivery models

There are three main cloud service models:

- Software as a service (SaaS)
- Infrastructure as a service (IaaS)
- Platform as a service (PaaS)

Each of these cloud delivery models are subject to the concept of shared responsibility. However, ownership of security tasks and functions varies depending on the delivery model in use.

Software as a service (SaaS): SaaS is a software delivery model wherein the vendor centrally hosts an application in the cloud that can be used by a subscriber. In this model, the provider is responsible for application security, as well as its maintenance and management.

Platform as a service (PaaS): PaaS is a platform delivery model that can be purchased and used to develop, run and manage applications. In the cloud platform model, the vendor provides both the hardware and software generally used by application developers; the service provider is also responsible for security of the platform and its infrastructure.

Infrastructure as a service (IaaS): IaaS is an infrastructure delivery model wherein a vendor provides a wide range of compute resources such as virtualized servers, storage and network equipment over the internet. In this model, the business is responsible for maintaining security of anything they own or install on the cloud infrastructure, such as the operating system, applications, middleware, containers, workloads, data and code.

Service Type	Vendor Responsibility	User Responsibility
SaaS	Application security	Endpoints, user and network sec misconfigurations, workloads and data
PaaS	Platform security, including hardware and software	Security of applications developed on the pla Endpoints, user and network security, and workloads
IaaS	Security of all infrastru components	Security of any application installed on the infrastru (e.g. OS, applications, middle Endpoints, user and network security, workloads, and

Element	SaaS	PaaS	IaaS
Application Security	CSP	User	User
Platform Security	CSP	CSP	User
Infrastructure	CSP	User	CSP
Endpoint Security	User	User	User
Data Security / Data Protection	User	User	User
Network Security	CSP	CSP	User
User Security	User	User	User
Containers and Cloud Workloads	User	User	User
APIs and Middleware	CSP	User	User
Code	User	User	User
Virtualization	CSP	CSP	User

The Shared Responsibility Model in practice

4.2.1 Direct Control

While the Shared Responsibility Model is based on the idea that two or more parties play a role in ensuring security of distinct elements within the public cloud environment, it is important to note that the customer and CSP do not share responsibility for the same asset.

Rather, the CSP or the customer has full and complete responsibility for the security of all assets under their direct control, regardless of the service model type.

For example, the customer will always have responsibility for data security, compliance and access regardless of whether they are following a SaaS, PaaS or IaaS model. Practically speaking, this is because CSPs have no visibility into data that is stored in the public cloud and therefore cannot effectively manage data security or access.

Customers are typically also responsible for:

- Identity Access and Management (IAM)
- User security and credentials
- Endpoint security
- Network security
- Security of workloads and containers
- Configurations
- APIs and middleware
- Code

Meanwhile, the cloud provider—such as Amazon, Microsoft or Google—are responsible for areas for which they possess direct control. This typically includes security of:

- The physical layer and all associated hardware and infrastructure
- The virtualization layer
- Network controls and provider services
- Facilities that run cloud resources

4.2.2 Divided responsibilities

In some IaaS and PaaS models, security responsibilities may vary depending on the cloud provider or the terms outlined in the service level agreement (SLA).

For example, when it comes to a network control like a firewall, the cloud service provider may be responsible for providing the firewall service. However, it is up to the user to manage all other aspects such as configuration, rules, monitoring and response. While both parties play a role in the security element, the responsibilities are still clearly defined and divided.

Likewise, if a customer is using a public cloud data storage service offered by a CSP, then the cloud provider is responsible for all aspects of that cloud datacenter, including security, monitoring, maintenance and updating. However, the customer is still wholly responsible for securing any data within the cloud environment, as well as ensuring only authorized users can access it.

Based on the concept of divided responsibility, no party has authority over another in terms of how they protect their assets. For example, a customer cannot dictate how or when their CSP performs monitoring and testing. That said, the service agreement should outline the steps the provider will take to protect customers, as well as how documentation for that activity will be shared. Typically, cloud vendors produce regular audit reports to confirm that they are taking the necessary and proper steps to protect their customers.

Shared Responsibility Model Advantages

While a shared security model is complex and requires careful consideration and coordination between the CSP and customer, the approach offers several important benefits to users. These include:

- **Efficiency**: Though the customer bears significant levels of responsibility under the Shared Responsibility Model, some key aspects of security – such as security of hardware, infrastructure and the virtualization layer – are almost always managed by the CSP. In a traditional on-premises model, these aspects were managed by the customer. The shift to the cloud frees up IT staff to refocus efforts on other tasks and needs, as well as dedicate available resources and investments to those areas for which they bear responsibility.
- **Enhanced protection**: Cloud service providers are hyper focused on the security of their cloud environment and typically dedicate significant resources to ensuring their customers are fully protected. As part of the service agreement, CSPs conduct robust monitoring and testing, as well as timely patching and updating.
- **Expertise**: CSPs often have a higher level of knowledge and expertise when it comes to the emerging field of cloud security. When customers engage a cloud vendor, they benefit from the partner organization's experience, assets and resources.Shared Responsibility Best Practices

As organizations shift to the cloud, many are defining their relationships with CSPs for the first time. As companies navigate this complex territory, we offer the following best practices:

- **Carefully review the SLA**. Security responsibilities will differ depending on the cloud model, cloud provider and other variables. It is critical for organizations to carefully review their SLA with their cloud vendor to ensure they are fully aware of their security responsibilities and to identify any potential gray areas that need to be clarified. In the event the organization is changing cloud providers or shifting to a new delivery model, they must carefully reevaluate their contract and identify any changes. While the SLA may appear to be very similar to a past agreement, even slight changes in wording can leave the organization vulnerable to serious security risks. Finally, for organizations operating a multi-cloud environment, it is crucial to review each SLA individually as the terms are not standard across vendors. Small differences across these agreements must be factored into the overarching cloud security strategy and architecture.

- **Embrace DevSecOps**. DevSecOps – short for Development Security and Operations – is the practice of integrating security continuously throughout the software and/or application development lifecycle in order to minimize security vulnerabilities and improve compliance – all without impacting speed of release cycles. A DevSecOps or is an absolute necessity for any IT organization that is leveraging containers or the cloud, both of which require new security guidelines, policies, practices, and tools.
- **Ensure robust identity and access management**. The cloud customer is also completely responsible for defining access rights to cloud-based resources and granting access to authorized users. These efforts should be incorporated into the organization's broader IAM policy and solution set.
- **Identify a trusted cybersecurity partner**. Cloud security is fundamentally different from securing on premises networks. Updating and adapting the cybersecurity strategy and toolset to address new cloud-based risks can be both overwhelming and complicated – especially if the organization is operating a hybrid or multi-cloud environment. A cybersecurity partner can assist the organization's internal security team in managing all aspects of cloud security – from selecting a CSP, to understanding their specific security responsibilities to deploying and integrating the tools and solutions that will protect the business.
- **Prioritize data security**. Cloud customers are always fully responsible for any data stored in the cloud or produced by applications in the cloud. As such, organizations must develop a robust data security strategy specifically designed to protect cloud-based data, whether it is in use, at rest or in motion.

CrowdStrike's Cloud Security Solutions

CrowdStrike has with the world's most advanced cloud-native platform that protects and enables the people, processes and technologies that drive modern enterprise. The industry continues to recognize CrowdStrike as a leader, most recently with CRN naming CrowdStrike a WPowered by the CrowdStrike Security Cloud, the CrowdStrike Falcon® Platform leverages real-time indicators of attack, threat intelligence, evolving adversary tradecraft and enriched telemetry from across the enterprise to deliver hyper-accurate detections, automated protection and remediation, elite threat hunting and prioritized observability of vulnerabilities.

Learn more about CrowdStrike's Cloud Security Solutions – including our services specific to AWS, GCP, and Azure, :



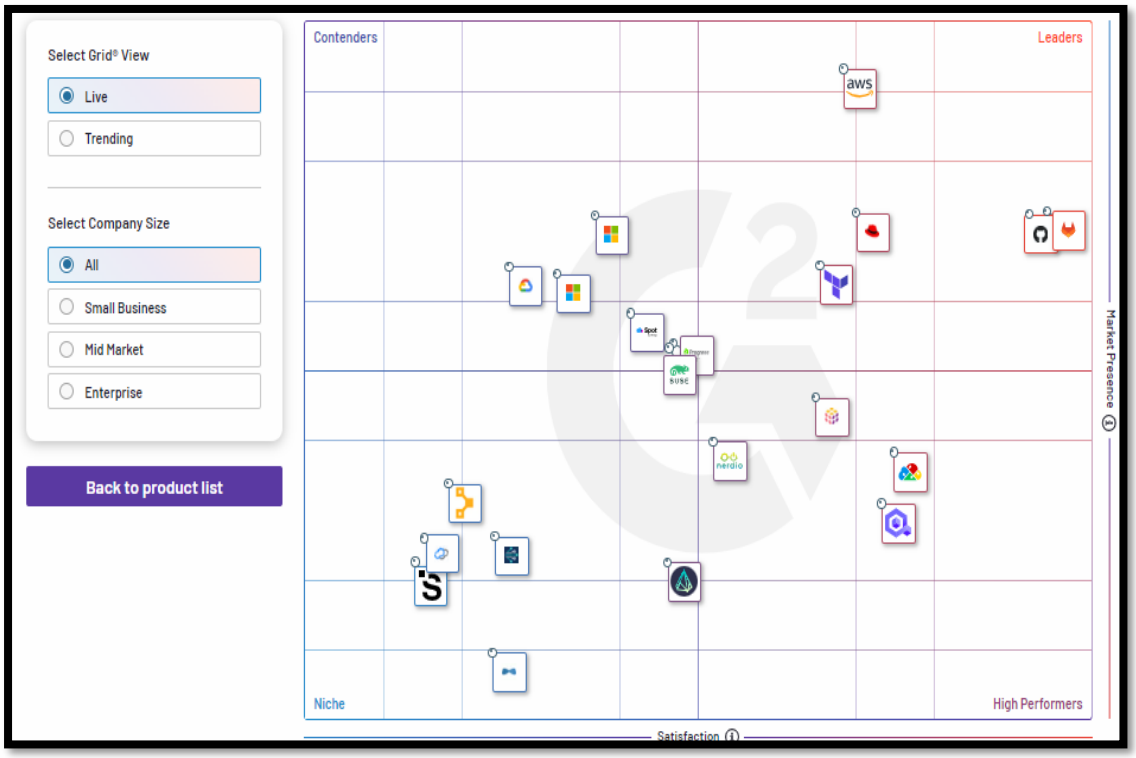
5. Cloud Infrastructure Automation Software

Cloud Infrastructure Automation User Satisfaction

"Ease of Admin", "Quality of Support", "Ease of Use", and "Has the product been a good partner in doing business?" are the top four factors that positively impact user satisfaction for Cloud Infrastructure Automation products. These factors are determined by an algorithm that selects the attributes that are most likely to predict user satisfaction within this category.

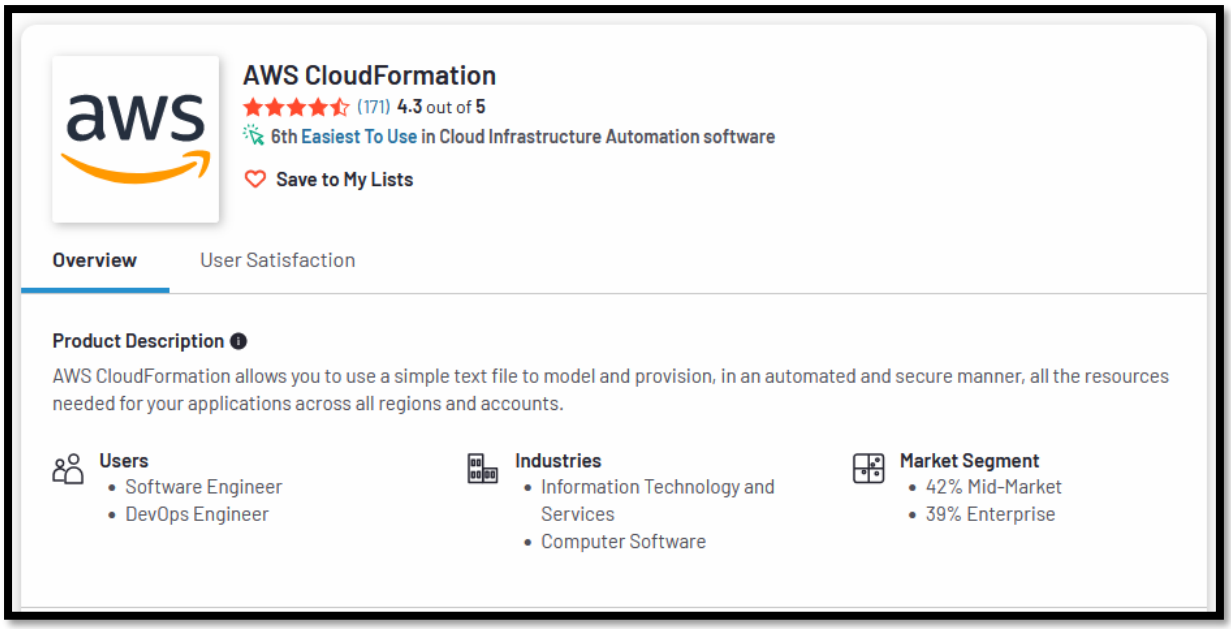
G2 Grid® for Cloud Infrastructure Automation:

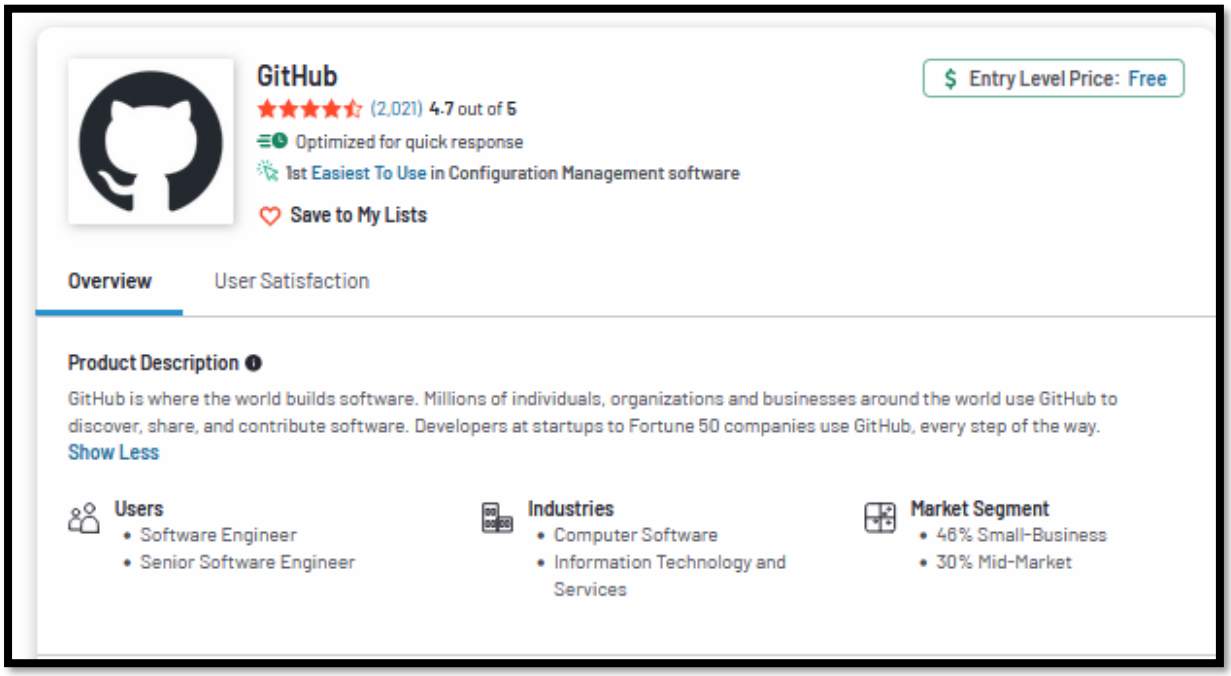
Check out the G2 Grid® for the top Cloud Infrastructure Automation Software products. G2 scores products and sellers based on reviews gathered from our user community, as well as data aggregated from online sources and social networks. Together, these scores are mapped on our proprietary G2 Grid®, which you can use to compare products, streamline the buying process, and quickly identify the best products based on the experiences of your peers. https://www.g2.com/categories/cloud-infrastructure-automation

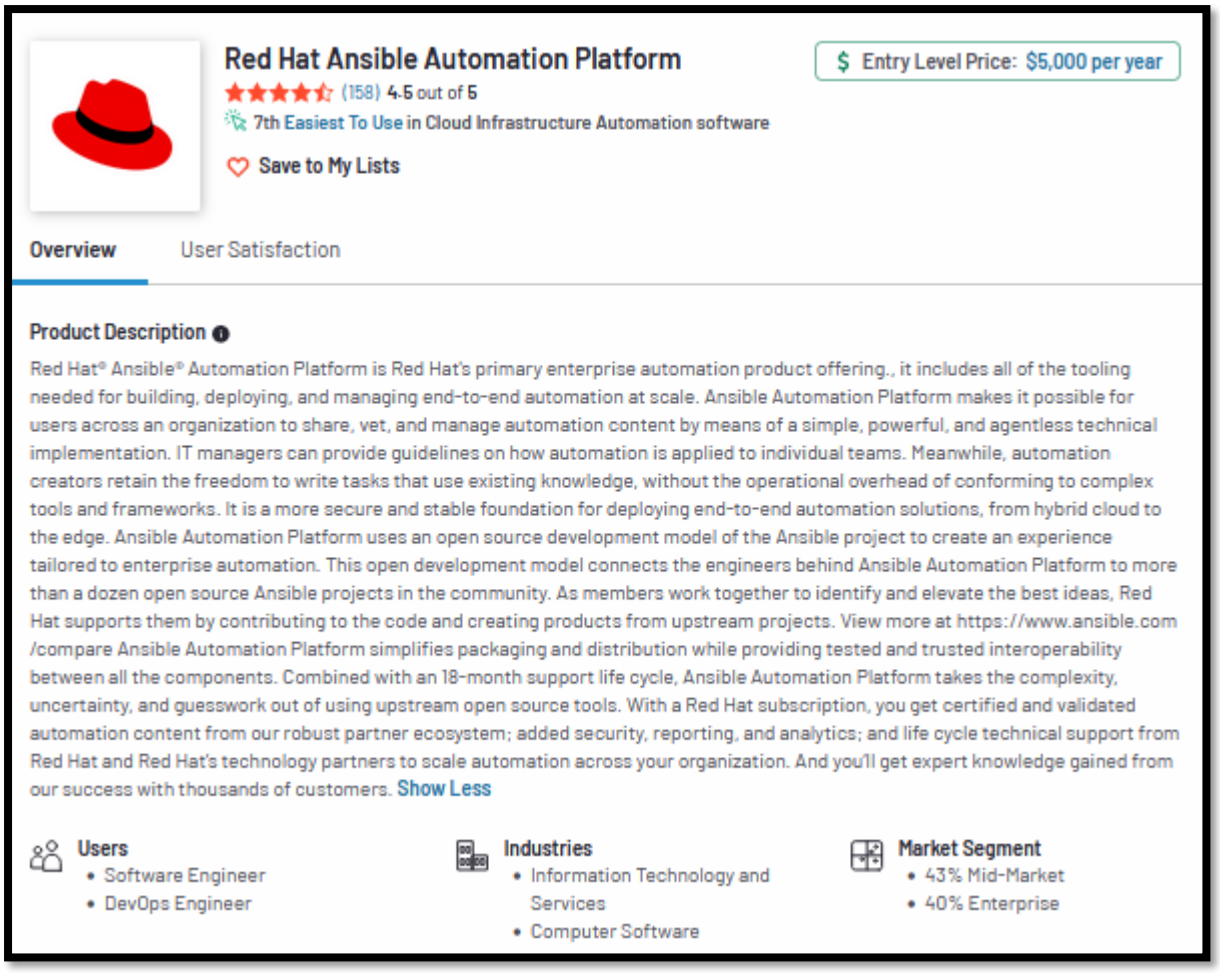


Ease of Admin 8.4 /10, Quality of Support, 8.9 /10, Ease of Use 8.7/10 As partner: 9.0/ 10

G2 takes pride in showing unbiased reviews on user satisfaction in our ratings and reports. We do not allow paid placements in any of our ratings, rankings, or reports.


AWS CloudFormation
(171) 4.3 out of 5
6th Easiest To Use in Cloud Infrastructure Automation software
Save to My Lists
Overview
User Satisfaction
Product Description
AWS CloudFormation allows you to use a simple text file to model and provision, in an automated and secure manner, all the resources needed for your applications across all regions and accounts.
Users
• Software Engineer
• DevOps Engineer
Industries
• Information Technology and Services
• Computer Software
Market Segment
• 42% Mid-Market
• 39% Enterprise


GitHub
(2,021) 4.7 out of 5
Optimized for quick response
1st Easiest To Use in Configuration Management software
Save to My Lists
Entry Level Price: Free
Overview
User Satisfaction
Product Description
GitHub is where the world builds software. Millions of individuals, organizations and businesses around the world use GitHub to discover, share, and contribute software. Developers at startups to Fortune 50 companies use GitHub, every step of the way.
Show Less
Users
• Software Engineer
• Senior Software Engineer
Industries
• Computer Software
• Information Technology and Services
Market Segment
• 46% Small-Business
• 30% Mid-Market



Red Hat Ansible Automation Platform

$ Entry Level Price: $5,000 per year

(158) 4.5 out of 5

7th Easiest To Use in Cloud Infrastructure Automation software

Save to My Lists

Overview | User Satisfaction

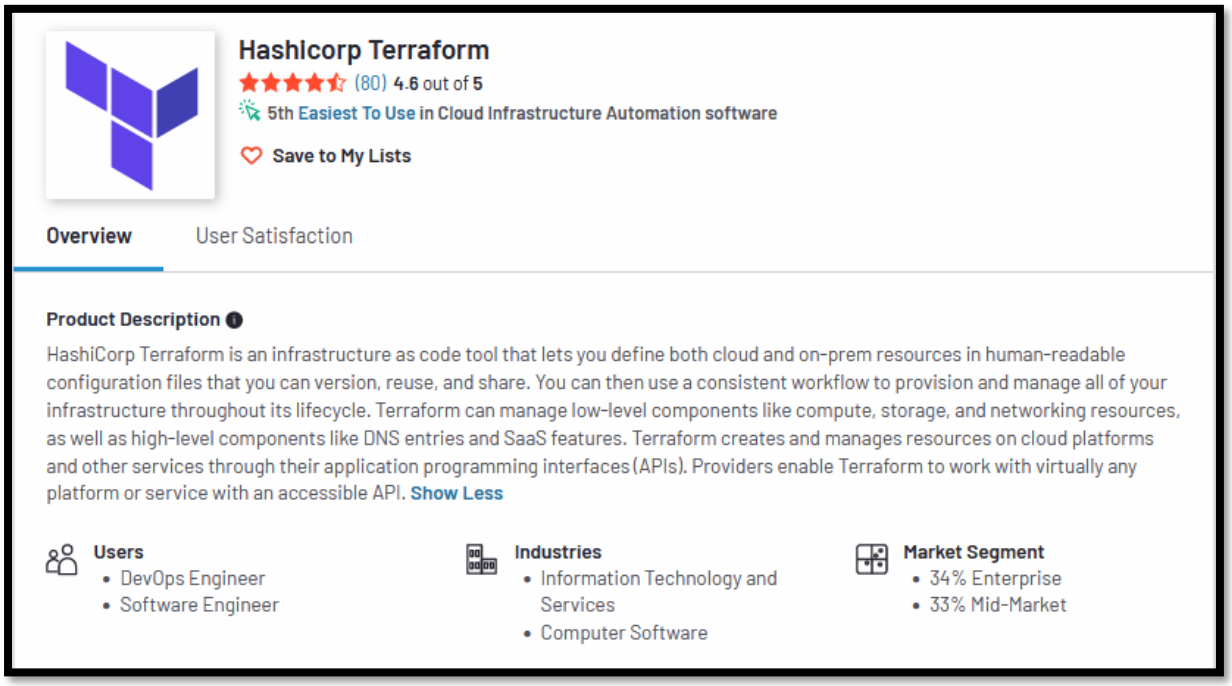
Product Description

Red Hat® Ansible® Automation Platform is Red Hat's primary enterprise automation product offering., it includes all of the tooling needed for building, deploying, and managing end-to-end automation at scale. Ansible Automation Platform makes it possible for users across an organization to share, vet, and manage automation content by means of a simple, powerful, and agentless technical implementation. IT managers can provide guidelines on how automation is applied to individual teams. Meanwhile, automation creators retain the freedom to write tasks that use existing knowledge, without the operational overhead of conforming to complex tools and frameworks. It is a more secure and stable foundation for deploying end-to-end automation solutions, from hybrid cloud to the edge. Ansible Automation Platform uses an open source development model of the Ansible project to create an experience tailored to enterprise automation. This open development model connects the engineers behind Ansible Automation Platform to more than a dozen open source Ansible projects in the community. As members work together to identify and elevate the best ideas, Red Hat supports them by contributing to the code and creating products from upstream projects. View more at https://www.ansible.com /compare Ansible Automation Platform simplifies packaging and distribution while providing tested and trusted interoperability between all the components. Combined with an 18-month support life cycle, Ansible Automation Platform takes the complexity, uncertainty, and guesswork out of using upstream open source tools. With a Red Hat subscription, you get certified and validated automation content from our robust partner ecosystem; added security, reporting, and analytics; and life cycle technical support from Red Hat and Red Hat's technology partners to scale automation across your organization. And you'll get expert knowledge gained from our success with thousands of customers. Show Less

Users
- Software Engineer
- DevOps Engineer

Industries
- Information Technology and Services
- Computer Software

Market Segment
- 43% Mid-Market
- 40% Enterprise



Hashicorp Terraform

(80) 4.6 out of 5

5th Easiest To Use in Cloud Infrastructure Automation software

Save to My Lists

Overview | User Satisfaction

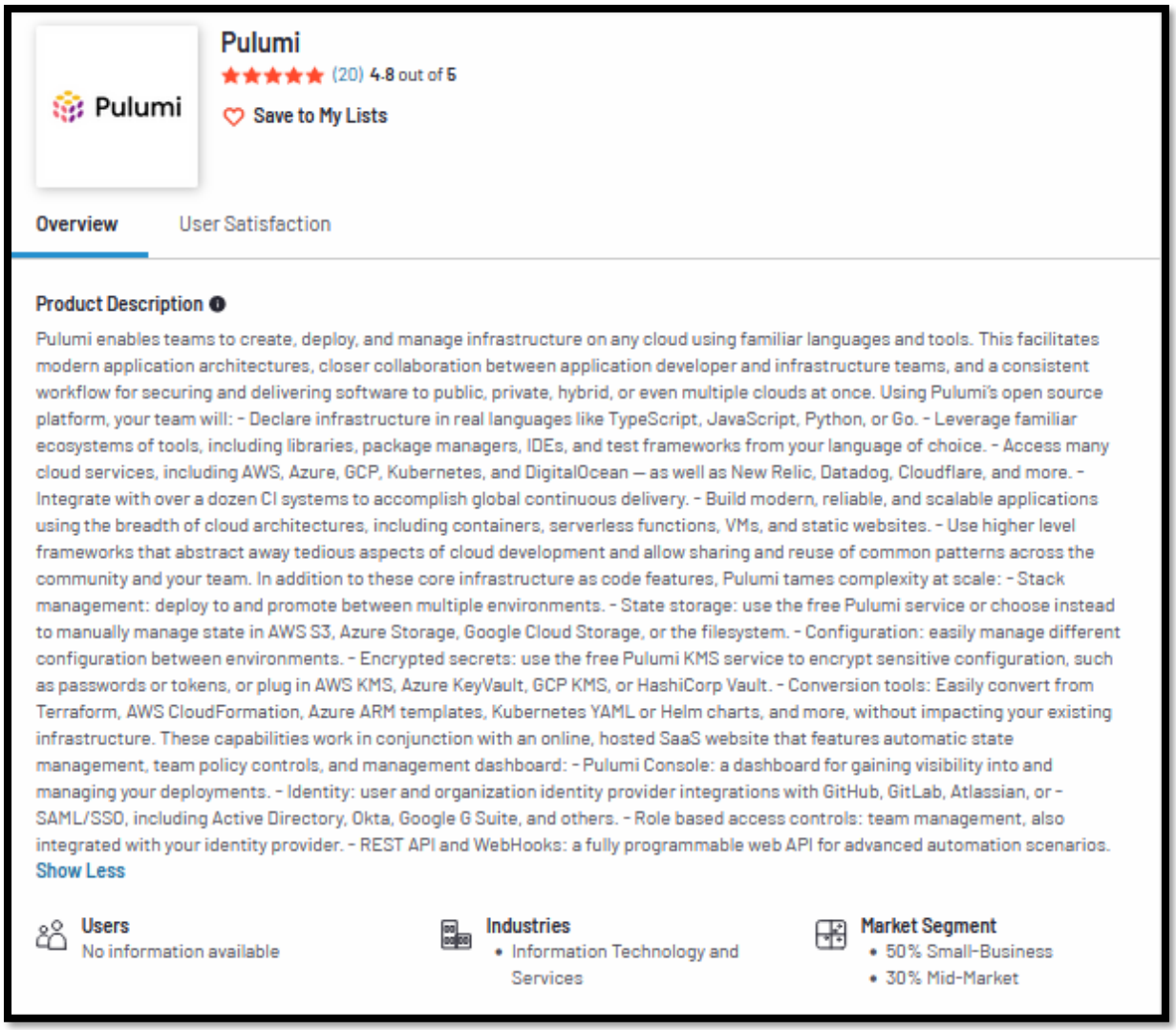
Product Description

HashiCorp Terraform is an infrastructure as code tool that lets you define both cloud and on-prem resources in human-readable configuration files that you can version, reuse, and share. You can then use a consistent workflow to provision and manage all of your infrastructure throughout its lifecycle. Terraform can manage low-level components like compute, storage, and networking resources, as well as high-level components like DNS entries and SaaS features. Terraform creates and manages resources on cloud platforms and other services through their application programming interfaces (APIs). Providers enable Terraform to work with virtually any platform or service with an accessible API. Show Less

Users
- DevOps Engineer
- Software Engineer

Industries
- Information Technology and Services
- Computer Software

Market Segment
- 34% Enterprise
- 33% Mid-Market



Pulumi

(20) 4.8 out of 5

Save to My Lists

Overview | User Satisfaction

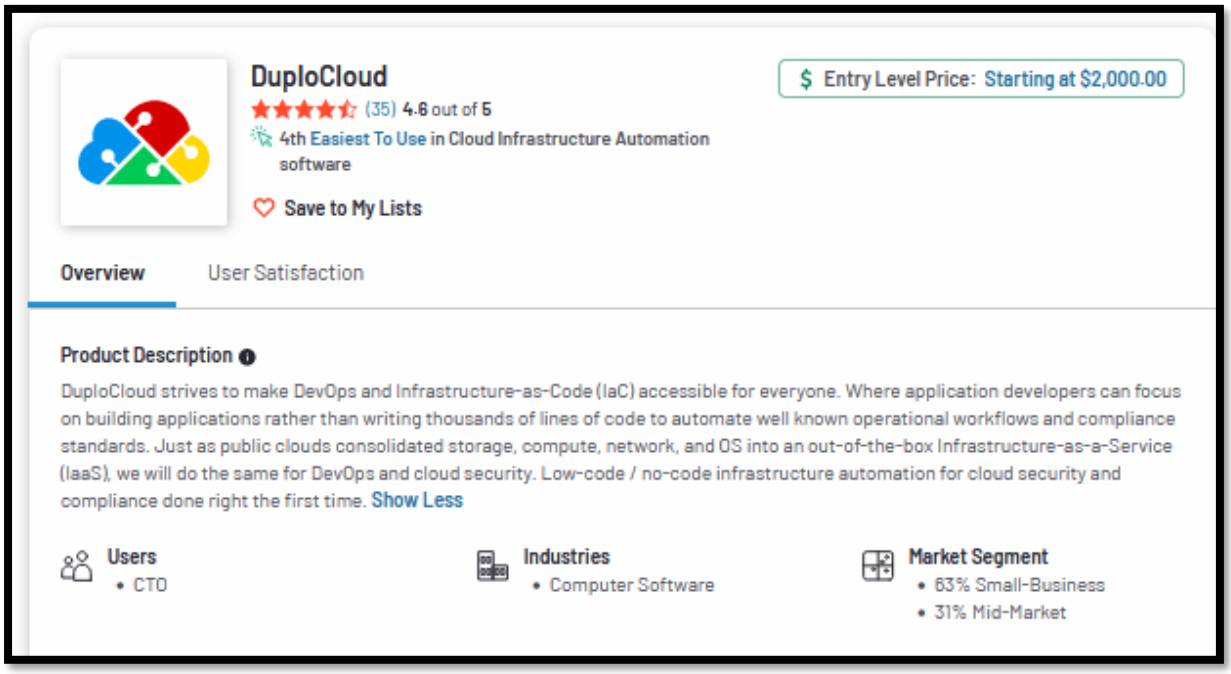
Product Description

Pulumi enables teams to create, deploy, and manage infrastructure on any cloud using familiar languages and tools. This facilitates modern application architectures, closer collaboration between application developer and infrastructure teams, and a consistent workflow for securing and delivering software to public, private, hybrid, or even multiple clouds at once. Using Pulumi's open source platform, your team will: - Declare infrastructure in real languages like TypeScript, JavaScript, Python, or Go. - Leverage familiar ecosystems of tools, including libraries, package managers, IDEs, and test frameworks from your language of choice. - Access many cloud services, including AWS, Azure, GCP, Kubernetes, and DigitalOcean — as well as New Relic, Datadog, Cloudflare, and more. - Integrate with over a dozen CI systems to accomplish global continuous delivery. - Build modern, reliable, and scalable applications using the breadth of cloud architectures, including containers, serverless functions, VMs, and static websites. - Use higher level frameworks that abstract away tedious aspects of cloud development and allow sharing and reuse of common patterns across the community and your team. In addition to these core infrastructure as code features, Pulumi tames complexity at scale: - Stack management: deploy to and promote between multiple environments. - State storage: use the free Pulumi service or choose instead to manually manage state in AWS S3, Azure Storage, Google Cloud Storage, or the filesystem. - Configuration: easily manage different configuration between environments. - Encrypted secrets: use the free Pulumi KMS service to encrypt sensitive configuration, such as passwords or tokens, or plug in AWS KMS, Azure KeyVault, GCP KMS, or HashiCorp Vault. - Conversion tools: Easily convert from Terraform, AWS CloudFormation, Azure ARM templates, Kubernetes YAML or Helm charts, and more, without impacting your existing infrastructure. These capabilities work in conjunction with an online, hosted SaaS website that features automatic state management, team policy controls, and management dashboard: - Pulumi Console: a dashboard for gaining visibility into and managing your deployments. - Identity: user and organization identity provider integrations with GitHub, GitLab, Atlassian, or - SAML/SSO, including Active Directory, Okta, Google G Suite, and others. - Role based access controls: team management, also integrated with your identity provider. - REST API and WebHooks: a fully programmable web API for advanced automation scenarios.
Show Less

Users
No information available

Industries
- Information Technology and Services

Market Segment
- 50% Small-Business
- 30% Mid-Market



DuplоCloud

(35) 4.6 out of 5

4th Easiest To Use in Cloud Infrastructure Automation software

Save to My Lists

$ Entry Level Price: Starting at $2,000.00

Overview | User Satisfaction

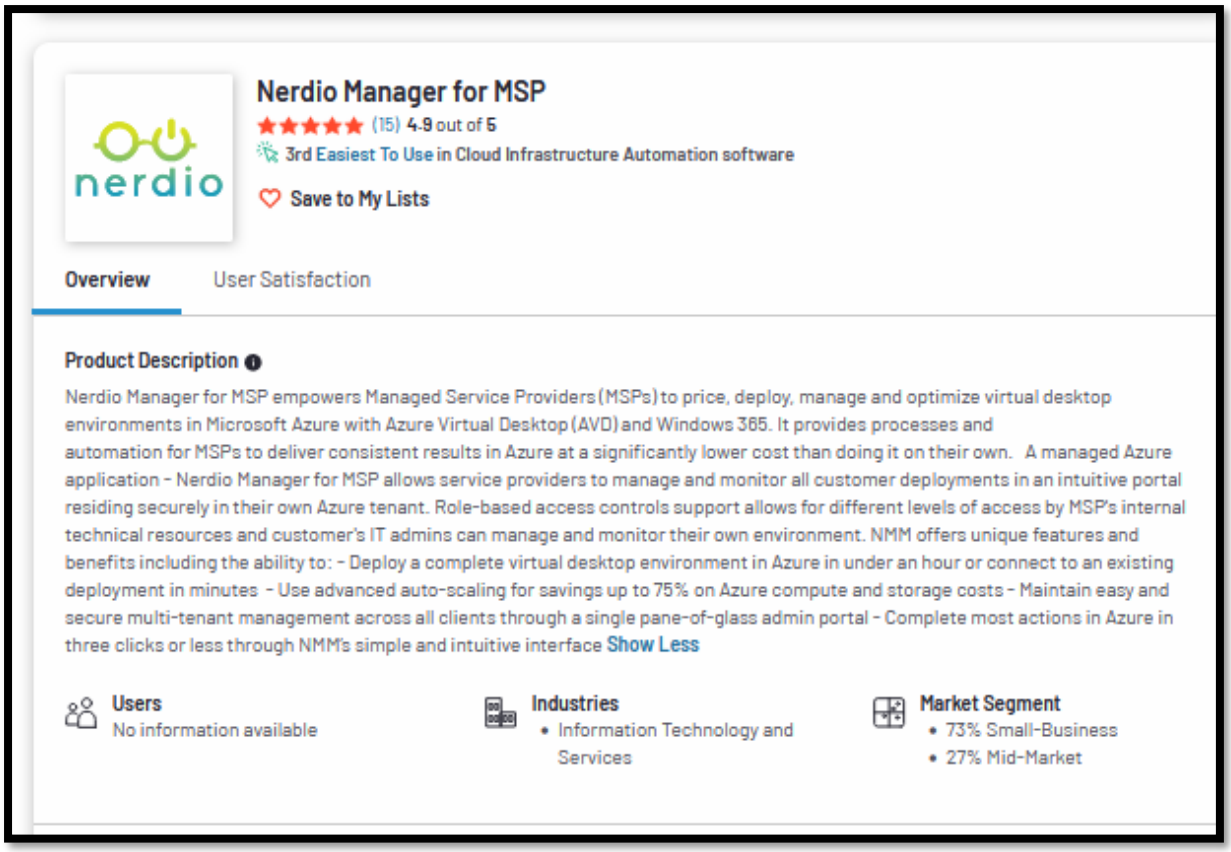
Product Description

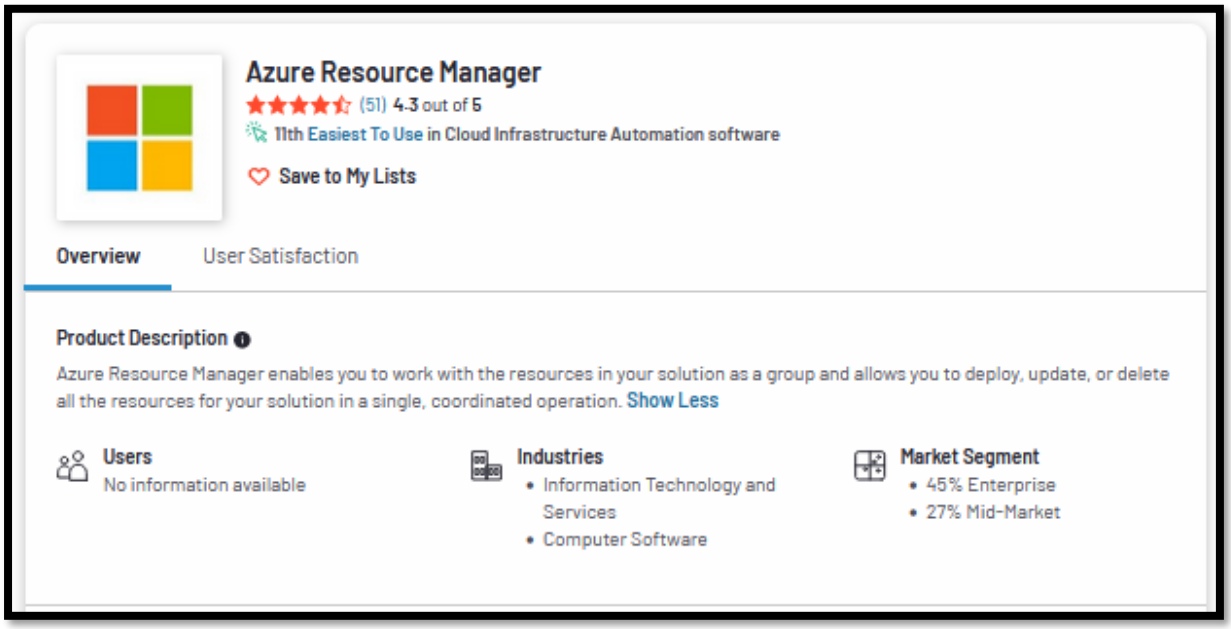
DuploCloud strives to make DevOps and Infrastructure-as-Code (IaC) accessible for everyone. Where application developers can focus on building applications rather than writing thousands of lines of code to automate well known operational workflows and compliance standards. Just as public clouds consolidated storage, compute, network, and OS into an out-of-the-box Infrastructure-as-a-Service (IaaS), we will do the same for DevOps and cloud security. Low-code / no-code infrastructure automation for cloud security and compliance done right the first time. Show Less

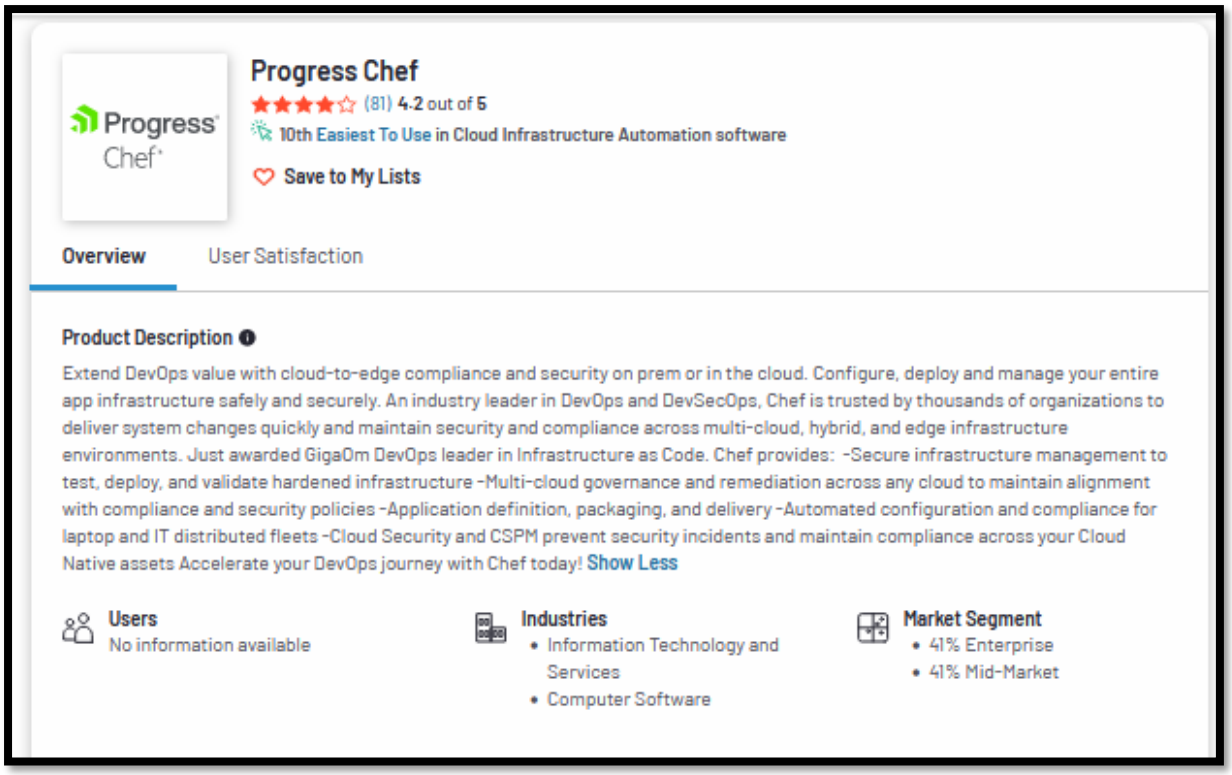
Users
- CTO

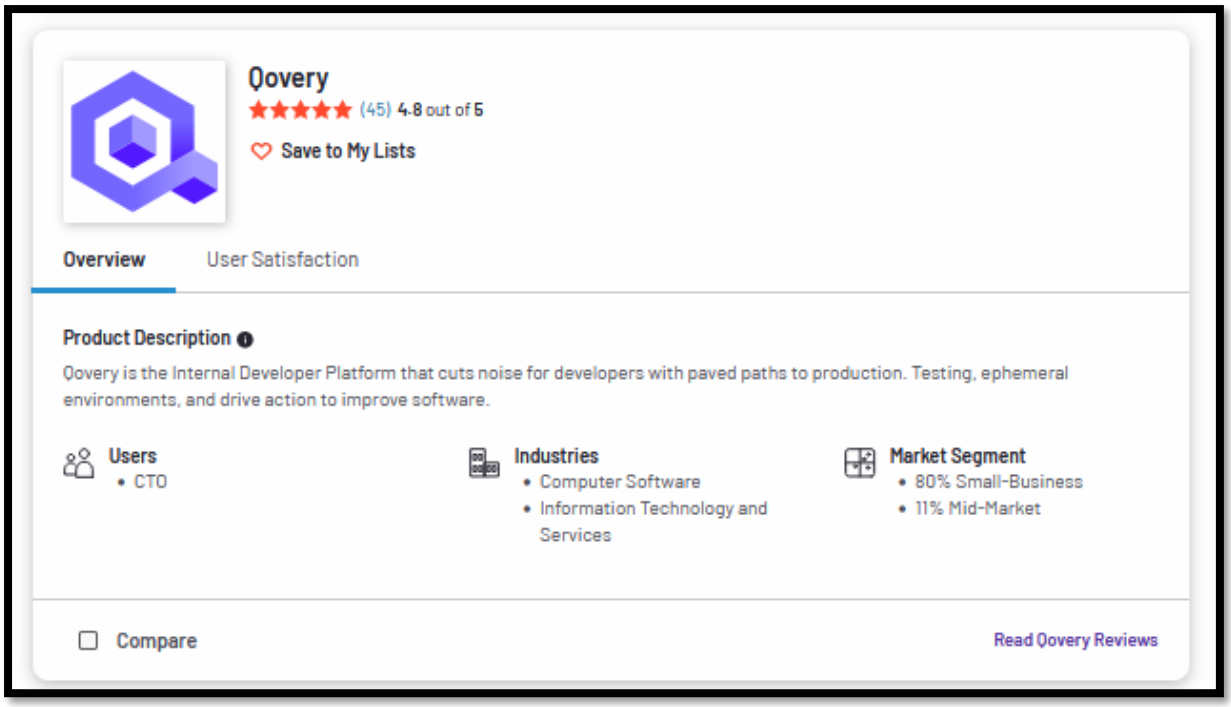
Industries
- Computer Software

Market Segment
- 63% Small-Business
- 31% Mid-Market


Nerdio Manager for MSP
(15) 4.9 out of 5
3rd Easiest To Use in Cloud Infrastructure Automation software
Save to My Lists
Overview
User Satisfaction
Product Description
Nerdio Manager for MSP empowers Managed Service Providers (MSPs) to price, deploy, manage and optimize virtual desktop environments in Microsoft Azure with Azure Virtual Desktop (AVD) and Windows 365. It provides processes and automation for MSPs to deliver consistent results in Azure at a significantly lower cost than doing it on their own. A managed Azure application - Nerdio Manager for MSP allows service providers to manage and monitor all customer deployments in an intuitive portal residing securely in their own Azure tenant. Role-based access controls support allows for different levels of access by MSP's internal technical resources and customer's IT admins can manage and monitor their own environment. NMM offers unique features and benefits including the ability to: - Deploy a complete virtual desktop environment in Azure in under an hour or connect to an existing deployment in minutes - Use advanced auto-scaling for savings up to 75% on Azure compute and storage costs - Maintain easy and secure multi-tenant management across all clients through a single pane-of-glass admin portal - Complete most actions in Azure in three clicks or less through NMM's simple and intuitive interface Show Less
Users
No information available
Industries
Information Technology and Services
Market Segment
73% Small-Business
27% Mid-Market


Azure Resource Manager
(51) 4.3 out of 5
11th Easiest To Use in Cloud Infrastructure Automation software
Save to My Lists
Overview
User Satisfaction
Product Description
Azure Resource Manager enables you to work with the resources in your solution as a group and allows you to deploy, update, or delete all the resources for your solution in a single, coordinated operation. Show Less
Users
No information available
Industries
Information Technology and Services
Computer Software
Market Segment
45% Enterprise
27% Mid-Market


Progress Chef
(81) 4.2 out of 5
10th Easiest To Use in Cloud Infrastructure Automation software
Save to My Lists
Overview
User Satisfaction
Product Description
Extend DevOps value with cloud-to-edge compliance and security on prem or in the cloud. Configure, deploy and manage your entire app infrastructure safely and securely. An industry leader in DevOps and DevSecOps, Chef is trusted by thousands of organizations to deliver system changes quickly and maintain security and compliance across multi-cloud, hybrid, and edge infrastructure environments. Just awarded GigaOm DevOps leader in Infrastructure as Code. Chef provides: -Secure infrastructure management to test, deploy, and validate hardened infrastructure -Multi-cloud governance and remediation across any cloud to maintain alignment with compliance and security policies -Application definition, packaging, and delivery -Automated configuration and compliance for laptop and IT distributed fleets -Cloud Security and CSPM prevent security incidents and maintain compliance across your Cloud Native assets Accelerate your DevOps journey with Chef today! Show Less
Users
No information available
Industries
Information Technology and Services
Computer Software
Market Segment
41% Enterprise
41% Mid-Market



Qovery

(45) 4.8 out of 5

Save to My Lists

Overview | User Satisfaction

Product Description

Qovery is the Internal Developer Platform that cuts noise for developers with paved paths to production. Testing, ephemeral environments, and drive action to improve software.

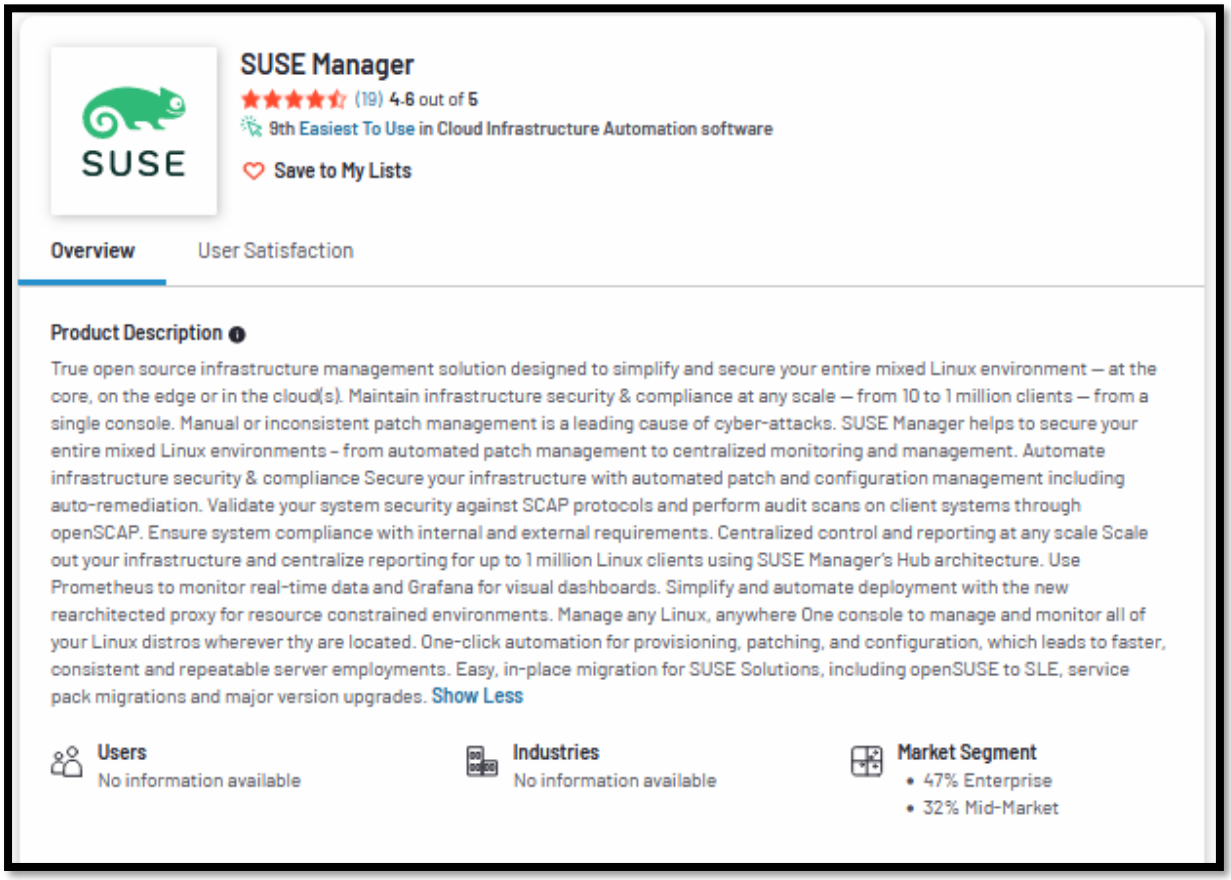
Users
- CTO

Industries
- Computer Software
- Information Technology and Services

Market Segment
- 80% Small-Business
- 11% Mid-Market

Compare

Read Qovery Reviews



SUSE Manager

(19) 4.6 out of 5

9th Easiest To Use in Cloud Infrastructure Automation software

Save to My Lists

Overview | User Satisfaction

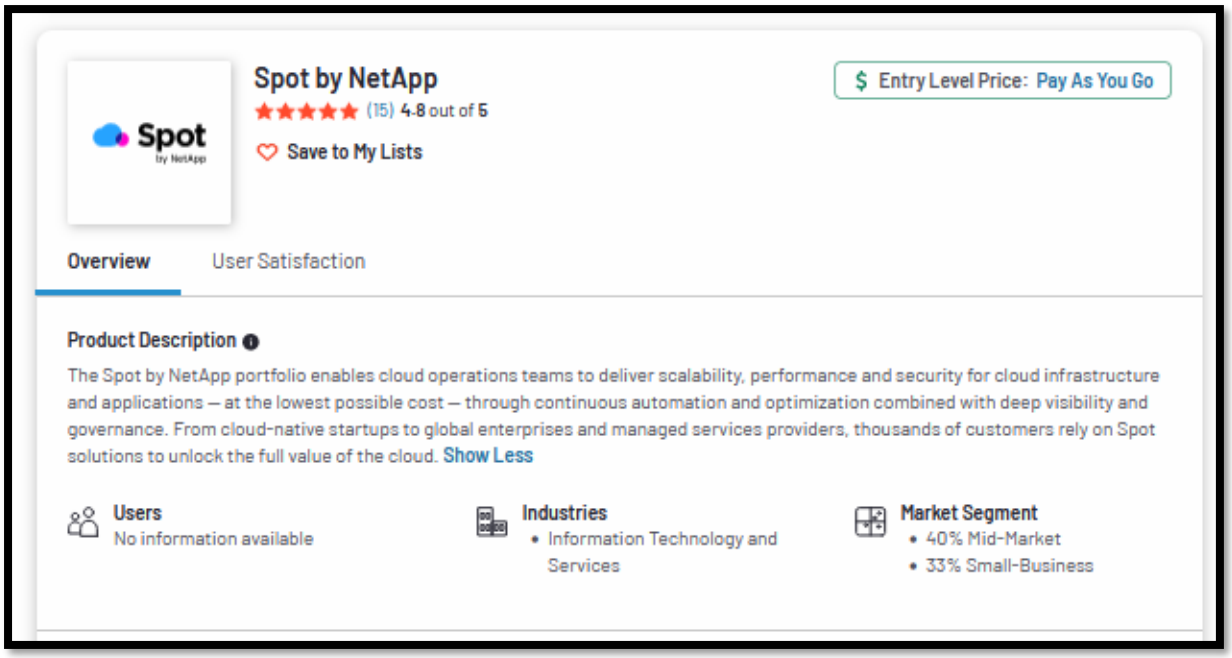
Product Description

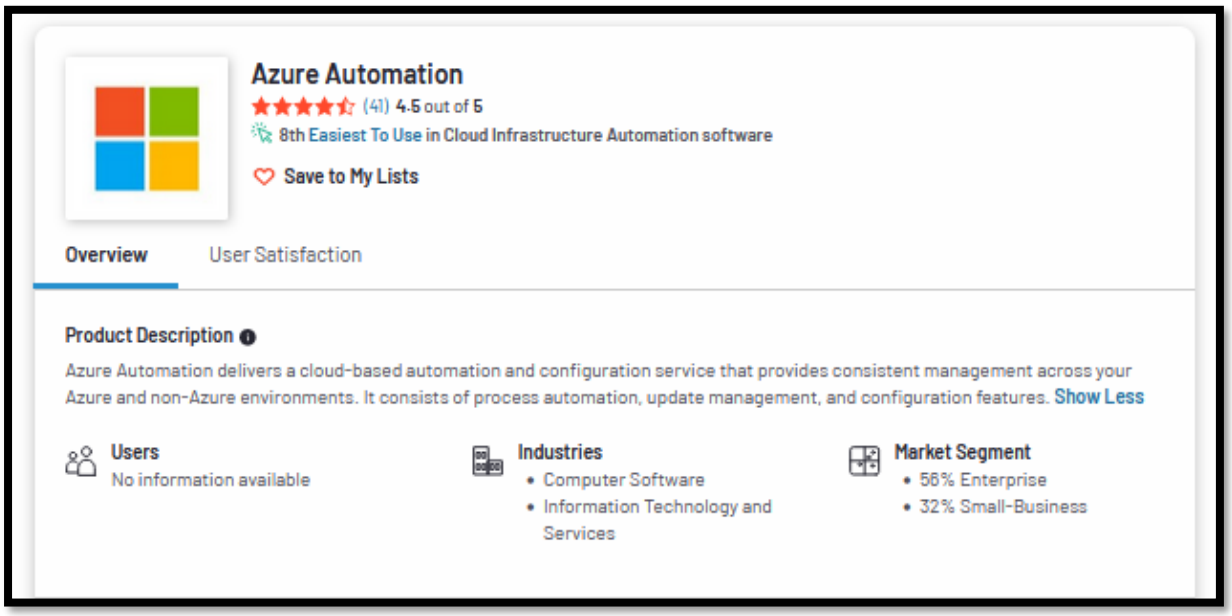
True open source infrastructure management solution designed to simplify and secure your entire mixed Linux environment — at the core, on the edge or in the cloud(s). Maintain infrastructure security & compliance at any scale — from 10 to 1 million clients — from a single console. Manual or inconsistent patch management is a leading cause of cyber-attacks. SUSE Manager helps to secure your entire mixed Linux environments – from automated patch management to centralized monitoring and management. Automate infrastructure security & compliance Secure your infrastructure with automated patch and configuration management including auto-remediation. Validate your system security against SCAP protocols and perform audit scans on client systems through openSCAP. Ensure system compliance with internal and external requirements. Centralized control and reporting at any scale Scale out your infrastructure and centralize reporting for up to 1 million Linux clients using SUSE Manager's Hub architecture. Use Prometheus to monitor real-time data and Grafana for visual dashboards. Simplify and automate deployment with the new rearchitected proxy for resource constrained environments. Manage any Linux, anywhere One console to manage and monitor all of your Linux distros wherever thy are located. One-click automation for provisioning, patching, and configuration, which leads to faster, consistent and repeatable server employments. Easy, in-place migration for SUSE Solutions, including openSUSE to SLE, service pack migrations and major version upgrades. Show Less

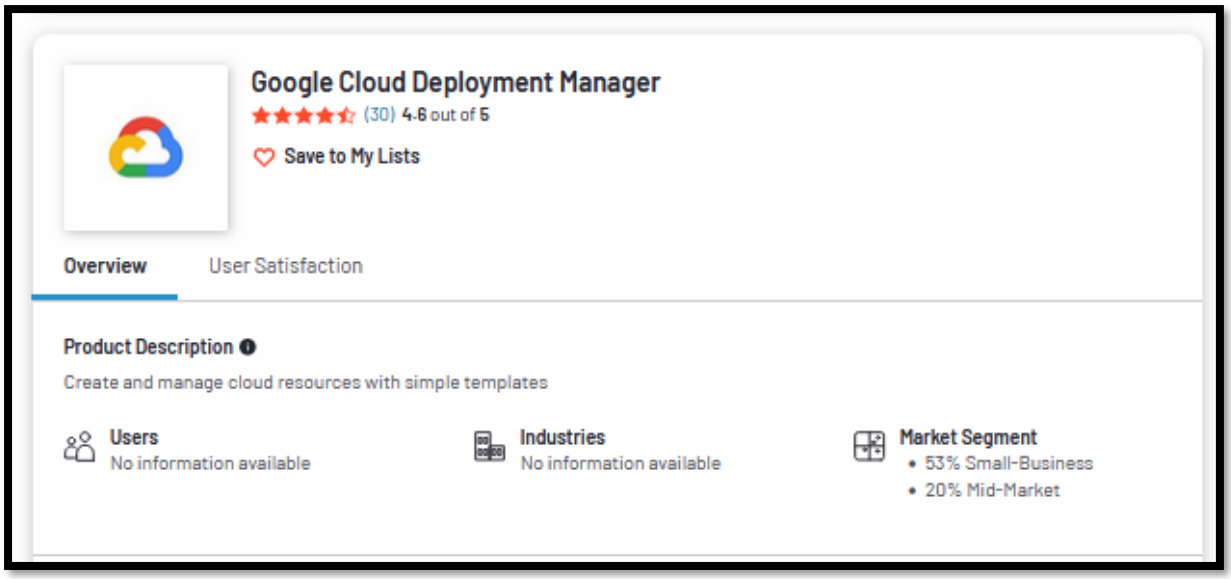
Users
No information available

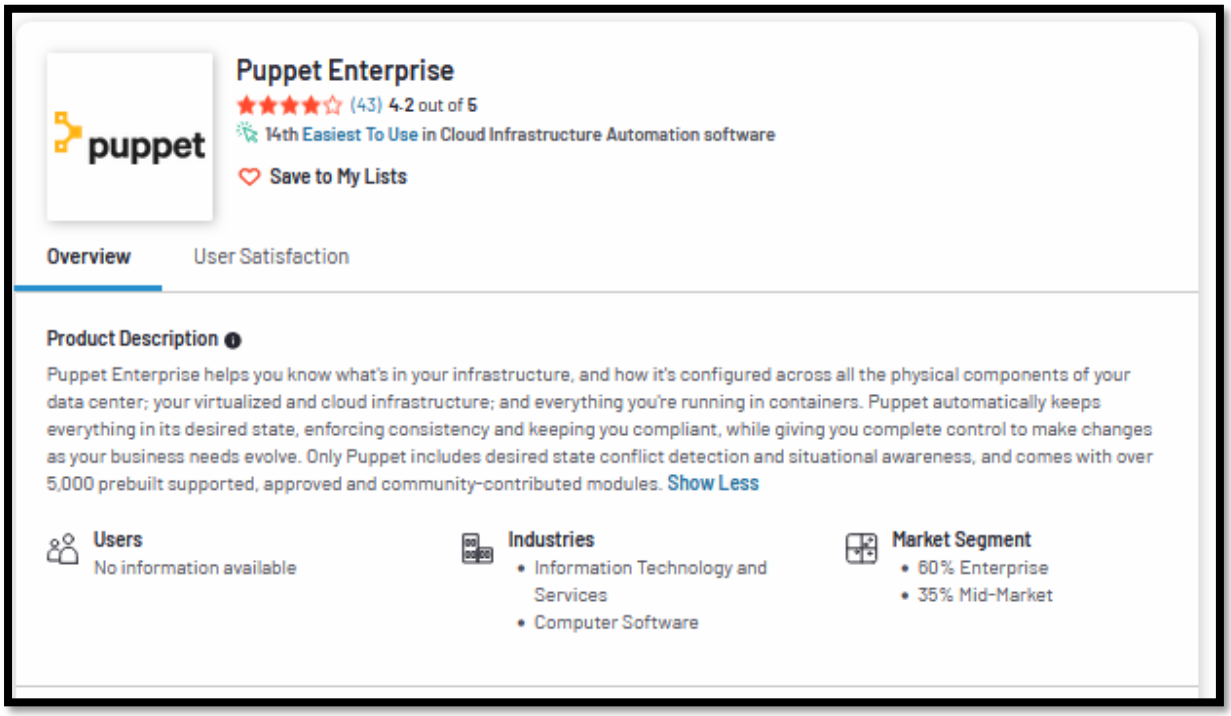
Industries
No information available

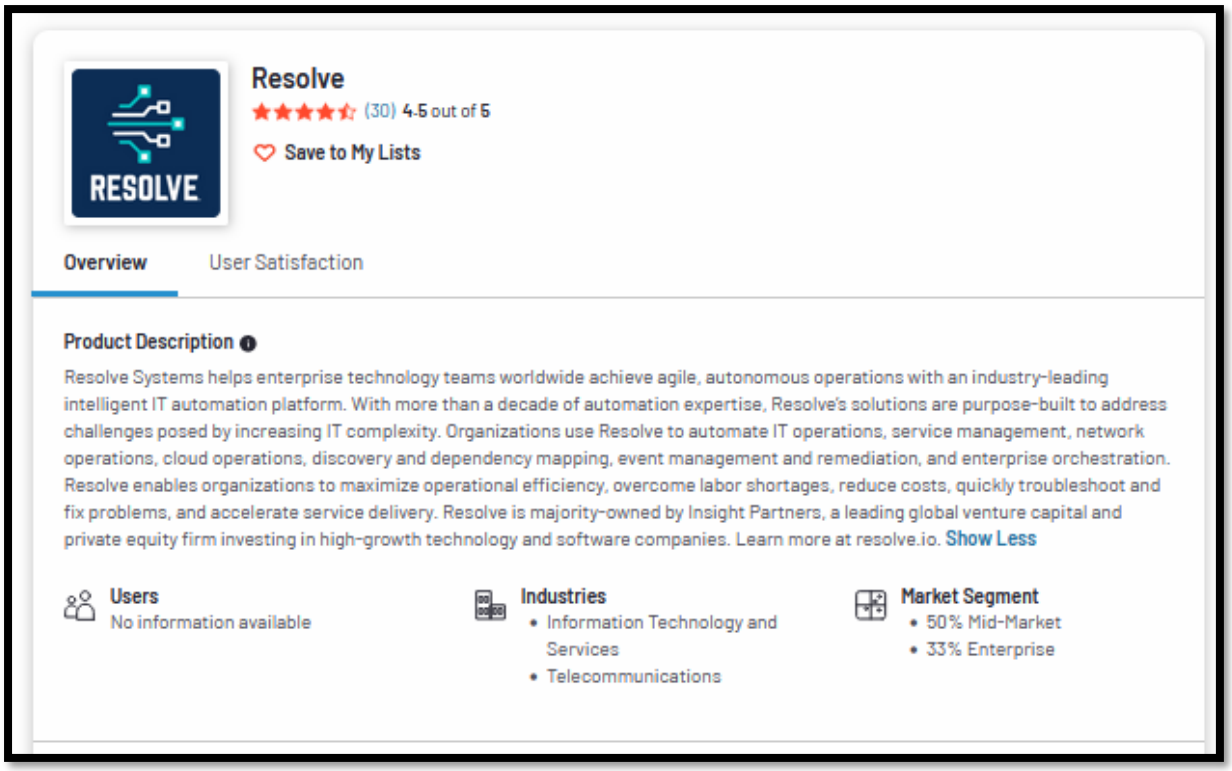
Market Segment
- 47% Enterprise
- 32% Mid-Market


Spot by NetApp
(15) 4.8 out of 5
Save to My Lists
$ Entry Level Price: Pay As You Go
Overview
User Satisfaction
Product Description
The Spot by NetApp portfolio enables cloud operations teams to deliver scalability, performance and security for cloud infrastructure and applications — at the lowest possible cost — through continuous automation and optimization combined with deep visibility and governance. From cloud-native startups to global enterprises and managed services providers, thousands of customers rely on Spot solutions to unlock the full value of the cloud. Show Less
Users
No information available
Industries
• Information Technology and Services
Market Segment
• 40% Mid-Market
• 33% Small-Business


Azure Automation
(41) 4.5 out of 5
8th Easiest To Use in Cloud Infrastructure Automation software
Save to My Lists
Overview
User Satisfaction
Product Description
Azure Automation delivers a cloud-based automation and configuration service that provides consistent management across your Azure and non-Azure environments. It consists of process automation, update management, and configuration features. Show Less
Users
No information available
Industries
• Computer Software
• Information Technology and Services
Market Segment
• 56% Enterprise
• 32% Small-Business


Google Cloud Deployment Manager
(30) 4.6 out of 5
Save to My Lists
Overview
User Satisfaction
Product Description
Create and manage cloud resources with simple templates
Users
No information available
Industries
No information available
Market Segment
• 53% Small-Business
• 20% Mid-Market


puppet
Puppet Enterprise
(43) 4.2 out of 5
14th Easiest To Use in Cloud Infrastructure Automation software
Save to My Lists
Overview
User Satisfaction
Product Description
Puppet Enterprise helps you know what's in your infrastructure, and how it's configured across all the physical components of your data center; your virtualized and cloud infrastructure; and everything you're running in containers. Puppet automatically keeps everything in its desired state, enforcing consistency and keeping you compliant, while giving you complete control to make changes as your business needs evolve. Only Puppet includes desired state conflict detection and situational awareness, and comes with over 5,000 prebuilt supported, approved and community-contributed modules. Show Less
Users
No information available
Industries
• Information Technology and Services
• Computer Software
Market Segment
• 60% Enterprise
• 35% Mid-Market



Resolve

(30) 4.5 out of 5

Save to My Lists

Overview | User Satisfaction

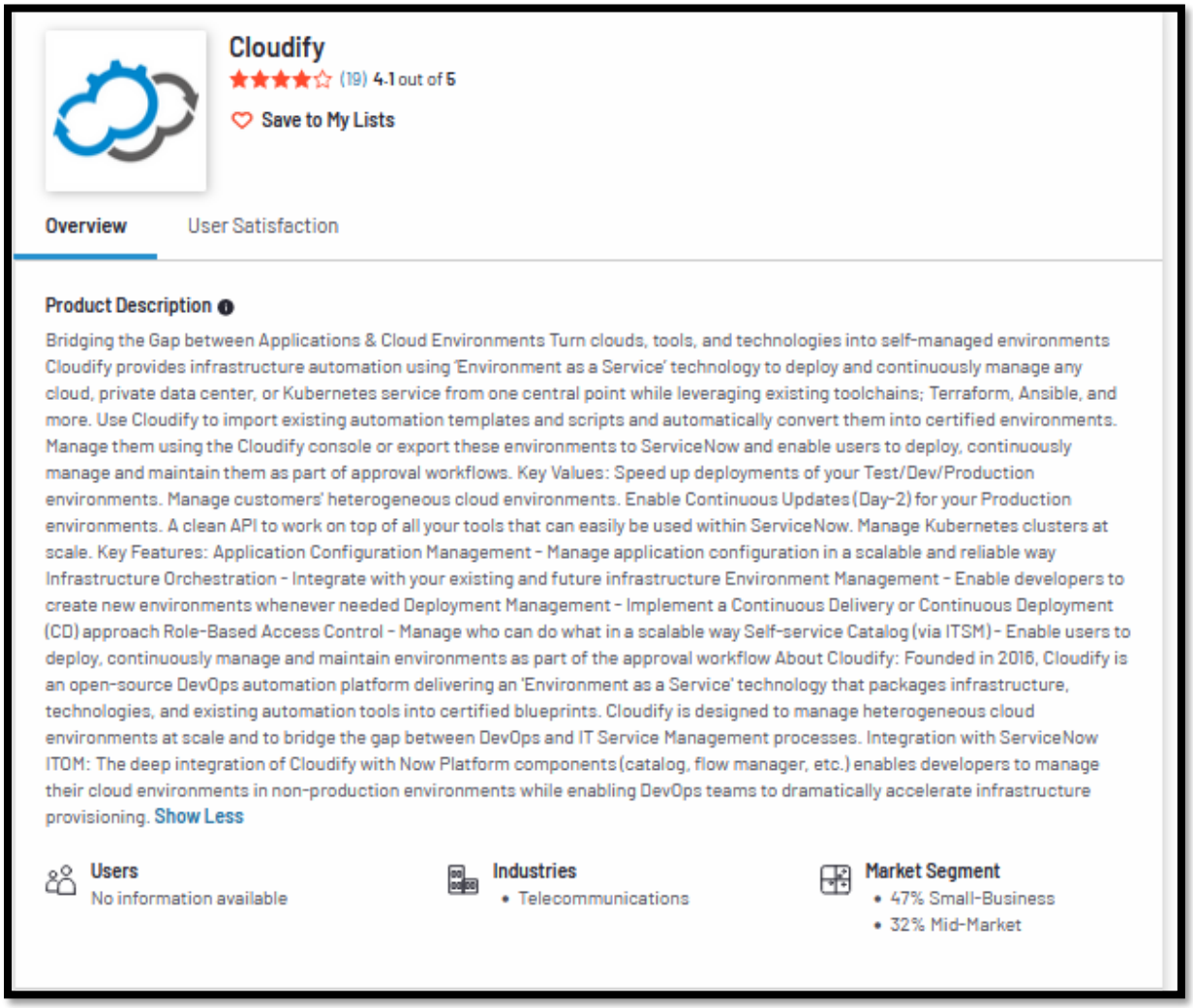
Product Description

Resolve Systems helps enterprise technology teams worldwide achieve agile, autonomous operations with an industry-leading intelligent IT automation platform. With more than a decade of automation expertise, Resolve's solutions are purpose-built to address challenges posed by increasing IT complexity. Organizations use Resolve to automate IT operations, service management, network operations, cloud operations, discovery and dependency mapping, event management and remediation, and enterprise orchestration. Resolve enables organizations to maximize operational efficiency, overcome labor shortages, reduce costs, quickly troubleshoot and fix problems, and accelerate service delivery. Resolve is majority-owned by Insight Partners, a leading global venture capital and private equity firm investing in high-growth technology and software companies. Learn more at resolve.io. Show Less

Users
No information available

Industries
- Information Technology and Services
- Telecommunications

Market Segment
- 50% Mid-Market
- 33% Enterprise



Cloudify

(19) 4.1 out of 5

Save to My Lists

Overview | User Satisfaction

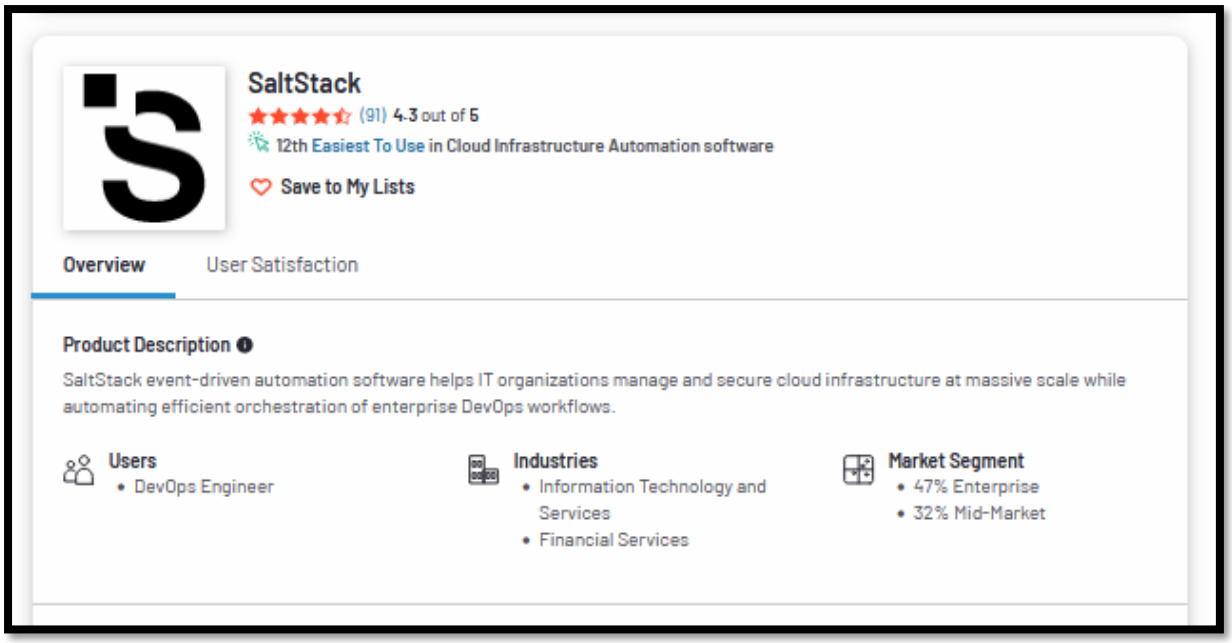
Product Description

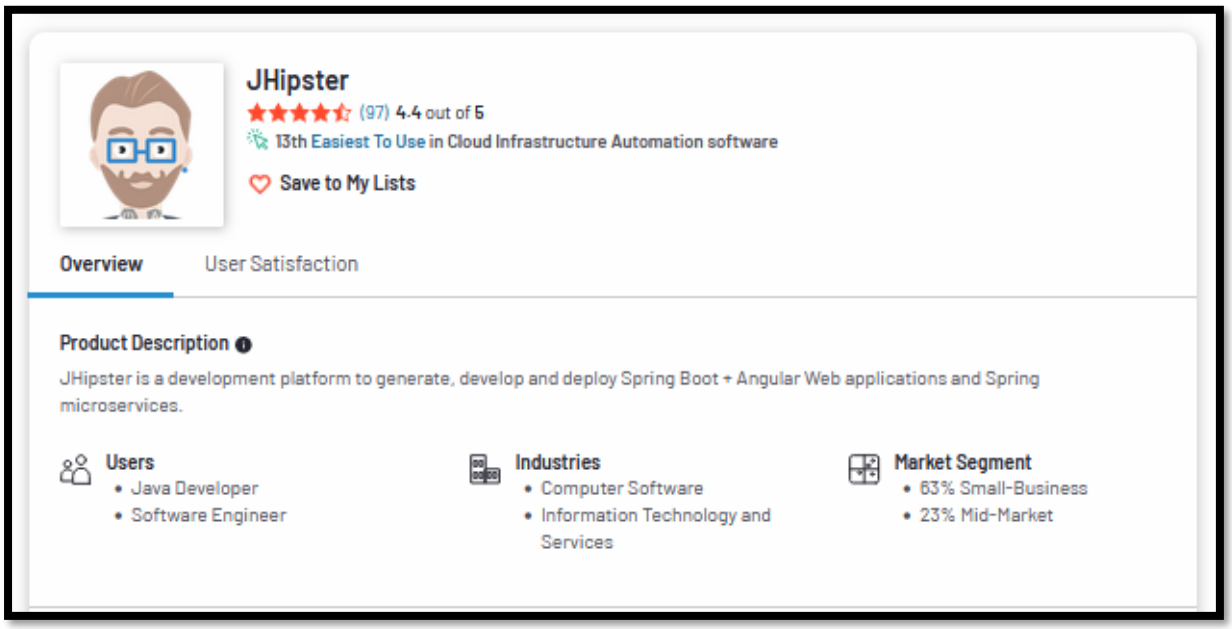
Bridging the Gap between Applications & Cloud Environments Turn clouds, tools, and technologies into self-managed environments Cloudify provides infrastructure automation using 'Environment as a Service' technology to deploy and continuously manage any cloud, private data center, or Kubernetes service from one central point while leveraging existing toolchains; Terraform, Ansible, and more. Use Cloudify to import existing automation templates and scripts and automatically convert them into certified environments. Manage them using the Cloudify console or export these environments to ServiceNow and enable users to deploy, continuously manage and maintain them as part of approval workflows. Key Values: Speed up deployments of your Test/Dev/Production environments. Manage customers' heterogeneous cloud environments. Enable Continuous Updates (Day-2) for your Production environments. A clean API to work on top of all your tools that can easily be used within ServiceNow. Manage Kubernetes clusters at scale. Key Features: Application Configuration Management - Manage application configuration in a scalable and reliable way Infrastructure Orchestration - Integrate with your existing and future infrastructure Environment Management - Enable developers to create new environments whenever needed Deployment Management - Implement a Continuous Delivery or Continuous Deployment (CD) approach Role-Based Access Control - Manage who can do what in a scalable way Self-service Catalog (via ITSM) - Enable users to deploy, continuously manage and maintain environments as part of the approval workflow About Cloudify: Founded in 2016, Cloudify is an open-source DevOps automation platform delivering an 'Environment as a Service' technology that packages infrastructure, technologies, and existing automation tools into certified blueprints. Cloudify is designed to manage heterogeneous cloud environments at scale and to bridge the gap between DevOps and IT Service Management processes. Integration with ServiceNow ITOM: The deep integration of Cloudify with Now Platform components (catalog, flow manager, etc.) enables developers to manage their cloud environments in non-production environments while enabling DevOps teams to dramatically accelerate infrastructure provisioning. Show Less

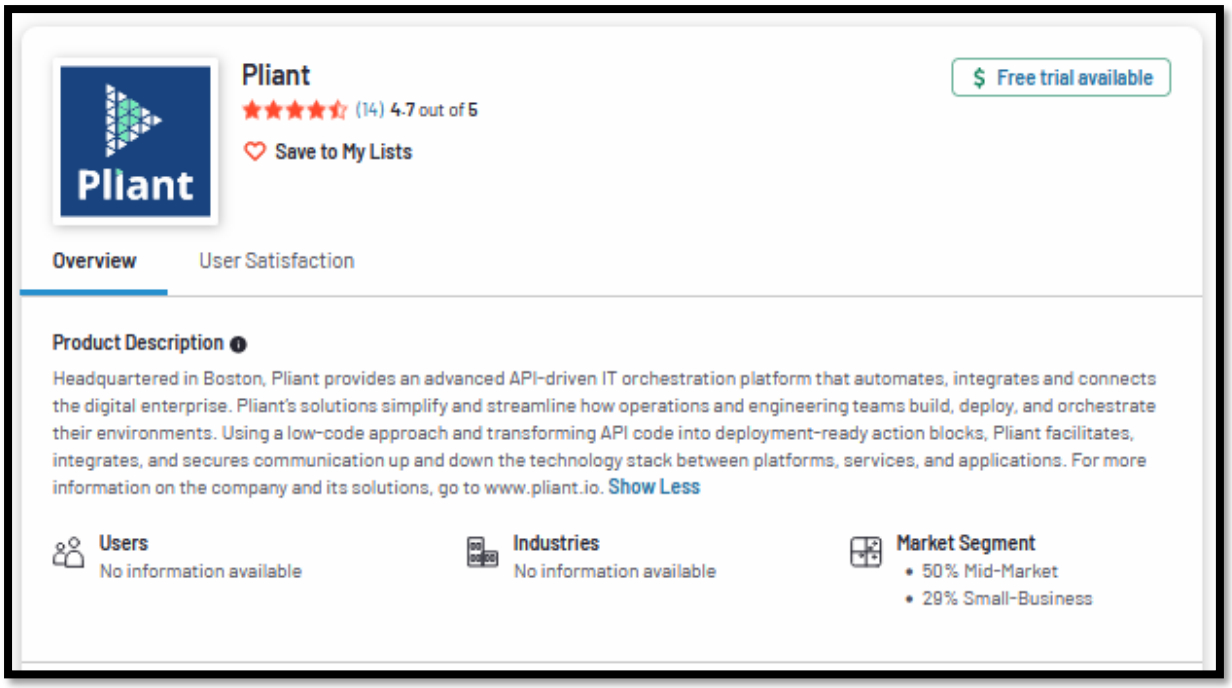
Users
No information available

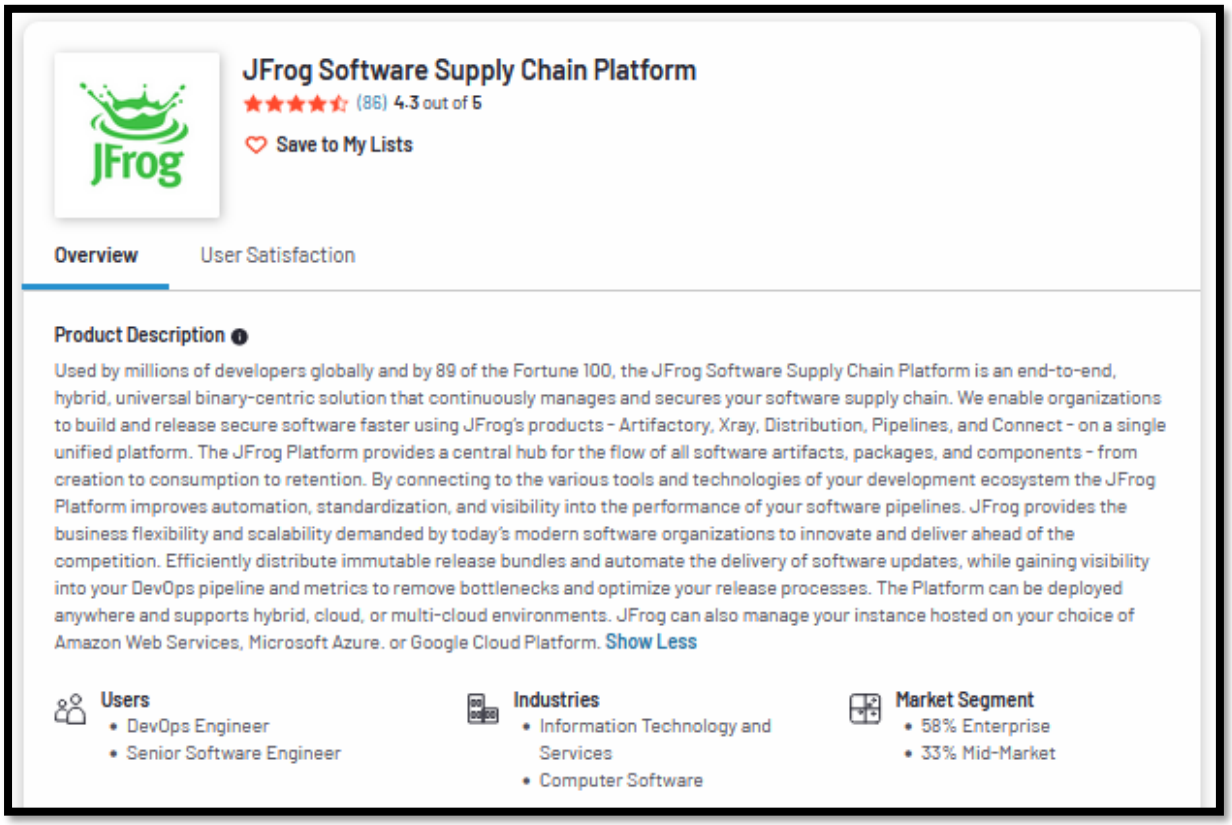
Industries
- Telecommunications

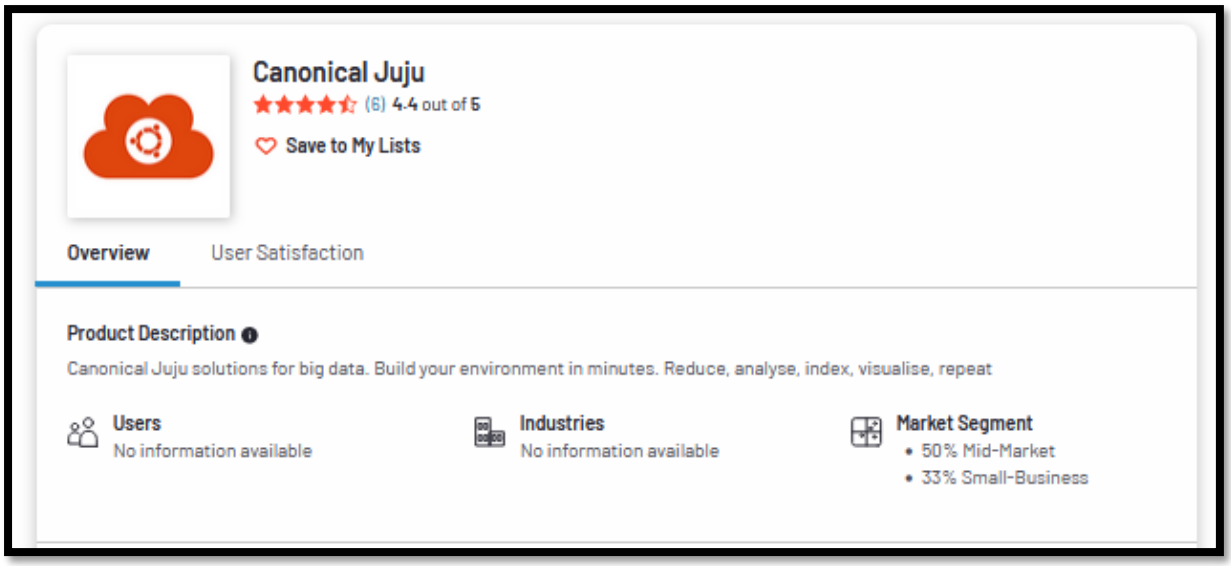
Market Segment
- 47% Small-Business
- 32% Mid-Market

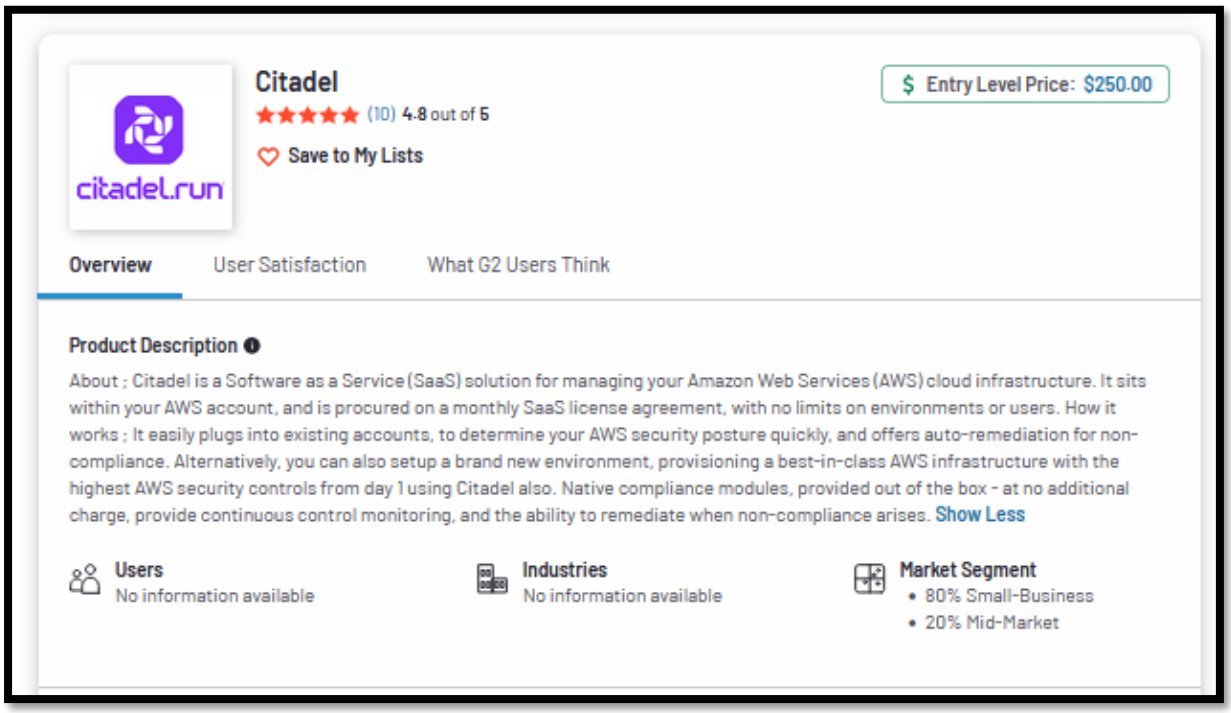

SaltStack
(91) 4.3 out of 5
12th Easiest To Use in Cloud Infrastructure Automation software
Save to My Lists
Overview
User Satisfaction
Product Description
SaltStack event-driven automation software helps IT organizations manage and secure cloud infrastructure at massive scale while automating efficient orchestration of enterprise DevOps workflows.
Users
• DevOps Engineer
Industries
• Information Technology and Services
• Financial Services
Market Segment
• 47% Enterprise
• 32% Mid-Market

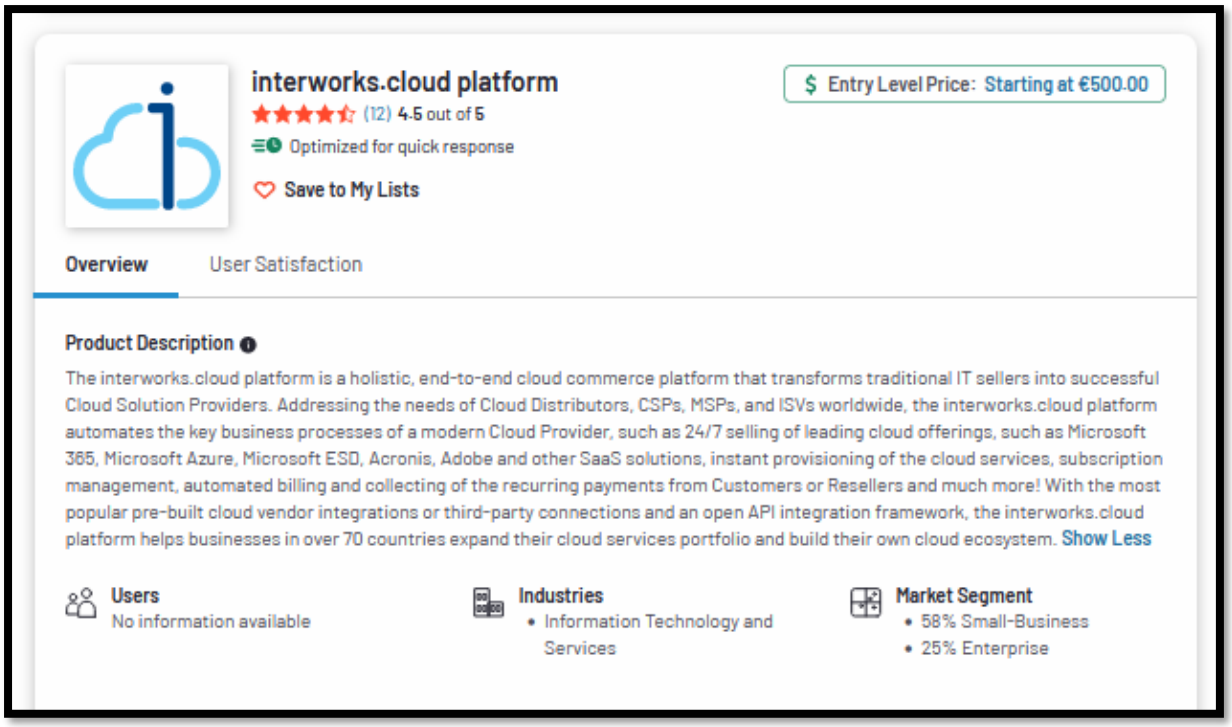

JHipster
(97) 4.4 out of 5
13th Easiest To Use in Cloud Infrastructure Automation software
Save to My Lists
Overview
User Satisfaction
Product Description
JHipster is a development platform to generate, develop and deploy Spring Boot + Angular Web applications and Spring microservices.
Users
• Java Developer
• Software Engineer
Industries
• Computer Software
• Information Technology and Services
Market Segment
• 63% Small-Business
• 23% Mid-Market

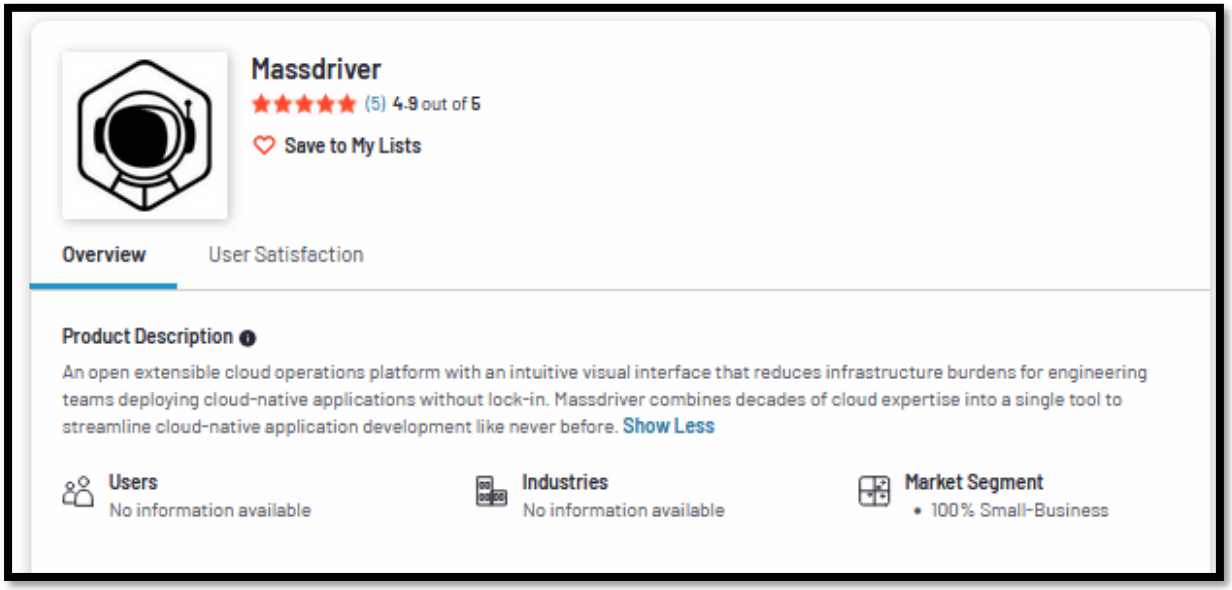

Pliant
Pliant
(14) 4.7 out of 5
Save to My Lists
$ Free trial available
Overview
User Satisfaction
Product Description
Headquartered in Boston, Pliant provides an advanced API-driven IT orchestration platform that automates, integrates and connects the digital enterprise. Pliant's solutions simplify and streamline how operations and engineering teams build, deploy, and orchestrate their environments. Using a low-code approach and transforming API code into deployment-ready action blocks, Pliant facilitates, integrates, and secures communication up and down the technology stack between platforms, services, and applications. For more information on the company and its solutions, go to www.pliant.io. Show Less
Users
No information available
Industries
No information available
Market Segment
• 50% Mid-Market
• 29% Small-Business

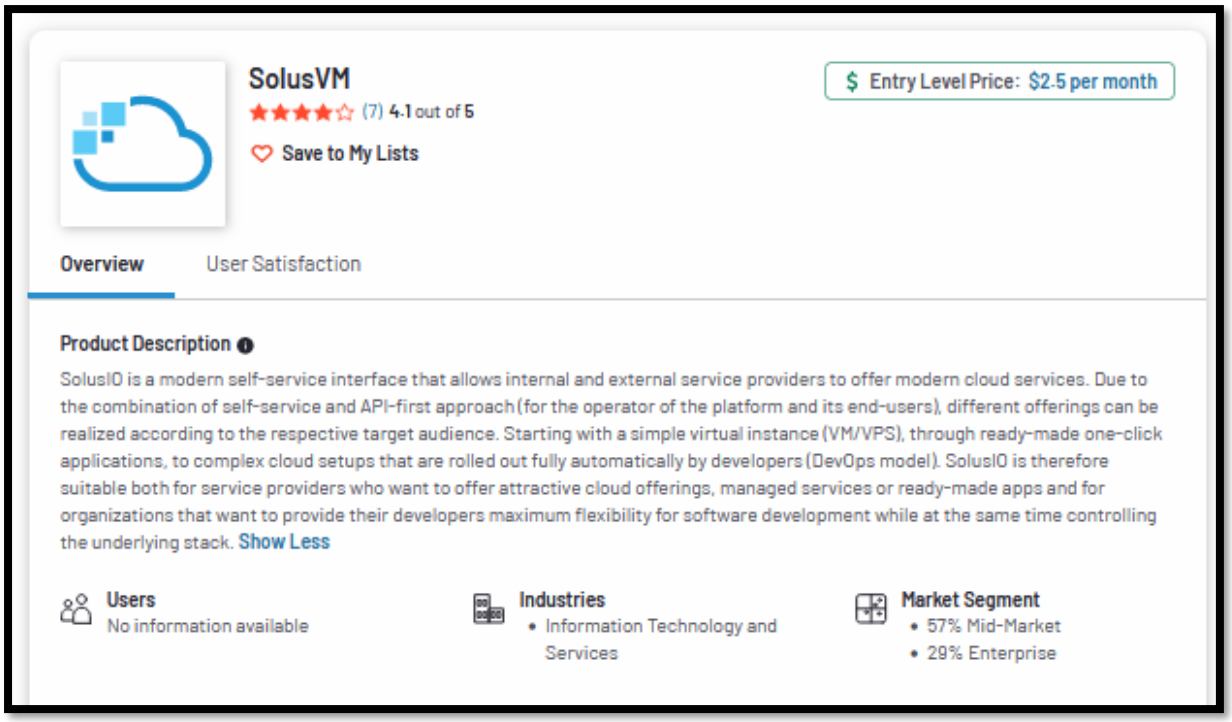

JFrog Software Supply Chain Platform
JFrog
(86) 4.3 out of 5
Save to My Lists
Overview
User Satisfaction
Product Description
Used by millions of developers globally and by 89 of the Fortune 100, the JFrog Software Supply Chain Platform is an end-to-end, hybrid, universal binary-centric solution that continuously manages and secures your software supply chain. We enable organizations to build and release secure software faster using JFrog's products - Artifactory, Xray, Distribution, Pipelines, and Connect - on a single unified platform. The JFrog Platform provides a central hub for the flow of all software artifacts, packages, and components - from creation to consumption to retention. By connecting to the various tools and technologies of your development ecosystem the JFrog Platform improves automation, standardization, and visibility into the performance of your software pipelines. JFrog provides the business flexibility and scalability demanded by today's modern software organizations to innovate and deliver ahead of the competition. Efficiently distribute immutable release bundles and automate the delivery of software updates, while gaining visibility into your DevOps pipeline and metrics to remove bottlenecks and optimize your release processes. The Platform can be deployed anywhere and supports hybrid, cloud, or multi-cloud environments. JFrog can also manage your instance hosted on your choice of Amazon Web Services, Microsoft Azure. or Google Cloud Platform. Show Less
Users
• DevOps Engineer
• Senior Software Engineer
Industries
• Information Technology and Services
• Computer Software
Market Segment
• 58% Enterprise
• 33% Mid-Market


Canonical Juju
(6) 4.4 out of 5
Save to My Lists
Overview
User Satisfaction
Product Description
Canonical Juju solutions for big data. Build your environment in minutes. Reduce, analyse, index, visualise, repeat
Users
No information available
Industries
No information available
Market Segment
• 50% Mid-Market
• 33% Small-Business


Citadel
(10) 4.8 out of 5
Save to My Lists
$ Entry Level Price: $250.00
Overview
User Satisfaction
What G2 Users Think
Product Description
About ; Citadel is a Software as a Service (SaaS) solution for managing your Amazon Web Services (AWS) cloud infrastructure. It sits within your AWS account, and is procured on a monthly SaaS license agreement, with no limits on environments or users. How it works ; It easily plugs into existing accounts, to determine your AWS security posture quickly, and offers auto-remediation for non-compliance. Alternatively, you can also setup a brand new environment, provisioning a best-in-class AWS infrastructure with the highest AWS security controls from day 1 using Citadel also. Native compliance modules, provided out of the box - at no additional charge, provide continuous control monitoring, and the ability to remediate when non-compliance arises. Show Less
Users
No information available
Industries
No information available
Market Segment
• 80% Small-Business
• 20% Mid-Market


interworks.cloud platform
(12) 4.5 out of 5
Optimized for quick response
Save to My Lists
$ Entry Level Price: Starting at €500.00
Overview
User Satisfaction
Product Description
The interworks.cloud platform is a holistic, end-to-end cloud commerce platform that transforms traditional IT sellers into successful Cloud Solution Providers. Addressing the needs of Cloud Distributors, CSPs, MSPs, and ISVs worldwide, the interworks.cloud platform automates the key business processes of a modern Cloud Provider, such as 24/7 selling of leading cloud offerings, such as Microsoft 365, Microsoft Azure, Microsoft ESD, Acronis, Adobe and other SaaS solutions, instant provisioning of the cloud services, subscription management, automated billing and collecting of the recurring payments from Customers or Resellers and much more! With the most popular pre-built cloud vendor integrations or third-party connections and an open API integration framework, the interworks.cloud platform helps businesses in over 70 countries expand their cloud services portfolio and build their own cloud ecosystem. Show Less
Users
No information available
Industries
• Information Technology and Services
Market Segment
• 58% Small-Business
• 25% Enterprise


Massdriver
(5) 4.9 out of 5
Save to My Lists
Overview
User Satisfaction
Product Description
An open extensible cloud operations platform with an intuitive visual interface that reduces infrastructure burdens for engineering teams deploying cloud-native applications without lock-in. Massdriver combines decades of cloud expertise into a single tool to streamline cloud-native application development like never before. Show Less
Users
No information available
Industries
No information available
Market Segment
• 100% Small-Business



SolusVM

(7) 4.1 out of 5

Save to My Lists

$ Entry Level Price: $2.5 per month

Overview | User Satisfaction

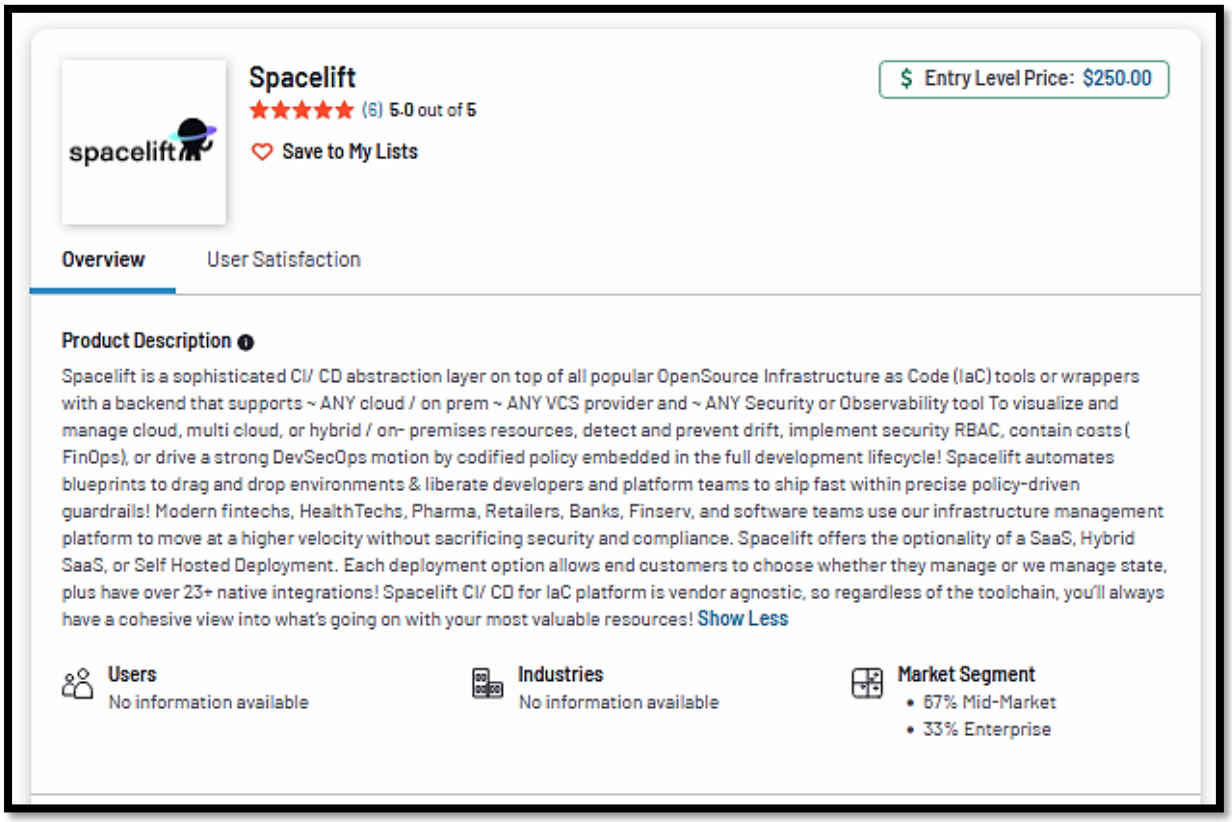
Product Description

SolusIO is a modern self-service interface that allows internal and external service providers to offer modern cloud services. Due to the combination of self-service and API-first approach (for the operator of the platform and its end-users), different offerings can be realized according to the respective target audience. Starting with a simple virtual instance (VM/VPS), through ready-made one-click applications, to complex cloud setups that are rolled out fully automatically by developers (DevOps model). SolusIO is therefore suitable both for service providers who want to offer attractive cloud offerings, managed services or ready-made apps and for organizations that want to provide their developers maximum flexibility for software development while at the same time controlling the underlying stack. Show Less

Users
No information available

Industries
- Information Technology and Services

Market Segment
- 57% Mid-Market
- 29% Enterprise



Spacelift

(6) 5.0 out of 5

Save to My Lists

$ Entry Level Price: $250.00

Overview | User Satisfaction

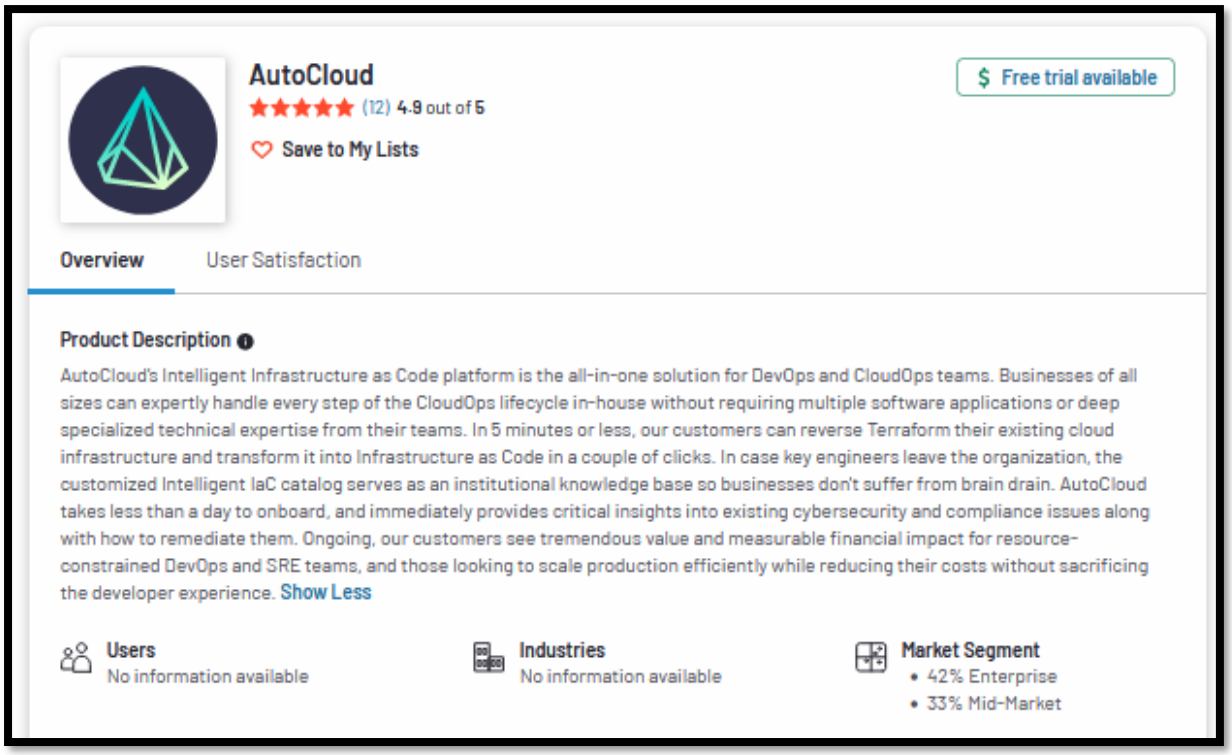
Product Description

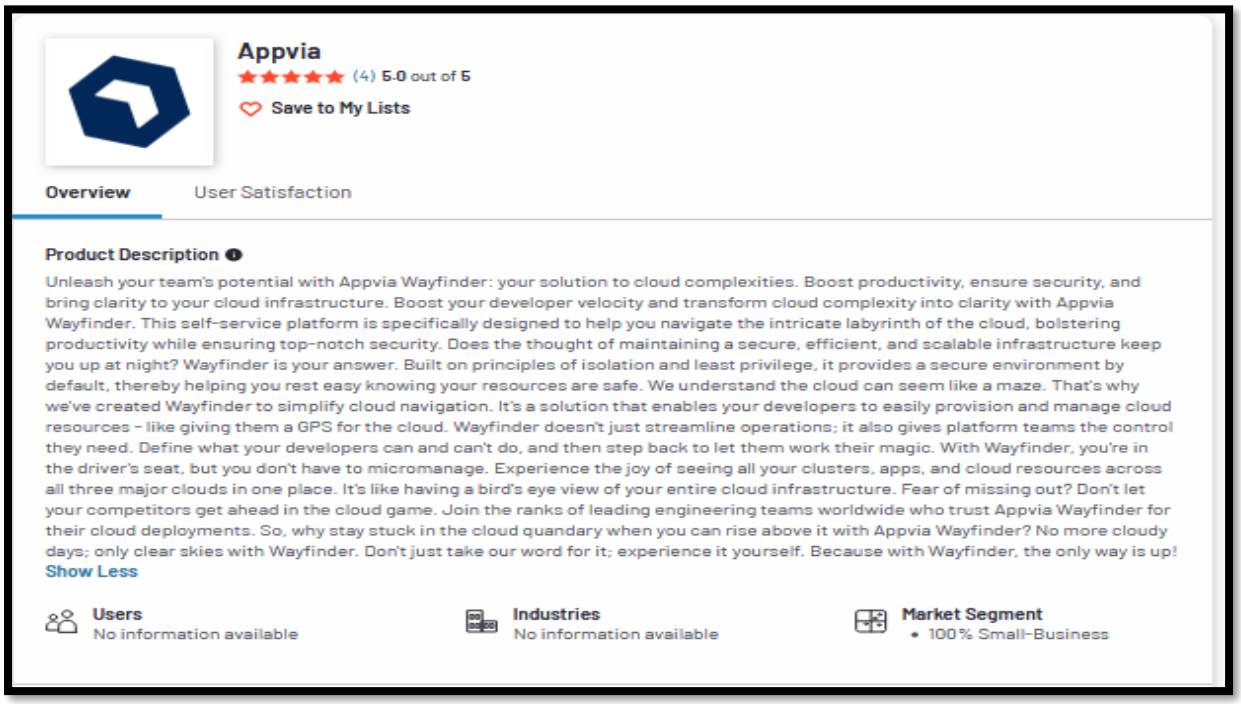
Spacelift is a sophisticated CI/ CD abstraction layer on top of all popular OpenSource Infrastructure as Code (IaC) tools or wrappers with a backend that supports ~ ANY cloud / on prem ~ ANY VCS provider and ~ ANY Security or Observability tool To visualize and manage cloud, multi cloud, or hybrid / on- premises resources, detect and prevent drift, implement security RBAC, contain costs (FinOps), or drive a strong DevSecOps motion by codified policy embedded in the full development lifecycle! Spacelift automates blueprints to drag and drop environments & liberate developers and platform teams to ship fast within precise policy-driven guardrails! Modern fintechs, HealthTechs, Pharma, Retailers, Banks, Finserv, and software teams use our infrastructure management platform to move at a higher velocity without sacrificing security and compliance. Spacelift offers the optionality of a SaaS, Hybrid SaaS, or Self Hosted Deployment. Each deployment option allows end customers to choose whether they manage or we manage state, plus have over 23+ native integrations! Spacelift CI/ CD for IaC platform is vendor agnostic, so regardless of the toolchain, you'll always have a cohesive view into what's going on with your most valuable resources! Show Less

Users
No information available

Industries
No information available

Market Segment
- 67% Mid-Market
- 33% Enterprise



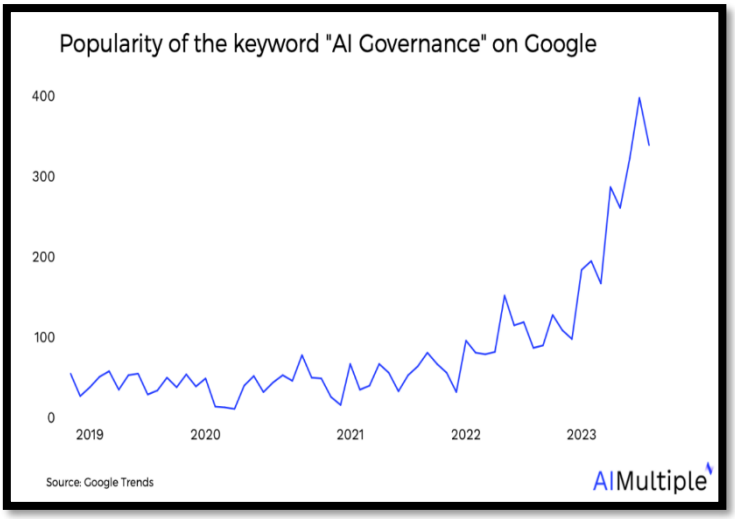


5.1 Artificial Intelligence

According to AI stats, 90% of commercial apps are expected to leverage AI by the end of 2025. Despite the increasing role of AI, business leaders are concerned due to:

The bias in data and algorithms leading to erroneous outcomes in 85% of AI projects. Authorities like the EU planning to add AI regulations. Similar regulations like the GDPR can cost 4% of a firm's worldwide annual revenue in case of infringements.

One recent answer to this problem is the adoption of AI governance tools (See Google Trends Graph). These tools can help deliver ethical and responsible AI. However, choosing an AI governance tool can be challenging for business leaders and data scientists since

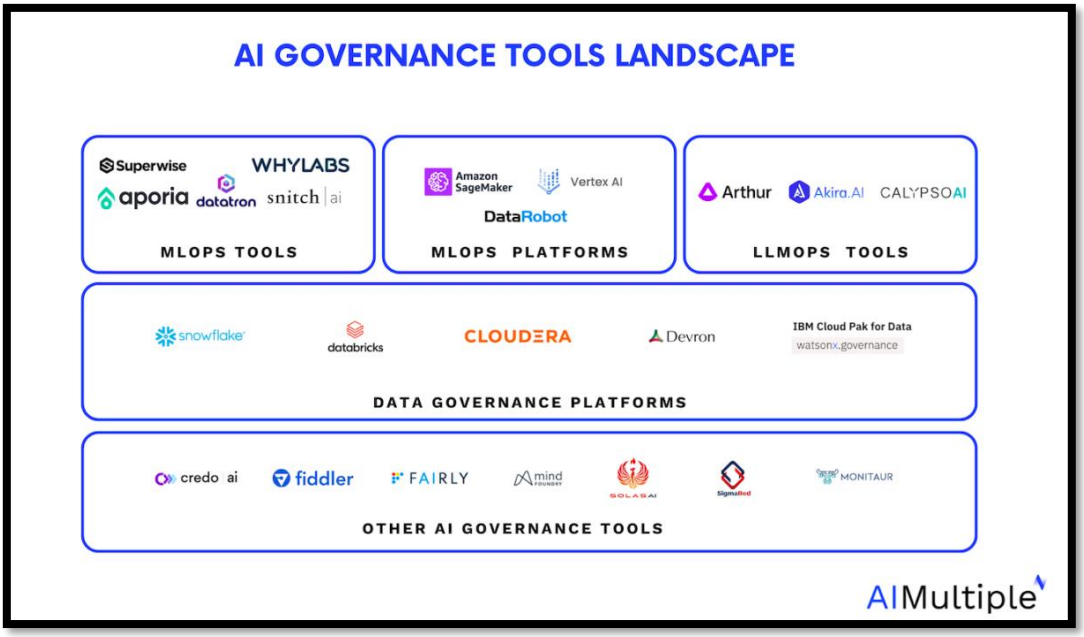


The AI governance software landscape is complex.

AI governance solutions are provided in MLOps, LLMOps tools and data platforms.

Therefore, this article provides an comprehensive view of AI governance software, enabling data-driven decision-making.

Compare AI Governance Software: We provide a list of AI governance solutions within a framework to highlight each tool's focus area. Depending on an organization's AI initiatives and governance needs, they can select tools from these categories to build a robust AI governance strategy that aligns with their goals and responsibilities.



AI governance tools landscape

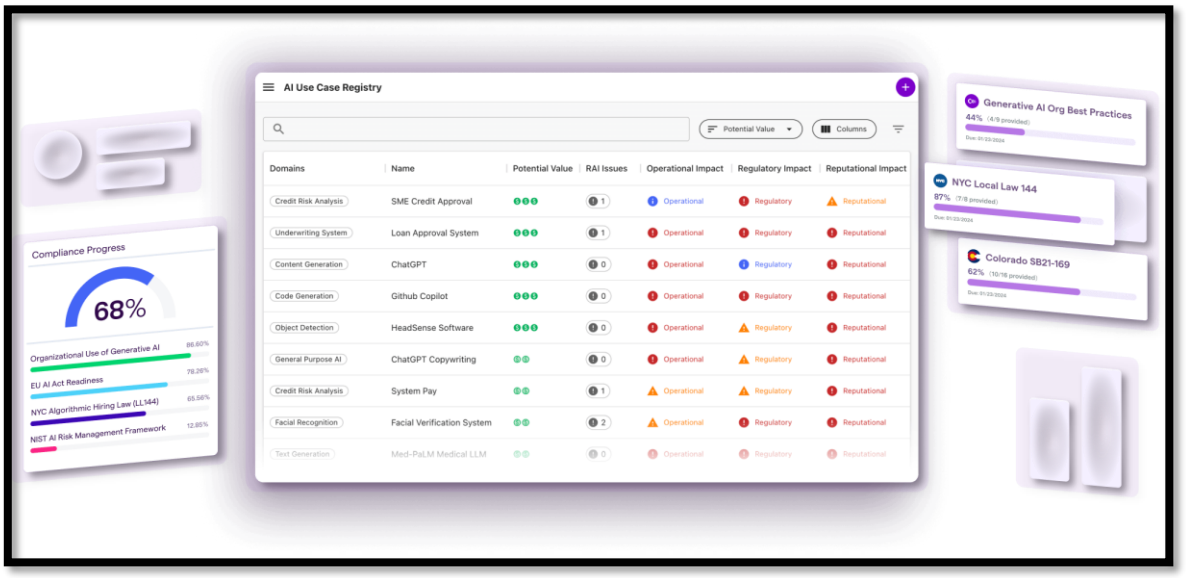
Other AI governance tools: These tools tend to focus on an aspect of AI governance, unlike platforms that manage the entire AI lifecycle. Such tools can be useful for small-scale projects or best-of-breed approaches. For example, they can focus on ensuring that AI systems comply with industry regulations and security standards. They help organizations mitigate AI risk by:

- Implementing security measures

- Staying in line with regulatory requirements and laws
- Managing model documentation.

Some of these tools include:

1.) **Credo AI:** Delivers AI model risk management, model governance and compliance assessments with an emphasis on generative AI to facilitate the adoption of AI technology.



Credo AI platform, an example from an AI governance tool

2.) **Fiddler AI:** An AI observability tool that provides ML model monitoring and relevant LLMOps and MLOPs features to build and deploy trustable AI, including generative AI.

3.) **Fairly AI:** Continuously monitors, governs and audits models to reduce risk and improve compliance.

4.) **Mind Foundry:** Monitor and validate AI models, maintain transparency in decision-making, and align AI behavior with ethical and regulatory standards, fostering responsible AI governance.

5.) **Monitaur**: Focuses on model monitoring for AI governance.

6.) **Sigma Red AI:** Detects and mitigates biases, ensuring model explainability and facilitating ethical AI practices.

7.) **Solas AI:** Checks for algorithmic discrimination to increase regulator and legal compliance.

Data Governance platforms

Data governance platforms contain various tools and toolkits primarily focused on data management to ensure the quality, privacy and compliance of data used in AI applications. They contribute to maintaining data integrity, security, and ethical use, which are crucial for responsible AI practices.

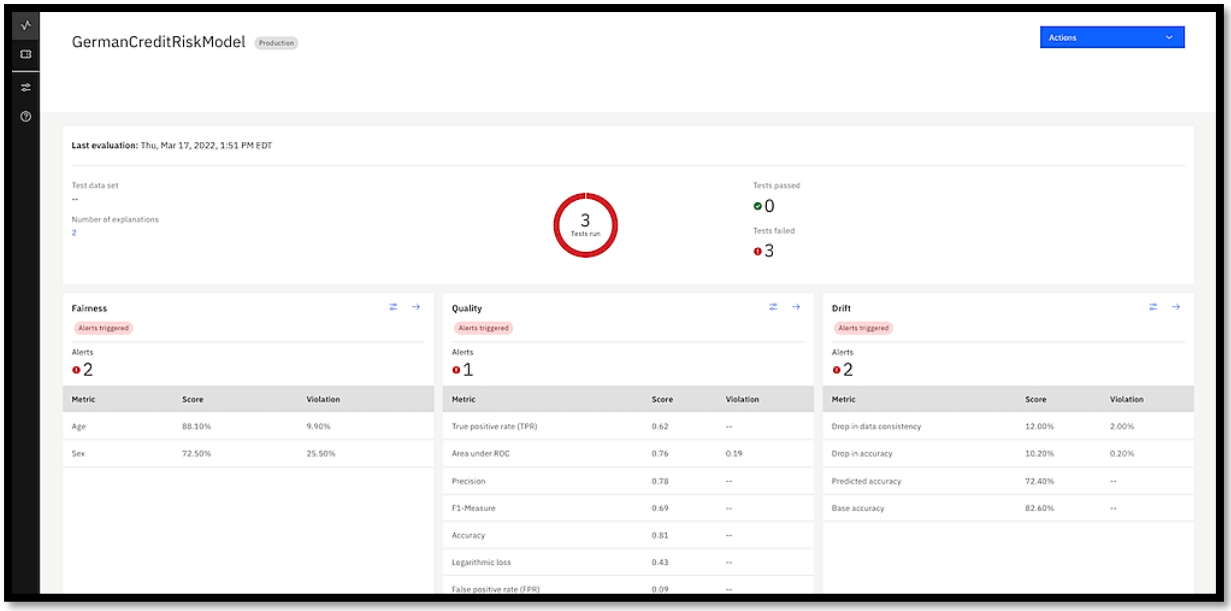
Some of these platforms can help check compliance and overall AI lifecycle management. These platforms can be valuable for organizations implementing comprehensive AI governance frameworks. Here are a few examples:

1.) **Cloudera:** A hybrid data platform that aims to improve the quality of data sets and ML models, focusing on data governance.

2.) **Databricks:** Combines data lakes and data warehouses in a platform that can also govern their structured and unstructured data, machine learning models, notebooks, dashboards and files on any cloud or platform.

3.) **Devron AI:** Offers a data science platform to build and train AI models and ensure that models meet governance policies and compliance requirements, including GDPR, CCPA and EU AI Act.

4.) **IBM Cloud Pak for Data**: IBM's comprehensive data and AI platform, offering end-to-end governance capabilities for AI projects:



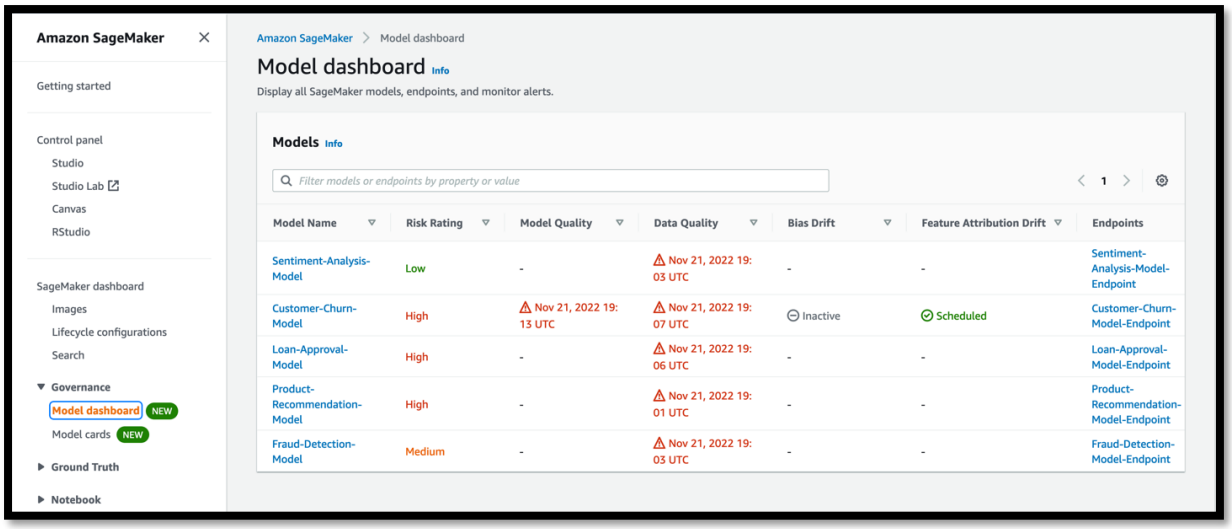
IBM Openscale, an example from a data governance tool

6.) **Snowflake:** Delivers a data cloud platform that can manage risk and improve operational efficiency through data management and security.

MLOps platforms:

MLOps (Machine learning operations) platforms include a wide range of tools and infrastructure to manage the entire machine learning lifecycle and assist model governance.

1.) **Amazon Sagemaker:** A fully managed machine learning service provided by Amazon Web Services (AWS). It simplifies the process of building, training, and deploying machine learning models, considering AI governance practices.



Amazon Sagemaker ML governance dashboard, an MLOps platform

2.) Datarobot: Delivers a single platform to deploy, monitor, manage, and govern all your models in production, including features like trusted AI and ML governance to provide an end-to-end AI lifecycle governance.

3.) Vertex AI: Offers a range of tools and services for building, training, and deploying machine learning models with AI governance techniques, such as model monitoring, fairness, and explainability features.

Compare more MLOPs platforms in our data-driven and comprehensive vendor list.

MLOps tools

MLOps tools are individual software tools that serve specific purposes within the entire machine learning process. For example, MLOps tools can focus on ML model development, monitoring or model deployment. A data science team can deliver responsible AI products by applying these tools to machine learning algorithms to:

- Monitor and detect biasses
- Check for availability and transparency
- Ensure ethical compliance and data privacy.

Some of these tools include:

1.) Aporia AI: Specialized in ML observability and monitoring to maintain the reliability and fairness of their machine learning models in production. It employs model performance tracking, bias detection, and data quality assurance.

2.) Datatron: Provides visibility into model performance, enables real-time monitoring, and Ensures compliance with ethical and regulatory standards, thus promoting responsible and accountable AI practices.

3.) **Snitch AI:** An ML observability and model validator which can track model performance, troubleshoot and continuously monitor.

4.) **Superwise AI:** Monitor AI models in real-time, detect biases, and explain model decisions, thereby promoting transparency, fairness, and accountability in AI systems.

5.) **Why Labs:** An LLMOps tool that monitors LLMs data and mode to identify issues.

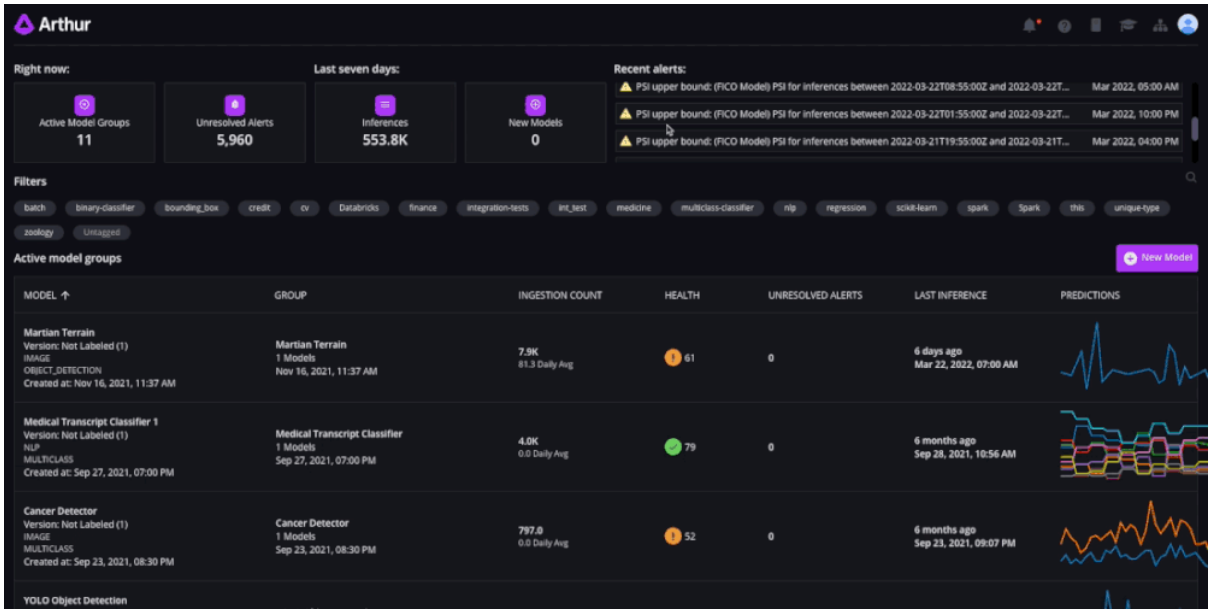
LLMOps tools

LLMOps tools include LLM monitoring solutions and tools that assist some aspects of LLM operations. These tools can deploy AI governance practices in LLMs by monitoring multiple models and detecting biases and unethical behavior in the model. Some of them include:

1.) **Akira AI:** Runs quality assurance to detect unethical behavior, bias or lack of robustness.

2.) **Calypso AI:** Delivers monitoring considering control, security and governance over generative AI models.

3.) **Arthur AI:** It tests LLMs, computer vision and NLPs (natural language processing) against established metrics to



Arthur AI, LLM governance tool, an example from dashboard

Compare more LLMOps tools in our data-driven and comprehensive vendor list.

What is AI governance?

AI governance refers to establishing rules, policies, and frameworks that guide the development, deployment, and use of artificial intelligence technologies. It aims to ensure ethical behavior, transparency, accountability, and societal benefit while mitigating potential risks and biases associated with AI systems.

Why do we need AI governance?

AI governance serves multiple essential purposes:

Ethical and Responsible AI: Ensures AI systems are designed, trained, and used ethically, preventing biased or harmful outcomes.

Transparency and Accountability: Promotes transparency in AI algorithms and decisions, making developers and organizations accountable for AI actions.

Data Privacy and Compliance: Helps organizations comply with data privacy regulations like GDPR and HIPAA, ensuring that data is collected and used legally and ethically.

Risk Mitigation: Identifies and mitigates various risks associated with AI, including legal, financial, and reputational risks, before they lead to negative consequences.

Fairness and Equity: Detects and rectifies bias in AI models, ensuring fair treatment of all individuals and groups.

Model Performance and Reliability: Continuously monitors AI models to maintain reliability, reducing errors and improving user satisfaction.

Public Trust: Builds public trust in AI technologies by emphasizing ethical behavior and transparency.

Alignment with Organizational Values: Allows organizations to align AI practices with their mission and values, demonstrating a commitment to ethics and responsibility.

Competitive Advantage: Ethical AI and responsible governance can provide a competitive edge by attracting customers, partners, and investors who value ethical AI solutions.

Key AI governance techniques & features

AI governance software employs common techniques to streamline building and deploying AI/ML models, such as:

Explainability and Interpretability: AI governance software employs visualizations and explanations for AI model outputs to provide insights into how AI models make decisions. These tools allow users to understand and predict complex model behaviour.

Transparency and Accountability: AI governance provides clear documentation of model training data and processes, which enables monitoring of model decisions for accountability.

Fairness and bias detection: AI governance practices mainly focus on identifying and quantifying biases in AI models and data. For example, AI governance tools can monitor model performance across different demographic groups, allowing to mitigate biases in real-time or during training. Two main ways to detect bias in the model is to ensure compliance with ethics and law:

- **Ethical AI compliance:** AI governance primarily aligns AI behavior with ethics by implementing guidelines and constraints. As a result, a data scientist can customize AI behavior to avoid harmful and offensive outputs of AI systems.
- **Regulatory compliance:** A major AI governance practice is to ensure adherence to legal and regulatory requirements, meet data privacy and security standards and help business users comply with industry-specific regulations.

Model lifecycle management: Once a model is ready, AI governance techniques can manage the deployment of the model in the production environment by monitoring models for drift, degradation, or unexpected behavior. Two features that can facilitate AI deployment include:

- Model validation and testing: Some AI governance tools can contain model validator features to test and verify models against benchmark datasets. Deploy these tools before production to detect potential issues.
- Model risk management: AI governance techniques provide insights to assess and mitigate risks for AI systems.
- Continual monitoring and auditing: Another common practice is tracking the model performance in production and behavior to ensure compliance and reliability in AI systems.

How to select the right AI governance software?

Identify your objective and scale: Consider the scale of your AI initiatives and the types of AI models and applications you are developing.

Research and evaluate available tools in the market: Look for vendors that specialize in the areas most relevant to your needs. Create a shortlist of promising tools based on their features, capabilities, and user reviews.

Benchmark the shortlisted tools based on the following:

- Each tool's features: Assess its ability to detect bias, ensure data privacy, provide transparency, and monitor compliance.
- Ease of integration: Assess how well the AI governance tool integrates with your existing AI development and deployment pipeline.
- Compatibility with your organization: Check for compatibility with the programming languages, frameworks, and platforms you use for AI development. Ensure the tool can work seamlessly with your data sources, storage solutions, and cloud providers.

User-friendly interface: How intuitive the tool is for seamless interaction.

Customization and flexibility: The extent to which the tool can be customized to match your requirements, allowing you to adjust settings and configurations.

Scalability: Consider the tool's scalability to accommodate your organization's growth in AI initiatives, such as increasing data volumes and workloads as your projects grow.

Quality of vendor support: Investigate the level of customer support, response time and assistance provided.

Training and resources: Review how comprehensive is the documentation, tutorials, user guides, online sources and training materials. Remember that adequate resources to help your team learn how to use the tool effectively.

Cost and budget: Evaluate the cost structure of the AI governance tool, including licensing fees, subscription costs, and implementation expenses. Calculate the long-term costs and benefits of the tool to ensure it provides value over time based on your financial resources.

Data security and privacy: Check compliance with data protection regulations, including encryption and access controls. Ensure the security and confidentiality of sensitive information.

Seek free trial and proof of concept (if applicable): Conduct a trial or proof of concept (PoC) with the selected AI governance software. You may use real or simulated AI projects to assess how well the tool addresses your governance needs. Involve key stakeholders, data scientists, and AI developers in the PoC to gather feedback on usability and effectiveness.

5.2 Cybersecurity

Complete 360° view of Cybersecurity

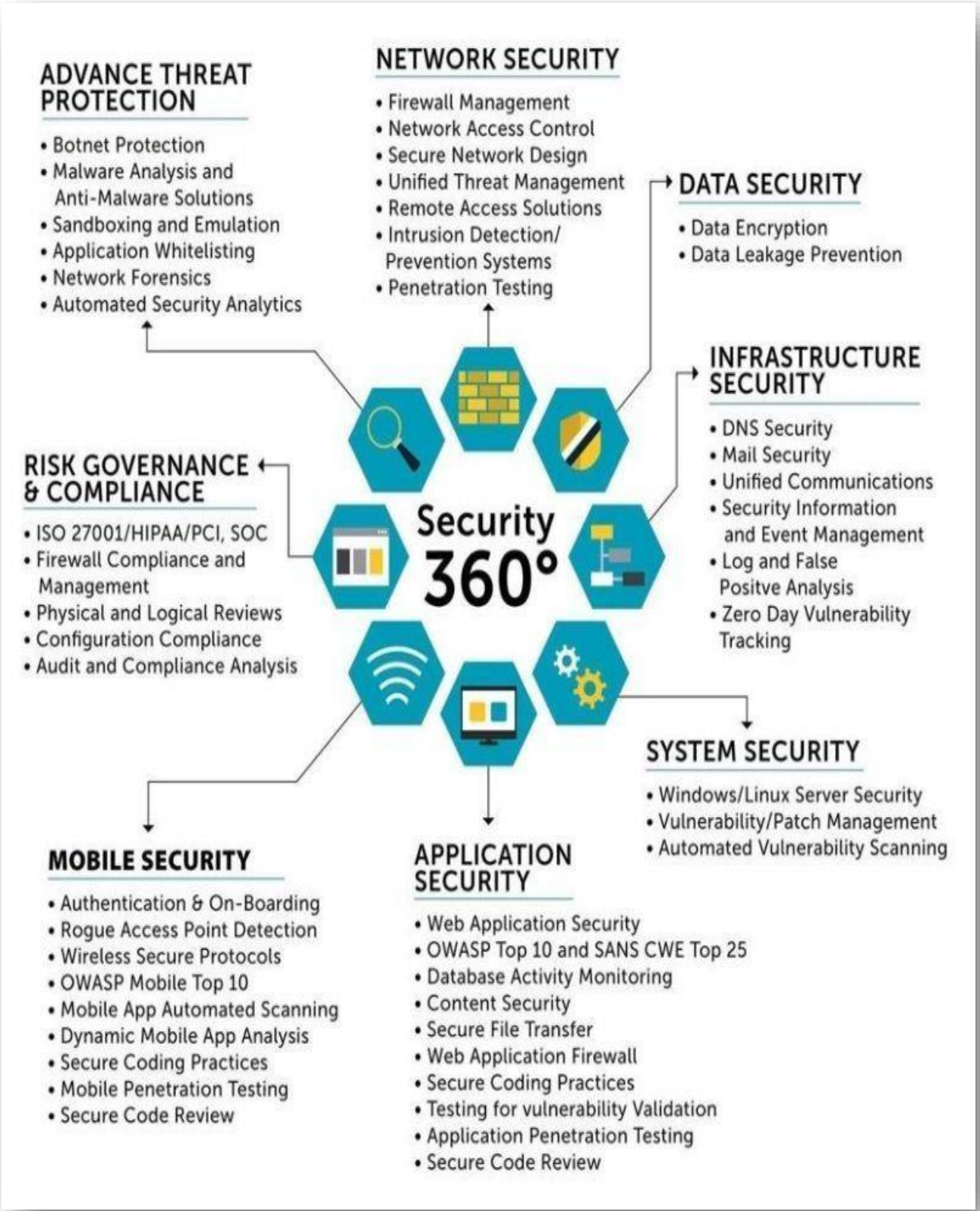
Regarding data security, there are many important factors to consider. As someone who works in this space, the following are essential, which are missing in the infographic.

- Data classification
- Data governance
- Data encryption
- Data masking/tokenization
- IAM/PAM
- Cryptokey management
- DLP
- EDRM
- SFTP
- FIM
- Data backup and recovery.

As technology advances, so does the need for enhanced security measures. It's important to stay up-to-date on the latest security trends to ensure your organization's protection. Here are six key areas to focus on:

- Cloud security
- Privacy Security Control
- Blockchain security
- Security by Design
- Security Architecture
- OT Security

By familiarizing yourself with these concepts, you can better protect yourself and your organization from potential security threats. Stay informed and stay secure!

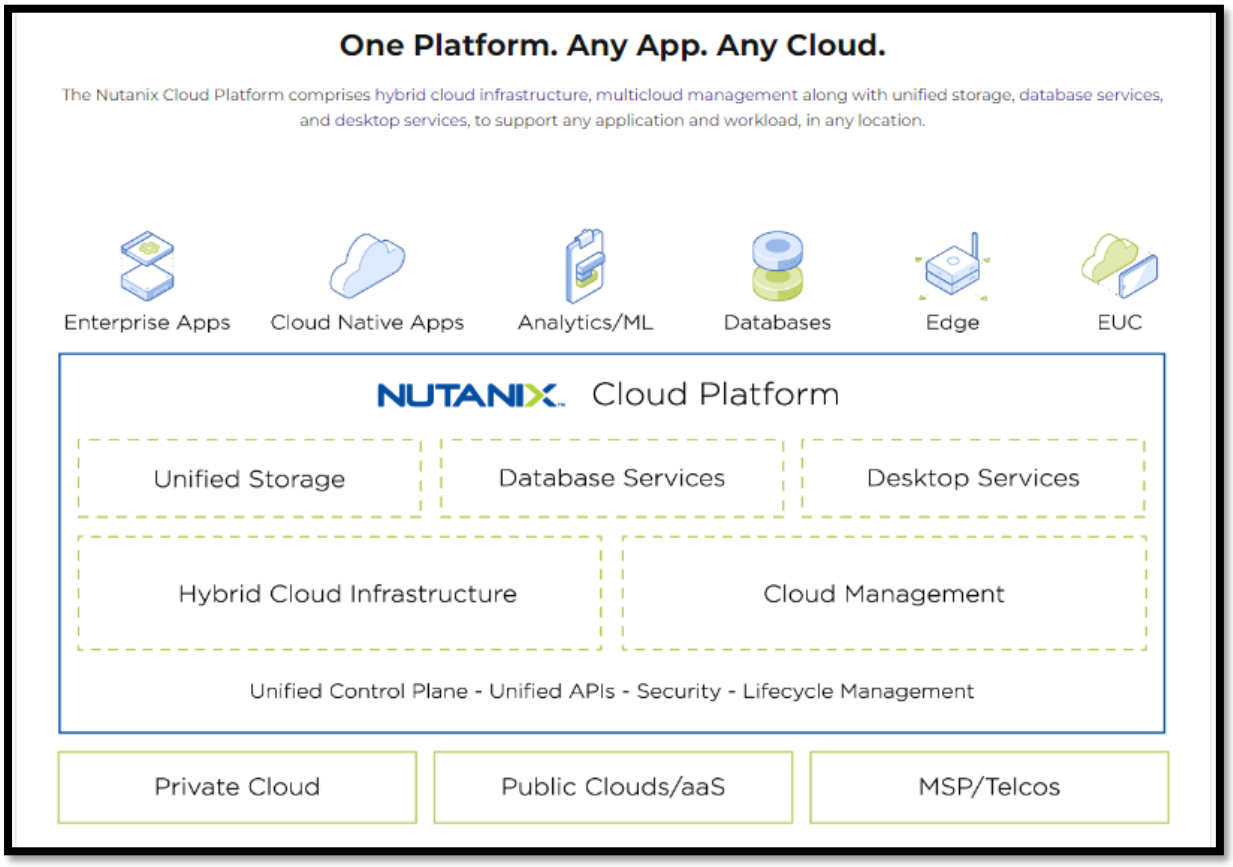

ADVANCE THREAT PROTECTION
• Botnet Protection
• Malware Analysis and Anti-Malware Solutions
• Sandboxing and Emulation
• Application Whitelisting
• Network Forensics
• Automated Security Analytics
NETWORK SECURITY
• Firewall Management
• Network Access Control
• Secure Network Design
• Unified Threat Management
• Remote Access Solutions
• Intrusion Detection/ Prevention Systems
• Penetration Testing
DATA SECURITY
• Data Encryption
• Data Leakage Prevention
INFRASTRUCTURE SECURITY
• DNS Security
• Mail Security
• Unified Communications
• Security Information and Event Management
• Log and False Positve Analysis
• Zero Day Vulnerability Tracking
RISK GOVERNANCE & COMPLIANCE
• ISO 27001/HIPAA/PCI, SOC
• Firewall Compliance and Management
• Physical and Logical Reviews
• Configuration Compliance
• Audit and Compliance Analysis
Security 360°
SYSTEM SECURITY
• Windows/Linux Server Security
• Vulnerability/Patch Management
• Automated Vulnerability Scanning
MOBILE SECURITY
• Authentication & On-Boarding
• Rogue Access Point Detection
• Wireless Secure Protocols
• OWASP Mobile Top 10
• Mobile App Automated Scanning
• Dynamic Mobile App Analysis
• Secure Coding Practices
• Mobile Penetration Testing
• Secure Code Review
APPLICATION SECURITY
• Web Application Security
• OWASP Top 10 and SANS CWE Top 25
• Database Activity Monitoring
• Content Security
• Secure File Transfer
• Web Application Firewall
• Secure Coding Practices
• Testing for vulnerability Validation
• Application Penetration Testing
• Secure Code Review

6. Some Leading companies facilitating Cloud Transformation

6.1 Nutanix

Nutanix, Inc. is an American cloud computing company that sells software, cloud services (such as desktops as a service, disaster recovery as a service, and cloud monitoring), and software-defined storage. Power any workload in any cloud

Nutanix Cloud Platform delivers the performance, resilience, and scalability to power all workloads—virtualized, container-based, and bare metal in the cloud of your choice.



Operations

Nutanix combines storage, computing, and virtualization. The company's software product families include Acropolis, Prism, Era, Frame, and Files. In 2015, Nutanix was reported to have built a Linux KVM based hypervisor, called AHV (Acropolis HyperVisor) in order to make managing computer infrastructure easier.[35]

Nutanix marketed its products as "infrastructure". In 2020, the company shifted to a subscription business model.[37]

Accelerate your business transformation



Start your cloud journey

Start your transformation with HCI as your cloud foundation and move from private cloud to a hybrid multicloud environment in a snap

Elevate your business ›



Deliver hybrid multicloud

Learn how enabling hybrid multicloud with Nutanix gives you the flexibility to pick the right resources for the right workload and adjust seamlessly as your business grows

Design your cloud ›



Manage IT costs

Find out which cloud - between public, private and hybrid - is right for your business based on current workloads and five-year total cost of ownership

Optimize your cloud ›



Simplify your cloud journey

Read more to see how Nutanix Hybrid Multicloud can help you accelerate your innovation

Enhance your operations ›

The Nutanix Cloud Platform advantage

Power of simplicity



1-click operations

Automated infrastructure management enables 1-click deploy, patching, provisioning, and scaling with a simple, intuitive UI



Self service IaaS

Give your users the power to provision their own apps and services with built-in role-based governance



Multicloud management

Experience a unified cloud control plane across private and public clouds for AI-powered application automation, self-service, cost and security governance

Simplify cloud

Data centricity



Data protection and security

Protect your data from hardware failure, disaster, and security threats with integrated protection



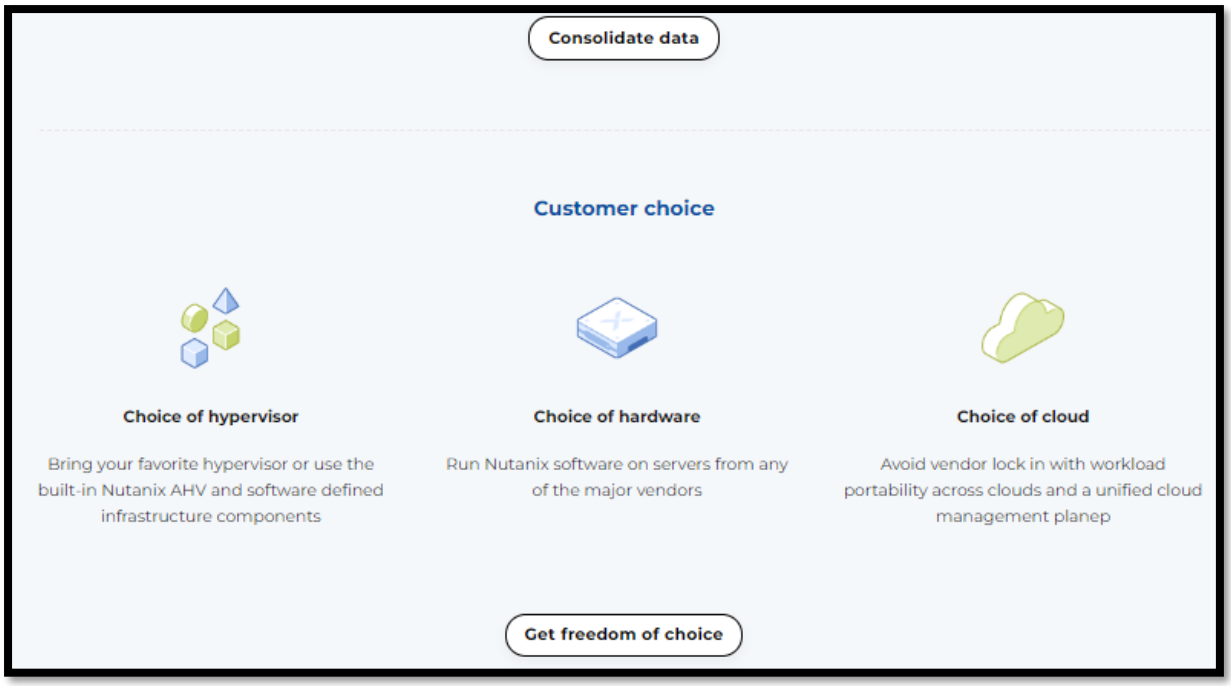
Storage for all workloads and data

Integrated storage for VMs & Containers, Files and Objects, and block storage for bare-metal workloads



Database automation

Simplify database operations across any cloud using a unified control plane



6.2 Trianz

Trianz enables digital transformations through effective strategies and excellence in execution. Collaborating with business and technology leaders, we help formulate and execute operational strategies to achieve intended business outcomes by bringing the best of consulting services, technology expertise, and execution models.

Powered by knowledge, research, and perspectives, we enable clients to transform their business ecosystems and achieve superior performance by leveraging infrastructure, cloud, analytics, digital, and security paradigms. Reach out to get in touch or learn more.



6.3 CrowdStrike

Single, universal agent for security consolidation

CrowdStrike Falcon gives you an intelligent, lightweight agent that consolidates point products and stops advanced attacks — both malware and malware-free — while capturing rich endpoint activity for industry-leading detection and response.

- All platform capabilities delivered via one agent that is effortless to deploy and manage
- Easy deployment with a cloud-native architecture for instant operationalization, with no reboots required after installation
- Single agent extends across on-premises, remote deployments, and cloud workloads for consistent visibility and protection across endpoint, clouds and identities
- Lightweight agent with minimal impact on endpoint performance and end-user productivity
- Simplified management from one unified console

The CrowdStrike Falcon platform is designed as a highly modular and extensible offering that helps customers solve new security challenges with a single click and without the need to re-architect or re-engineer the architecture, removing friction associated with security deployments.

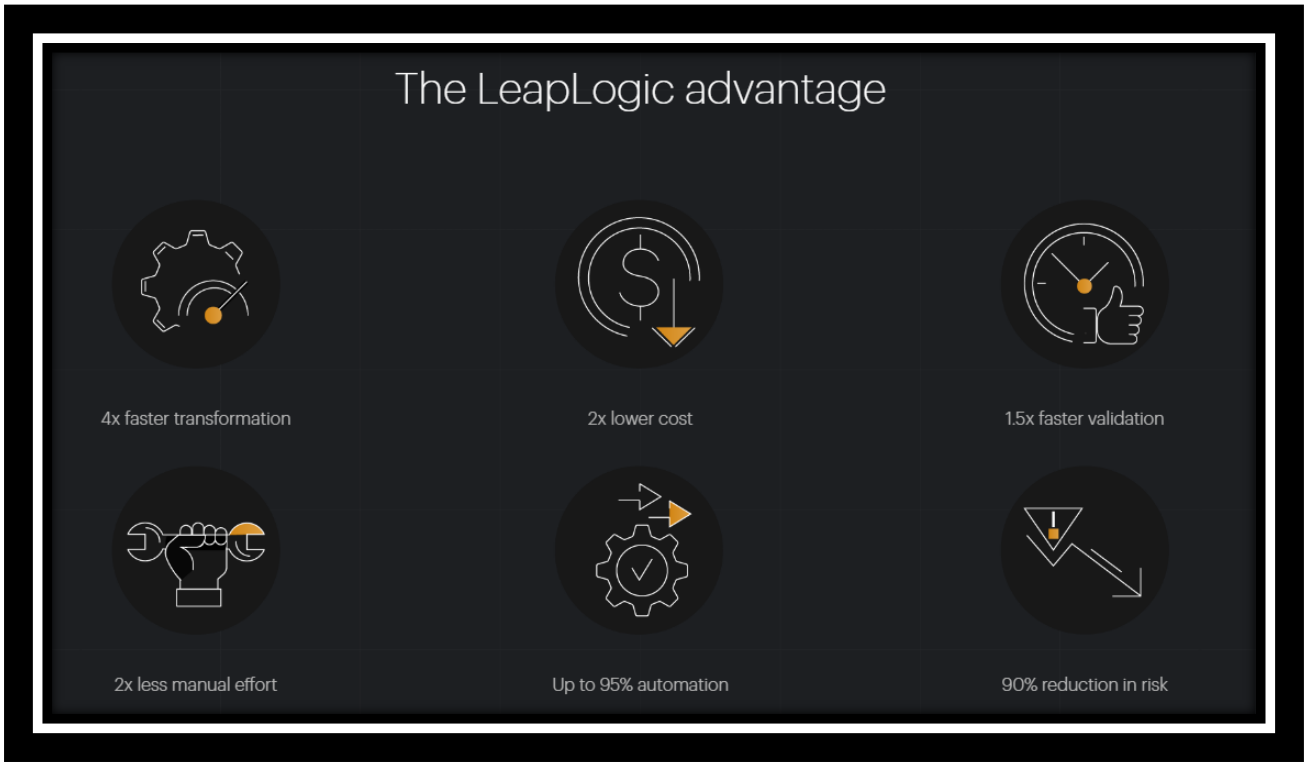
Endpoint Security & XDR Next-gen antivirus, EDR, and XDR. Cloud Security Unified cloud security, from endpoint to cloud. Identity Protection Stop identity-based attacks in /real-time.	Threat Intelligence Pioneering adversary intelligence to prepare and rapidly react to threats. Risk-based Vulnerability Management Reduce risk with unmatched real-time visibility into the devices, users and applications in your network. Observability & Log Management Centralize all log data for security, IT observability, and compliance.	Managed Detection and Response 24/7 expertise to close the cybersecurity gap. CrowdStrike Services Pre- and Post-Incident Response service CrowdStrike Marketplace Enterprise marketplace to discover, try buy and deploy trusted partner application extend their investment in the platform

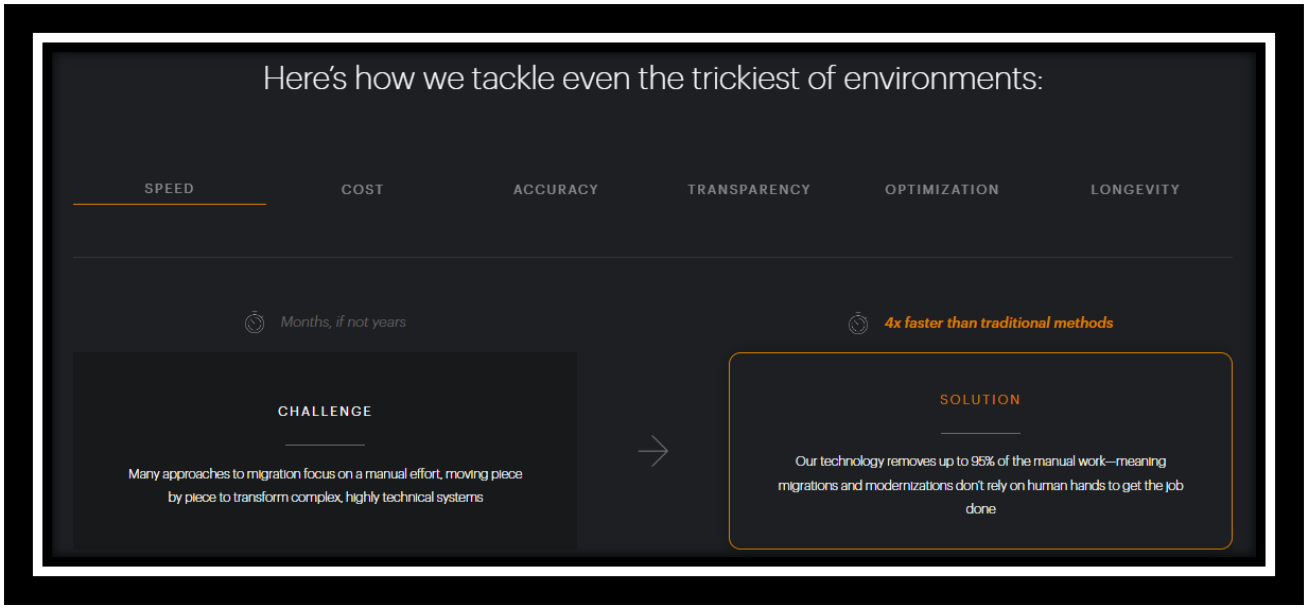
6.4 LeapLogic

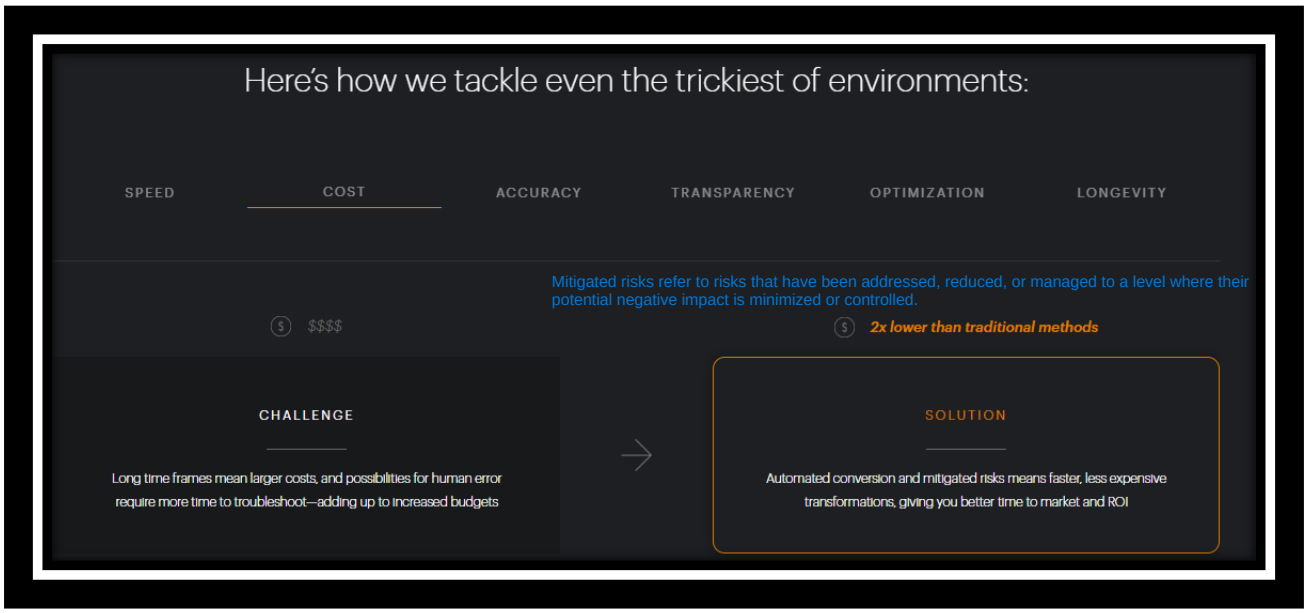
Modernize any legacy data platform to cloud-native stacks with up to 95% automation. Reimagine cloud migration of legacy DW, ETL, Hadoop, and analytics systems. Automated Cloud Migration. Zero Business Disruption. Modernize to any cloud. Migrate at 2x lower cost.

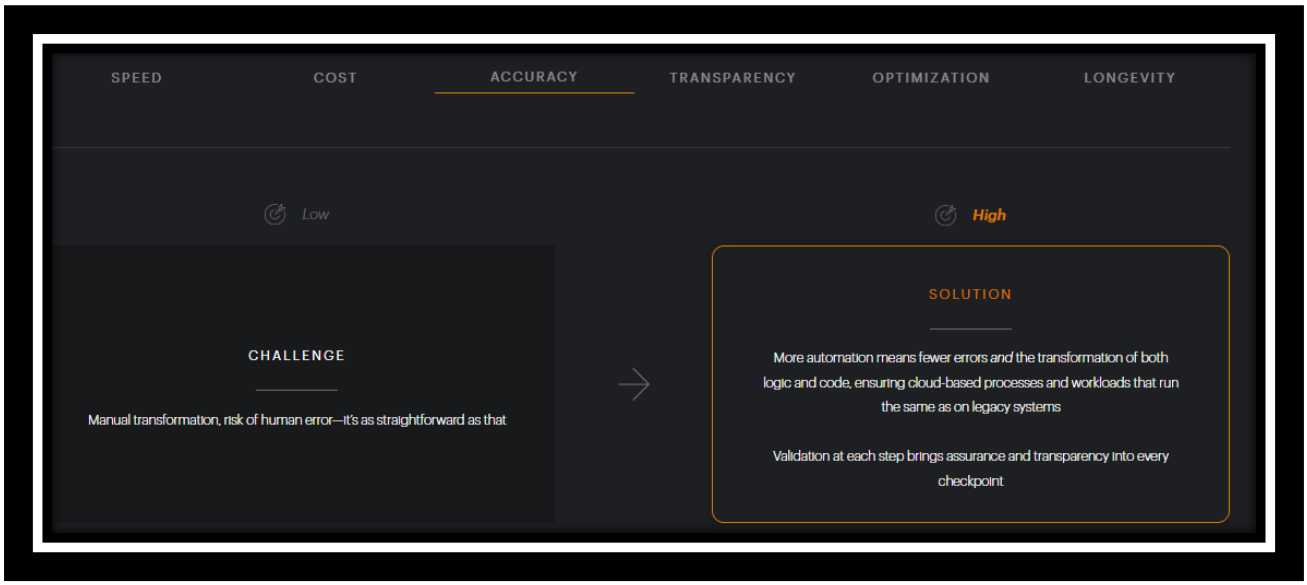
LeapLogic brings intelligence and automation to the modernization process for data warehouses, ETL, Hadoop, and analytics systems, improving speed and accuracy while lowering risks and costs

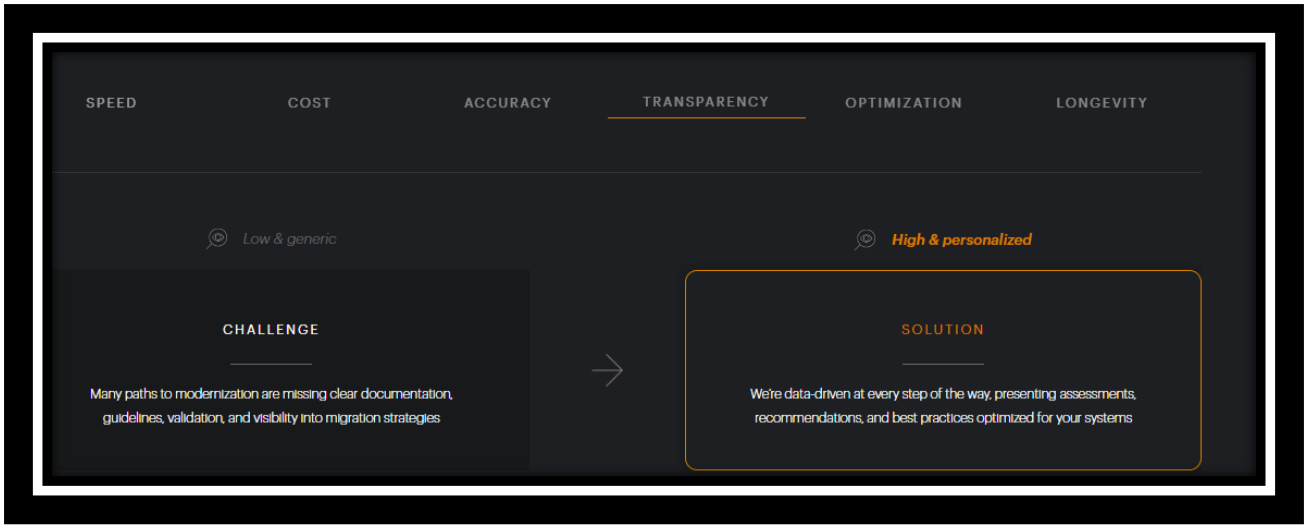

STEP 1
ASSESSMENT
Complete analysis of workloads, code profiling, and dependencies with actionable recommendations
STEP 2
TRANSFORMATION
End-to-end transformation, including core business logic to target-native equivalents
STEP 3
VALIDATION
Validation for pipelines, data, and row and cell-level queries
STEP 4
OPERATIONALIZATION
Target-specific executable packaging with optimal price-performance ratio

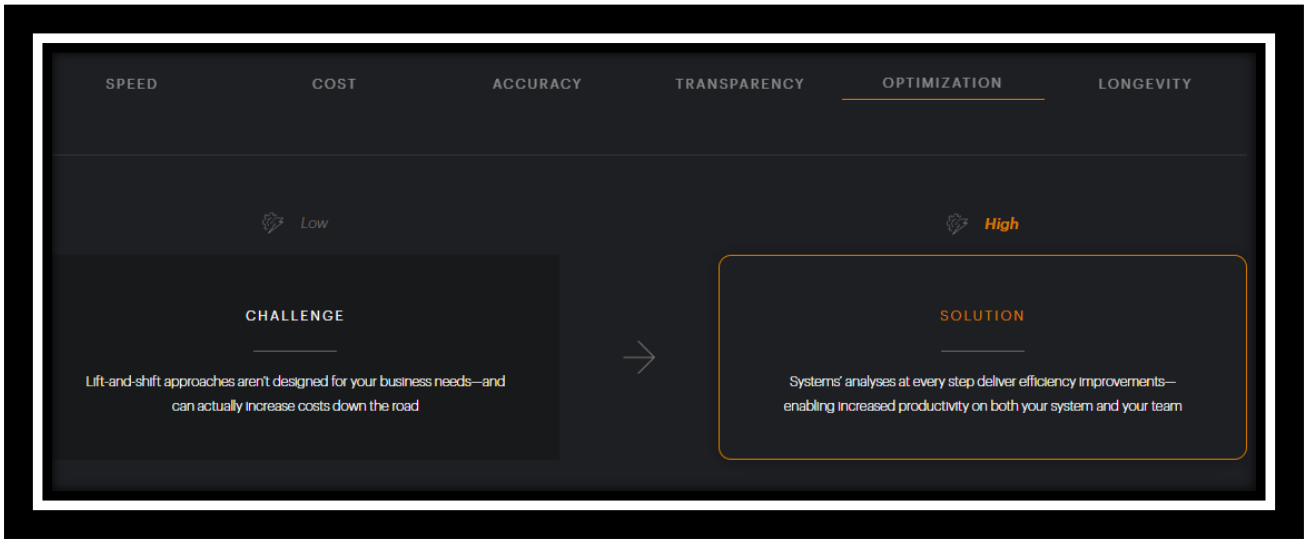

The LeapLogic advantage
4x faster transformation
2x lower cost
1.5x faster validation
2x less manual effort
Up to 95% automation
90% reduction in risk

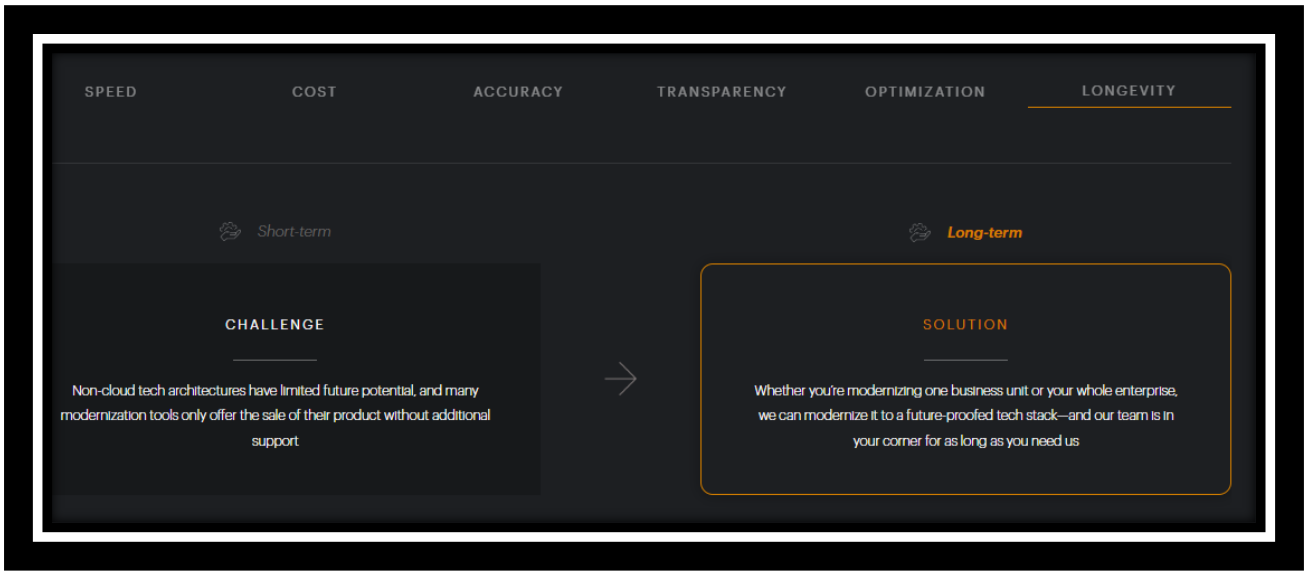

Here's how we tackle even the trickiest of environments:
SPEED
COST
ACCURACY
TRANSPARENCY
OPTIMIZATION
LONGEVITY
Months, if not years
4x faster than traditional methods
CHALLENGE
Many approaches to migration focus on a manual effort, moving piece by piece to transform complex, highly technical systems
SOLUTION
Our technology removes up to 95% of the manual work—meaning migrations and modernizations don't rely on human hands to get the job done


Here's how we tackle even the trickiest of environments:
SPEED
COST
ACCURACY
TRANSPARENCY
OPTIMIZATION
LONGEVITY
Mitigated risks refer to risks that have been addressed, reduced, or managed to a level where their potential negative impact is minimized or controlled.
$$$$
2x lower than traditional methods
CHALLENGE
Long time frames mean larger costs, and possibilities for human error require more time to troubleshoot—adding up to increased budgets
SOLUTION
Automated conversion and mitigated risks means faster, less expensive transformations, giving you better time to market and ROI


SPEED
COST
ACCURACY
TRANSPARENCY
OPTIMIZATION
LONGEVITY
Low
High
CHALLENGE
Manual transformation, risk of human error—it's as straightforward as that
SOLUTION
More automation means fewer errors and the transformation of both logic and code, ensuring cloud-based processes and workloads that run the same as on legacy systems
Validation at each step brings assurance and transparency into every checkpoint


SPEED
COST
ACCURACY
TRANSPARENCY
OPTIMIZATION
LONGEVITY
Low & generic
High & personalized
CHALLENGE
Many paths to modernization are missing clear documentation, guidelines, validation, and visibility into migration strategies
SOLUTION
We're data-driven at every step of the way, presenting assessments, recommendations, and best practices optimized for your systems


SPEED
COST
ACCURACY
TRANSPARENCY
OPTIMIZATION
LONGEVITY
Low
High
CHALLENGE
Lift-and-shift approaches aren't designed for your business needs—and can actually increase costs down the road
SOLUTION
Systems' analyses at every step deliver efficiency improvements—enabling increased productivity on both your system and your team


SPEED
COST
ACCURACY
TRANSPARENCY
OPTIMIZATION
LONGEVITY
Short-term
Long-term
CHALLENGE
Non-cloud tech architectures have limited future potential, and many modernization tools only offer the sale of their product without additional support
SOLUTION
Whether you're modernizing one business unit or your whole enterprise, we can modernize it to a future-proofed tech stack—and our team is in your corner for as long as you need us


Accelerate into modernization
Automation to deliver accuracy and efficiency
Prescriptive analysis to bring visibility to what your business needs
End-to-end optimization to ensure transformation without disruption